01. Introduction

This course will mainly focus on core elements of digital devices like Diodes, BJT, FET and Op-amp etc. Clipper, Clamper circuits, Rectifiers implementation using diodes, Biasing of BJT and FET, Small signal Analysis of BJT and FET, Integrator circuits and filters using Op-amp, adder circuit and Wien Bridge oscillator using Op-amp, Design an astable, Mono-stable and Bi-stable circuit using 555 timer IC.

02. Course Objectives

To learn about different Electronic Devices like Diodes, Zener Diodes, Bipolar Junction Transistors, Field Effect Transistors, Metal Oxide Semiconductor Field Effect Transistors, Operational Amplifier, 555 IC etc. and their applications.

03. Course Outcomes

Upon completion of this course students will be able to:

CO1: Investigate the performance of different digital devices, its characteristics and graphical analysis in electronic circuits**.**
CO2: Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.
CO3: Select and use modern hardware and software tools and devices
CO4: Assess societal, health, safety, legal and cultural issues involved with the mini project
CO5: Realize the impact of societal, environmental and sustainable development issues for the design solution of mini project.
CO6: Apply professional ethics and responsibilities in the implementation of mini project.
CO7: Work individually and in a team
CO8: Communicate and share knowledge, data, information, results etc. with others
CO9: Apply engineering project management knowledge and skill to implement the mini project.
CO10: Gather and apply knowledge, data and information from various multidisciplinary sources to analyze, design and implement the mini project.

04. Course Outcomes (COs), Program Outcomes (POs)

<table>
<tr><th>CO Statement</th><th>Corresponding PO</th><th>Domain / level of learning taxonomy</th><th>Delivery methods and activities</th><th rowspan="8">Assessment tools

☑ Lab tests
☑ Lab reports
☑ Final lab test

☐ Open ended lab
☑ Project show & project presentation</th></tr>
<tr><td>CO1: Investigate the performance of different digital devices, its characteristics and graphical analysis in electronic circuits</td><td>PO4–Investigation</td><td>Cognitive/ Evaluating</td><td rowspan="7">☑ Simulation
☑ Experiment
☑ Practice lab

☑ Group discussion
☐ Tutorial</td></tr>
<tr><td>CO2: Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.</td><td>PO4–Investigation</td><td>Cognitive/ Evaluating</td></tr>
<tr><td>CO3: Select and use modern hardware and software tools and devices</td><td>PO5–Modern tool usage</td><td>Affective domain/ analyzing level</td></tr>
<tr><td>CO4: Assess societal, health, safety, legal and cultural issues involved with the mini project</td><td>PO6–The engineer and society</td><td>Psychomotor</td></tr>
<tr><td>CO5: Realize the impact of societal, environmental and sustainable development issues for the design solution of mini project.</td><td>PO7–Environment and sustainability</td><td></td></tr>
<tr><td>CO6: Apply professional ethics and responsibilities in the implementation of mini project.</td><td>PO8–Ethics</td><td></td></tr>
<tr><td>CO7: Work individually and in a team</td><td>PO9–Individual work and teamwork</td><td></td></tr>
</table>

CO8: Communicate and share knowledge, data, information, results etc. with others	**PO10–Communication**			
CO9: Apply engineering project management knowledge and skill to implement the mini project.	**PO11–Project management and finance**			
CO10: Gather and apply knowledge, data and information from various multidisciplinary sources to analyze, design and implement the mini project.	**PO12–Life-long learning**			

05. Text Book

1.Electronic Devices and Circuit Theory by Robert L. Boylested. (11th Edition).

ISBN: 978-0-13-262226-4

2.

06. Reference Books

- Principles of Electronics by V.K. Mehta and Rohit Mehta.
- Operational Amplifiers and Linear Integrated Circuits by Frederick F. Driscoll and Robert Coughlin

LAB MANUAL
Electronic Circuit Lab
EEE-208

Experiment No.: 01

Name of the Experiment: Simulation of diode full wave rectifier circuit, BJT CE amplifier circuit, and Transient behavior of RC and RL circuits using Pspice/Proteus software.

Introduction: In this experiment we are going to use Proteus, a simulation software where we can simulate different circuits. It can be used to make new modules and design PCB's. The version we are going to use is Proteus 8 Professional (8.8/8.9).

Objective:
- To learn how to simulate circuit
- To learn how to analysis electric circuits

Course outcomes (COs), Program Outcomes (POs), and Assessment:

Exp. No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
01	**CO1: I**nvestigate the performance of different digital devices, its characteristics and graphical analysis in electronic circuits**.**	**PO4–** Investigation	Affective domain/ analyzing level	☑ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open-ended lab ☐ Project show & project presentation

Steps for using Proteus:

1) First, you need to open the software and select what mode we want to use. Launching the software, you'll get this menu where we have to select an existing project or open a new project. For starting on a new project, click on "New Project".

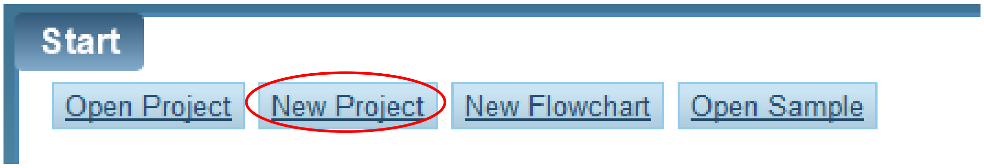


Figure: 1.1

2) After clicking, you'll see a pop-up menu where you have to put a name for the project.

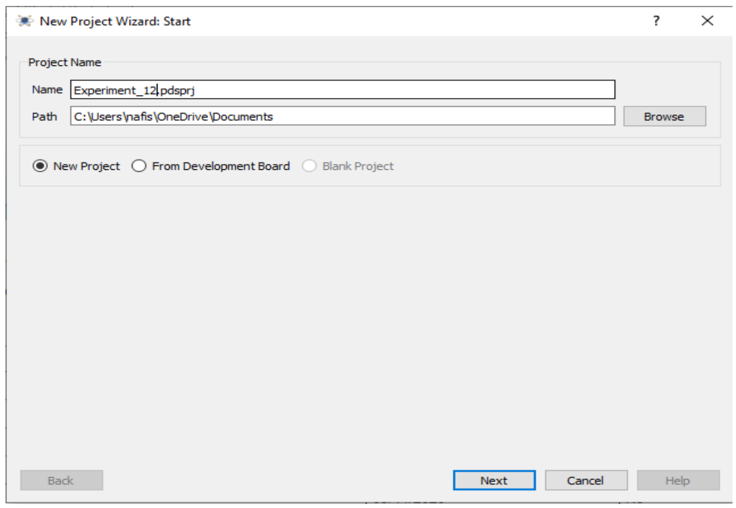
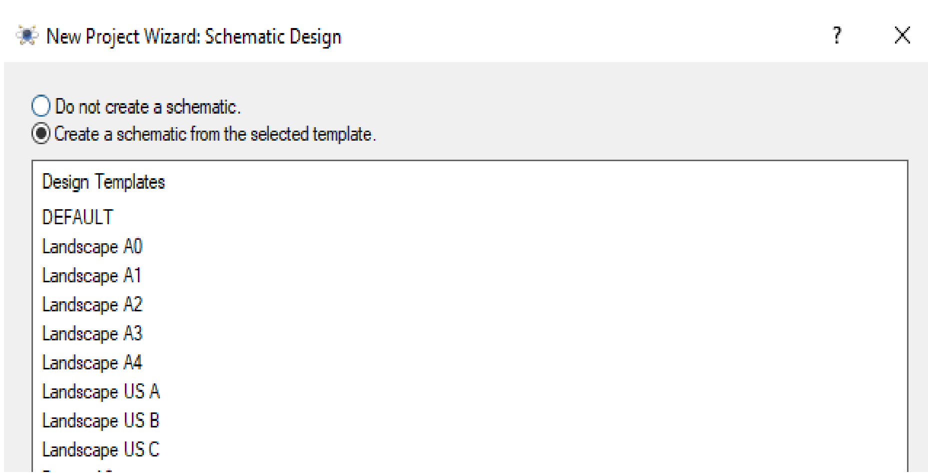


Figure: 1.2

3) After that, three pop-up menu will come where you have to select different options. You have to select these following options and click on next.



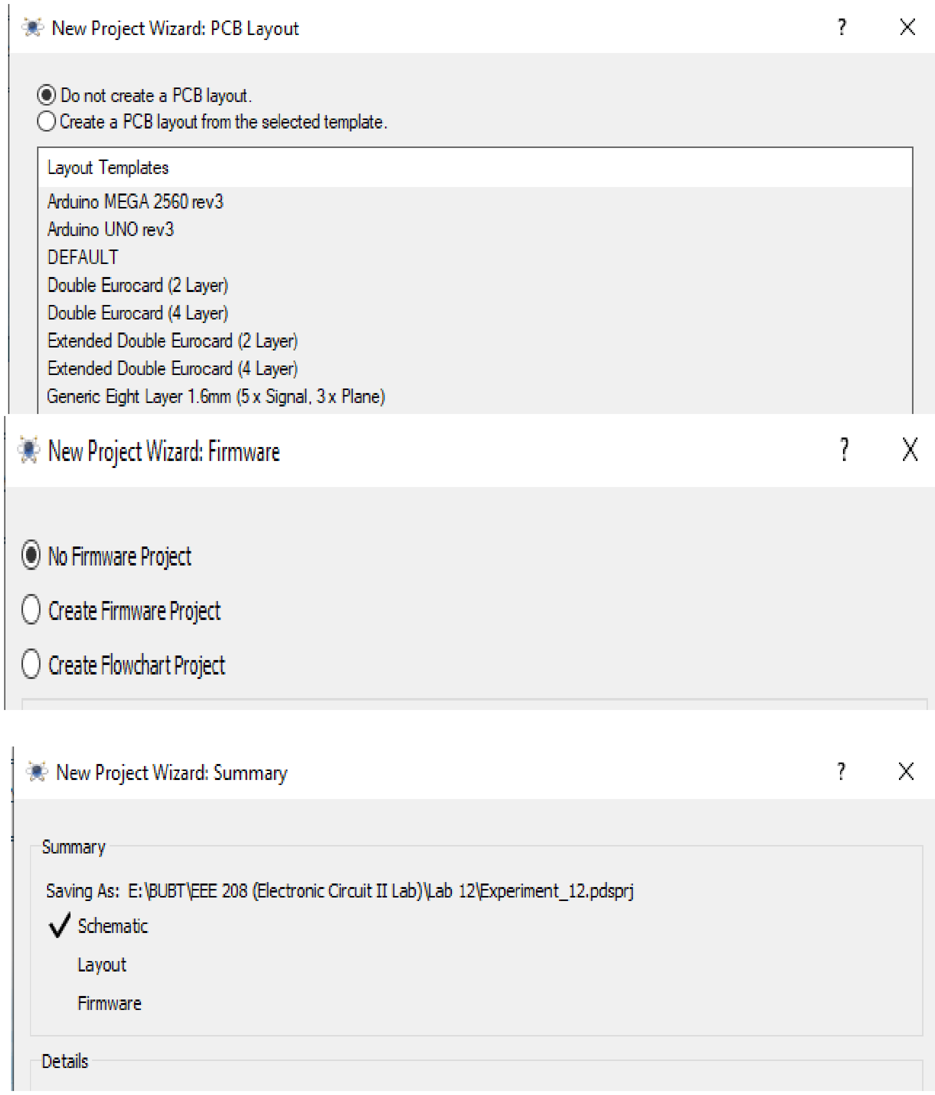


Figure: 1.3

4) After that you have to click on finish. Then you'll get a window like below.

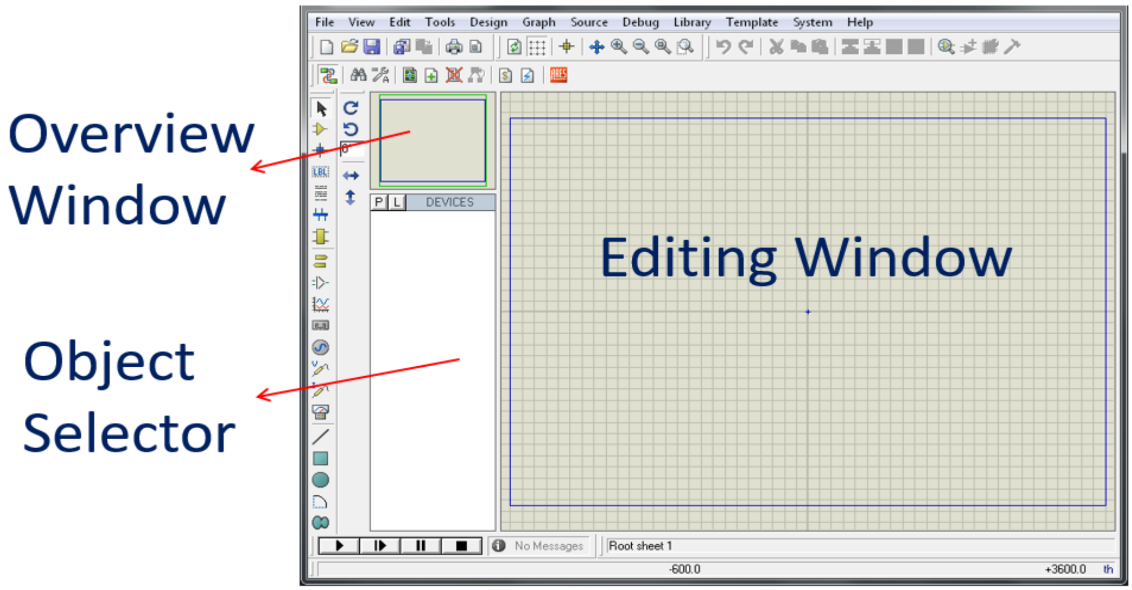


Figure:1.4

5) Tools and Buttons:

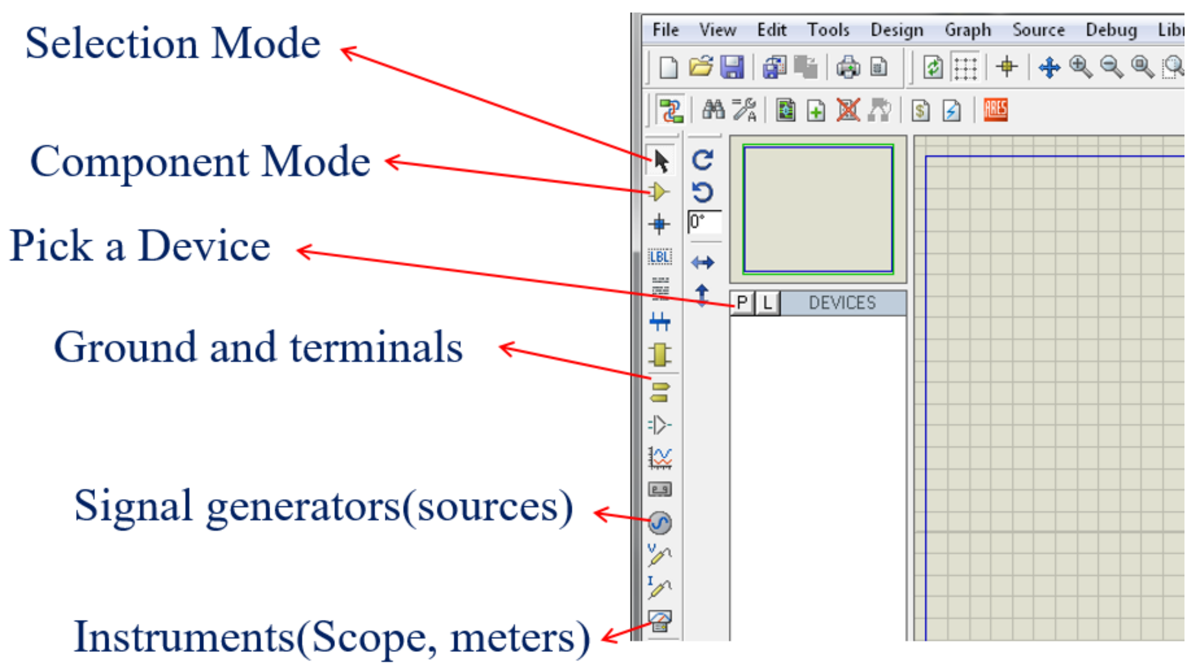


Figure: 1.5

6) Search for an equipment by searching the name or by category.

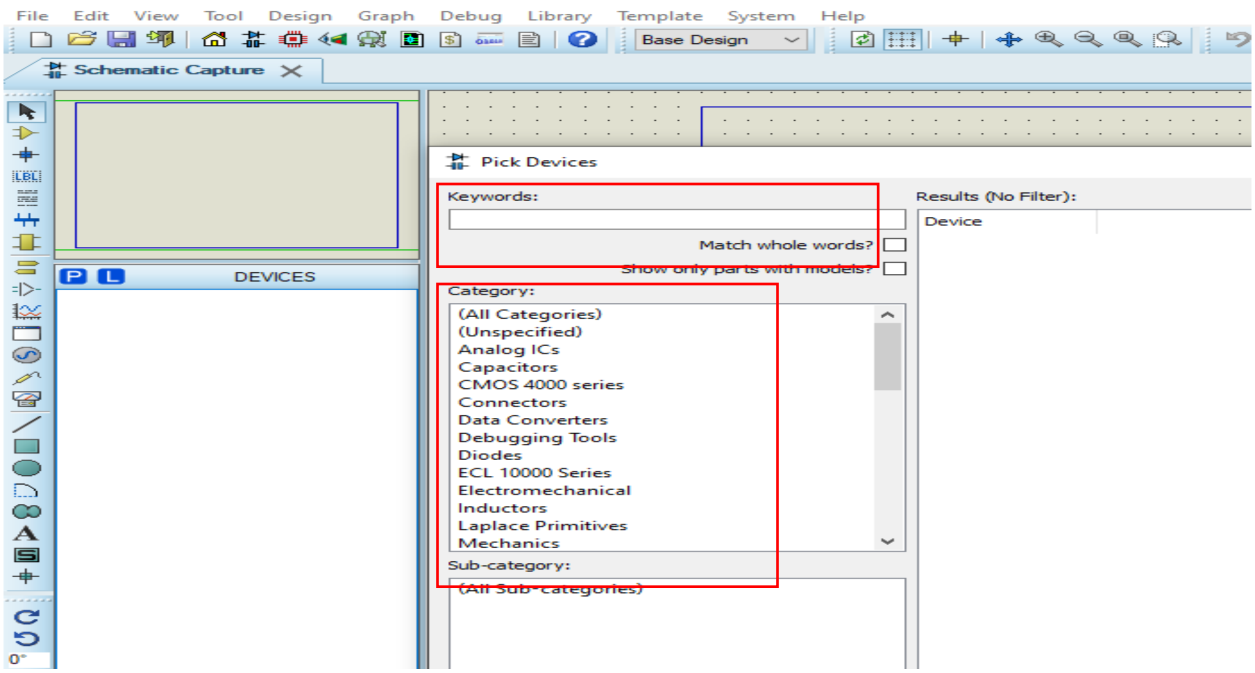


Figure: 1.6

7) You can double click on a component to add it to the "DEVICES" windows. Then use it to construct the circuit.

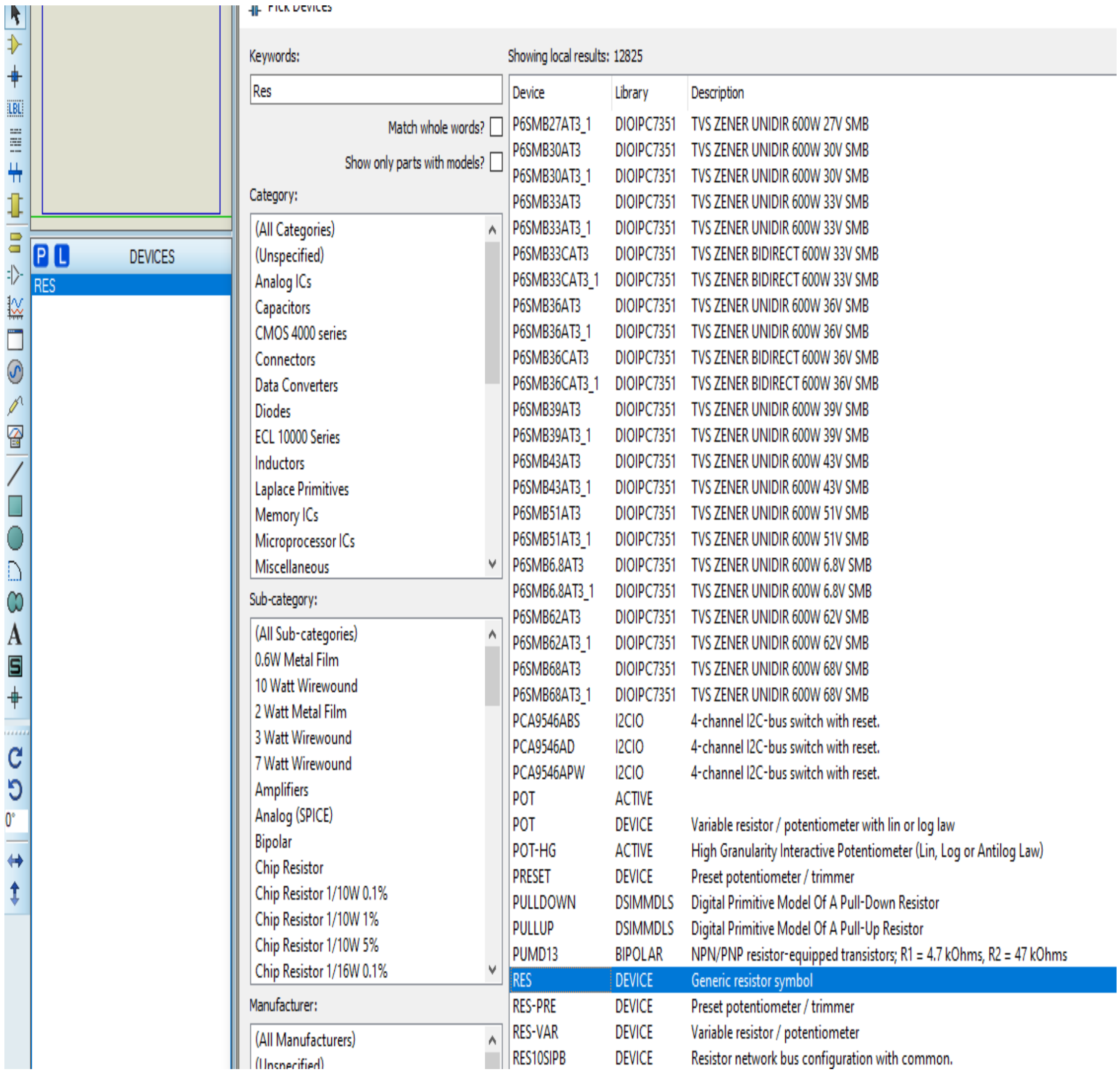


Figure: 1.7

Now you'll try to construct a common emitter circuit with voltage divider biasing and see the output voltage using an oscilloscope.

1. First, you have to choose components from the components section. You can do it by directly search for the desired component in the keyword section.
2. After collecting all the components in the devices section, You have to build the circuit in the editing window. Simple drag and drop method is used by first clicking on to the component and then placing it in the editing window.
3. After placing all the components, you have to fix the parameters. Right clicking on each of the components you can find edit parameters in a drop-down menu.

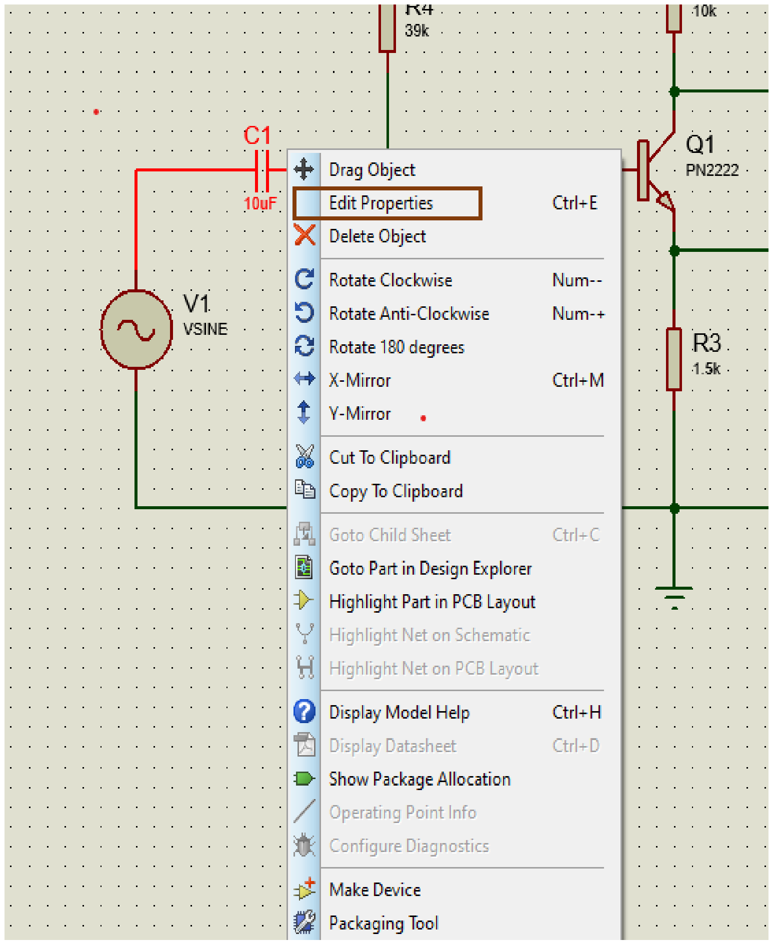


Figure: 1.8

4. Now you have to connect the wires and make the circuit complete. For that component mode in the far-left options.

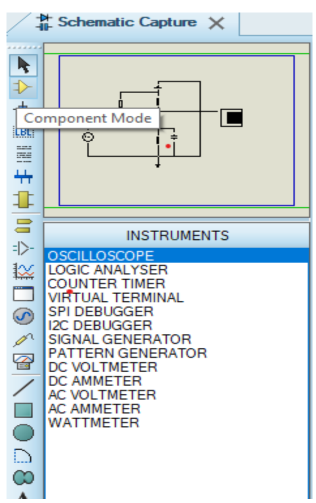


Figure: 1.9

5. After connecting all the wires, put the ground connections where necessary. It can be found in the left side menus named “terminal mode”.

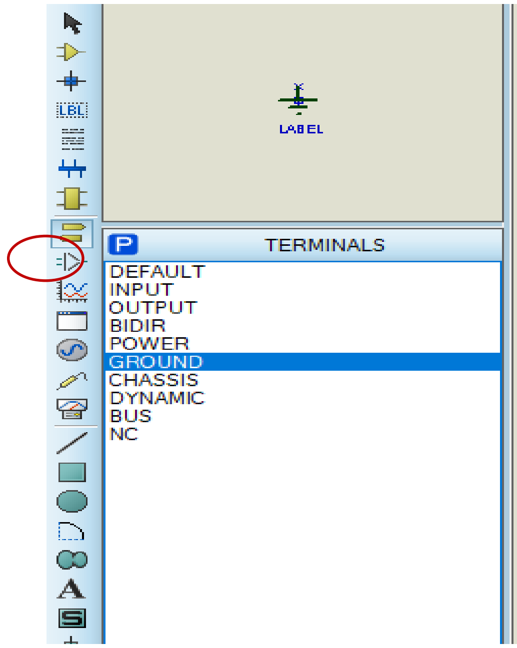


Figure: 1.10

6. Then go for simulation. You can use an oscilloscope to see the output waveform. Oscilloscope can be found in the menu instruments.

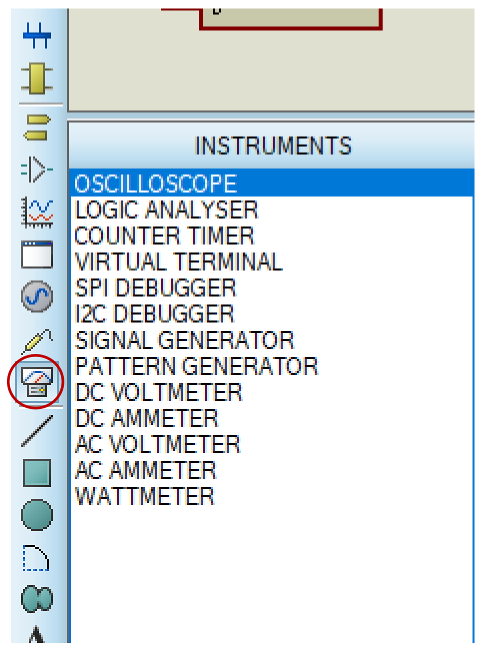


Figure: 1.11

7. After connecting the oscilloscope, the circuit will look like Figure 1.13.
8. You can set the value of VSINE by clicking on it and the properties are like the following window,

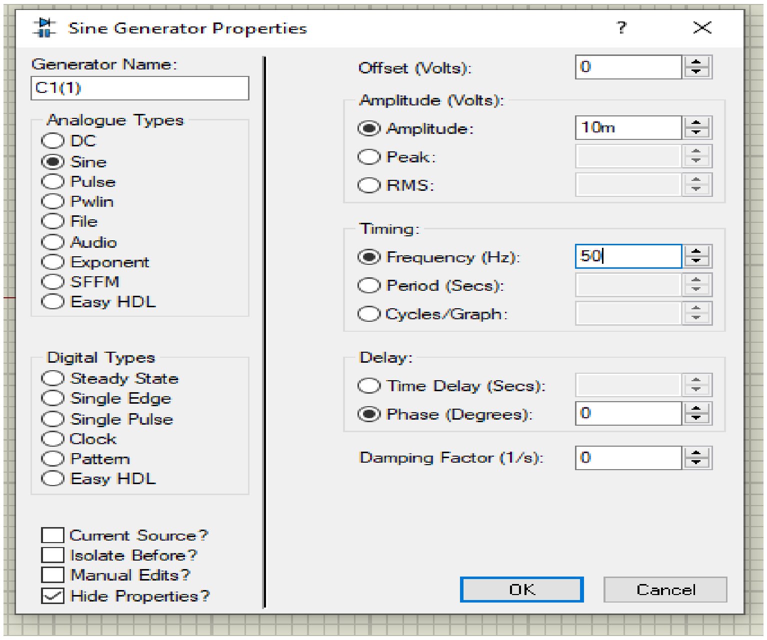


Figure: 1.12

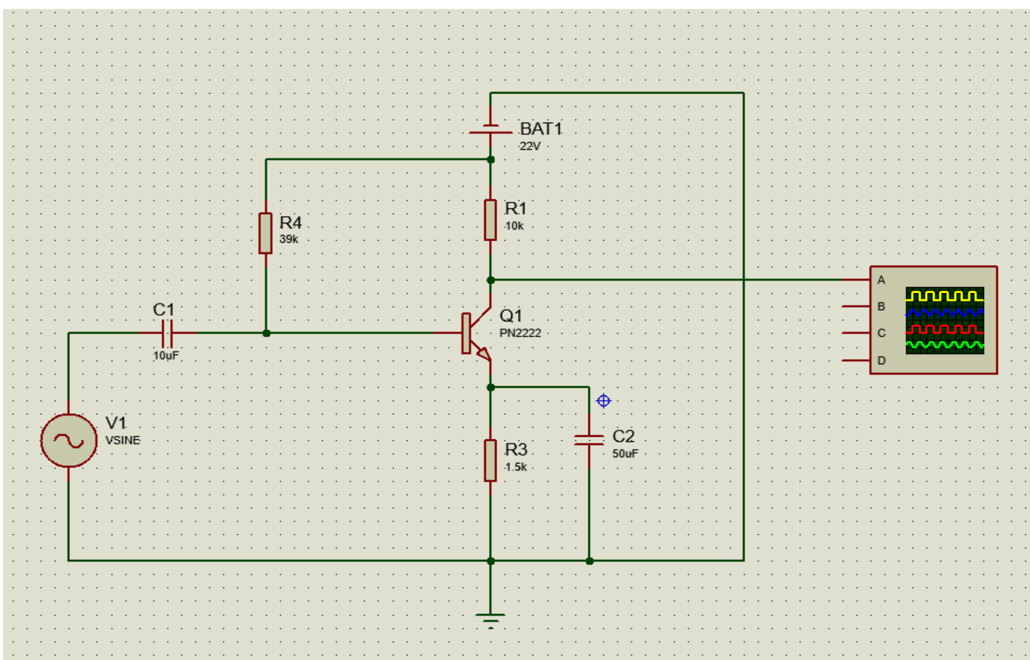


Figure: 1.13

9. Then you have to run it using the run button in the bottom of the window.

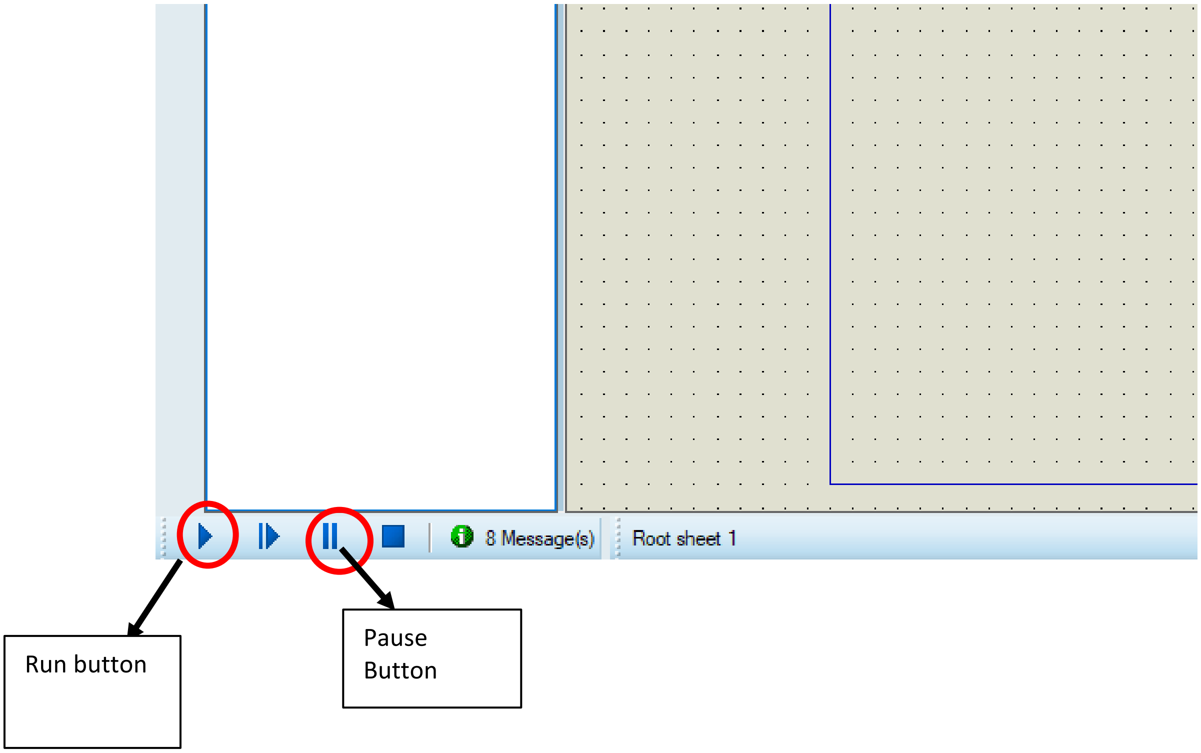


Figure: 1.14

10. After running you can see oscilloscope view popping up. Then you can see the output from the channel that is connected to the output.

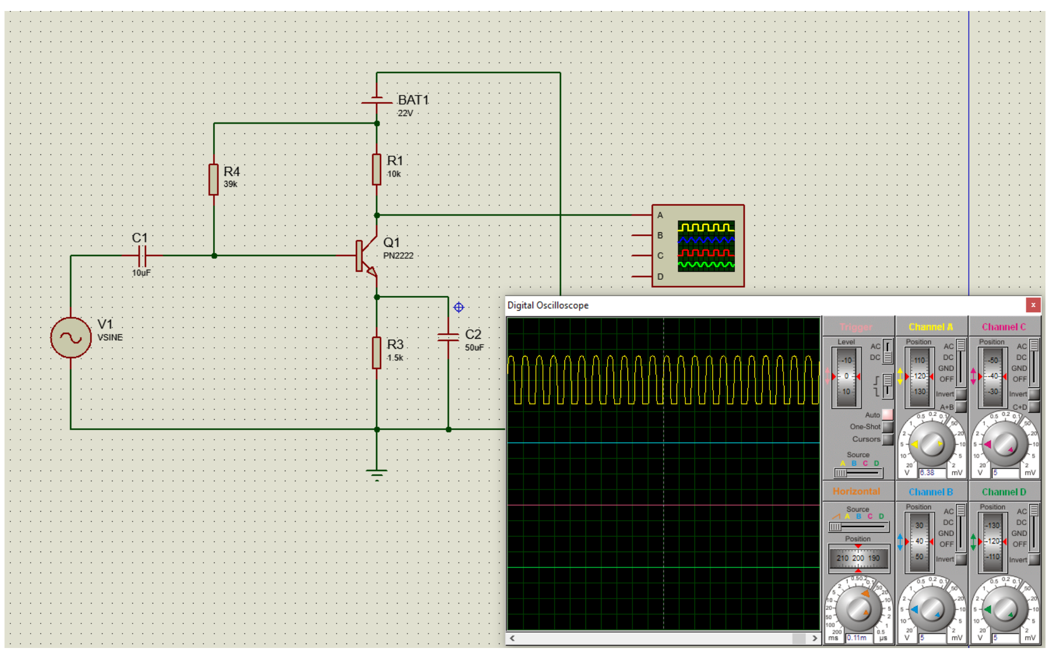


Figure: 1.15

Miscellaneous Functions and components:

1) You can find ammeter and voltmeter in the instrument section where you found the oscilloscope.
2) Voltage and current probes can be used to measure current and point voltages easily. It can be found in probe mode.

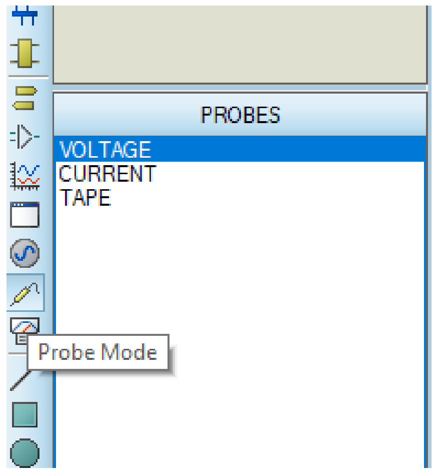


Figure: 1.16

3) In generator mode, you can see various types of signal that you can use in a circuit like, sinusoidal signal, pulse and triangular signal etc. It is handy to use in large circuits.

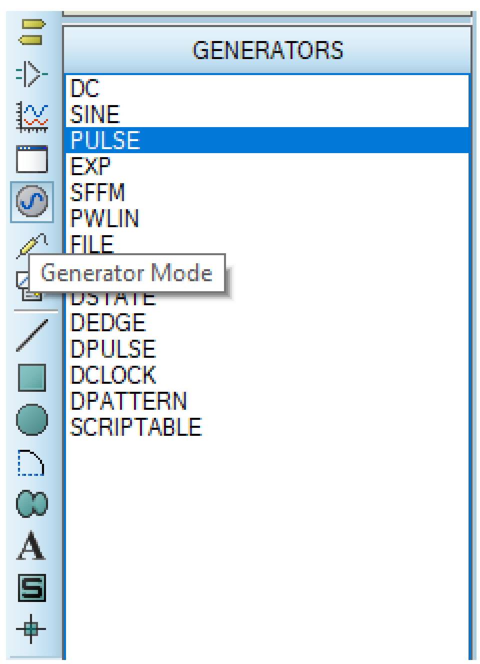


Figure: 1.17

4) The oscilloscope in Proteus bears a resemblance to the oscilloscope we use for hardware experiments. You can tune the **Voltage per division** and **Time per division** using appropriate nobs.

Transient Response of an RC Circuit

Objective:
The objective of this exercise is to study the transient response of a series RC circuit and understand the time constant concept using pulse waveforms.

Background:
In this exercise you will apply a pulse waveform to the RC circuit to analyze the transient response of the circuit. The pulse-width relative to a circuit's time constant determines how it is affected by an RC circuit.
Time Constant (τ): Denoted by the Greek letter tau, , it represents a measure of time required for certain changes in voltages and currents in RC and RL circuits. Generally, when the elapsed time exceeds five time constants (5 τ) after switching has occurred, the currents and voltages have reached their final value, which is also called steady-state response. The time constant of an RC circuit is the product of equivalent capacitance and the Thevenin resistance as viewed from the terminals of the equivalent capacitor.

$$\tau = RC \qquad \ldots\ldots\ldots(1)$$

A Pulse is a voltage or current that changes from one level to another and back again. If a waveform's high time equals its low time it is called a square wave. The length of each cycle of a pulse is its period (T).The pulse width t_p of an ideal square wave is equal to half the time period. The relation between pulse width and frequency is then given by,

$$f = \frac{1}{2f_p} \quad \ldots\ldots\ldots\ldots(2)$$

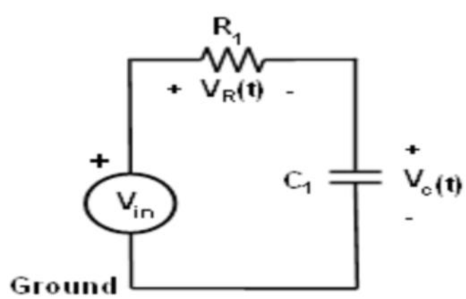


Figure: 1.18 Series *RC* Circuit.

From Kirchhoff's laws, it can be shown that the charging voltage V_C (t) across the capacitor is given by:

$$V_C(t) = V\left(1 - e^{-\frac{t}{RC}}\right) \qquad for\ t \geq 0 \qquad \ldots\ldots(3)$$

Where V is the applied source voltage to the circuit at time $t = 0$. The product RC is the time constant. The response curve is increasing and is shown in Figure 2.

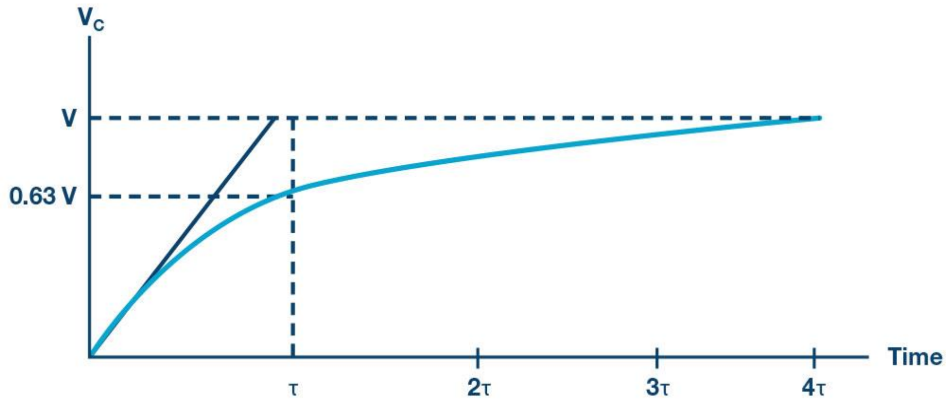


Figure 1.19: Capacitor Charging for Series *RC* Circuit with a Step Input.
(Time Axis Normalized by τ)

The discharge voltage for the capacitor is given by:

$$V_C(t) = V_0 e^{-\frac{t}{RC}} \quad for\ t \geq 0 \quad \ldots\ldots\ldots\ldots\ldots\ldots (4)$$

Where V_o is the initial voltage stored in the capacitor at $t = 0$. The product RC is often referred to as the so-called time constant τ. The response curve is a decaying exponential as shown in Figure 1.19.

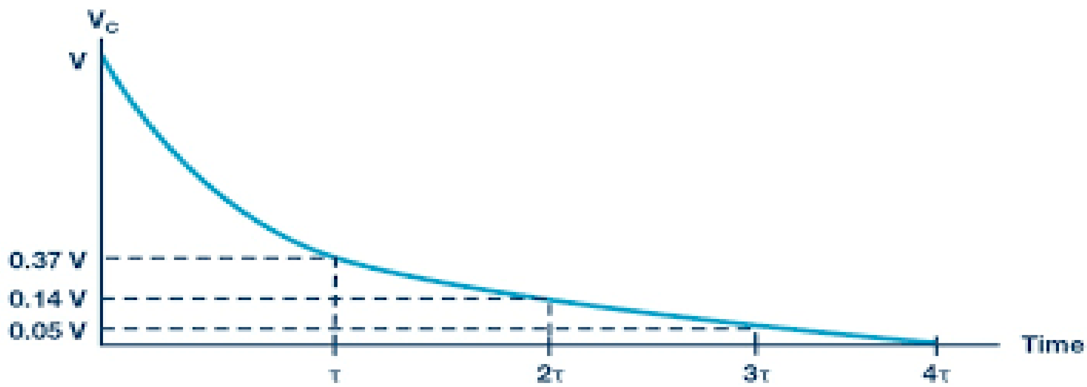


Figure 1.20: Capacitor Discharging for Series *RC* Circuit.

Simulation in Proteus:

1. First, find all the equipment needed for the experiment from the equipment section. A resistor, a capacitor and an inductor.
2. Then you have to construct the circuit in the editing window. The constructed circuit will something like the figure given below.

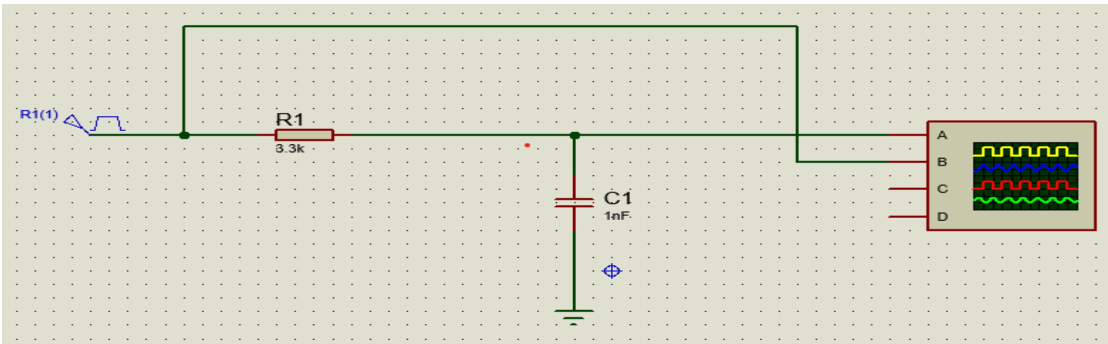


Figure 1.20: Circuit Diagram for RL transient response.

3. Here R1(1) is a pulse voltage source that can be found in the source section. Oscilloscope is connected as per the picture.
4. Then set the value of pulse source by following the figure 1.21,
5. You can now run the simulation and get the waveforms from the oscilloscope. Which can be seen in figure 1.23.

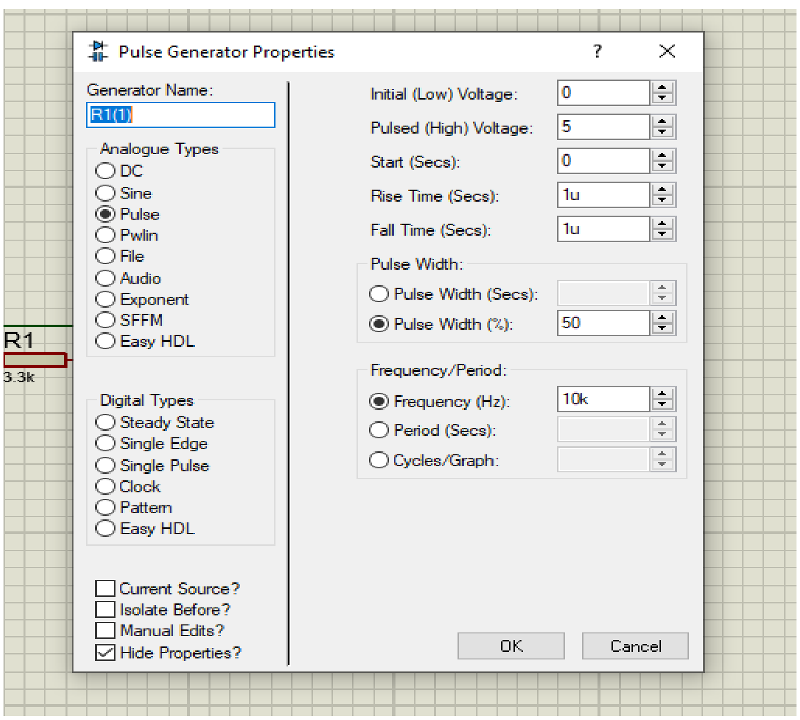


Figure 1.21

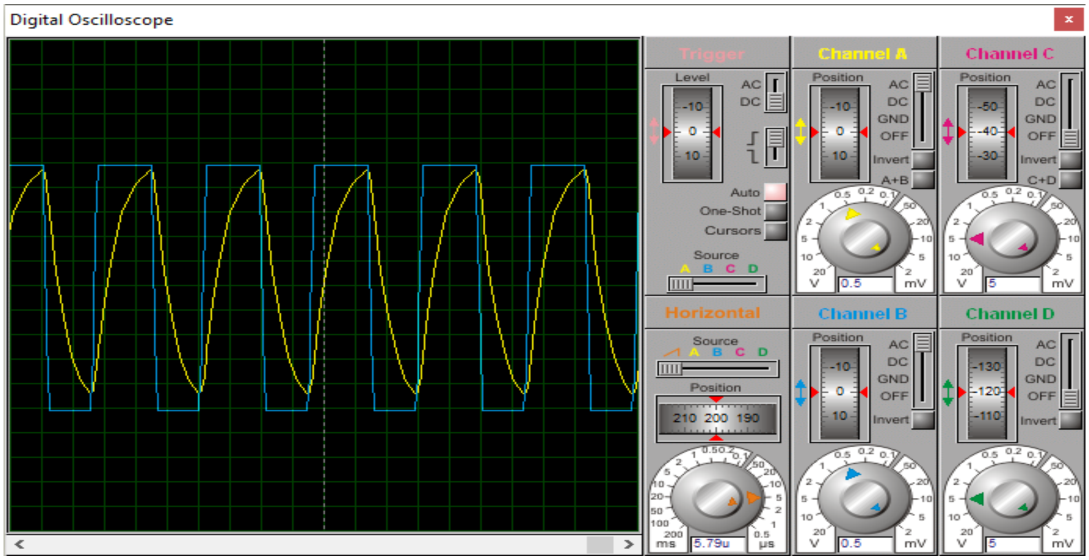


Figure 1.22: Transient Response of RC circuit.

6. You can also use signal generator instead of pulse voltage source and the connection of signal generator is like figure 1.22

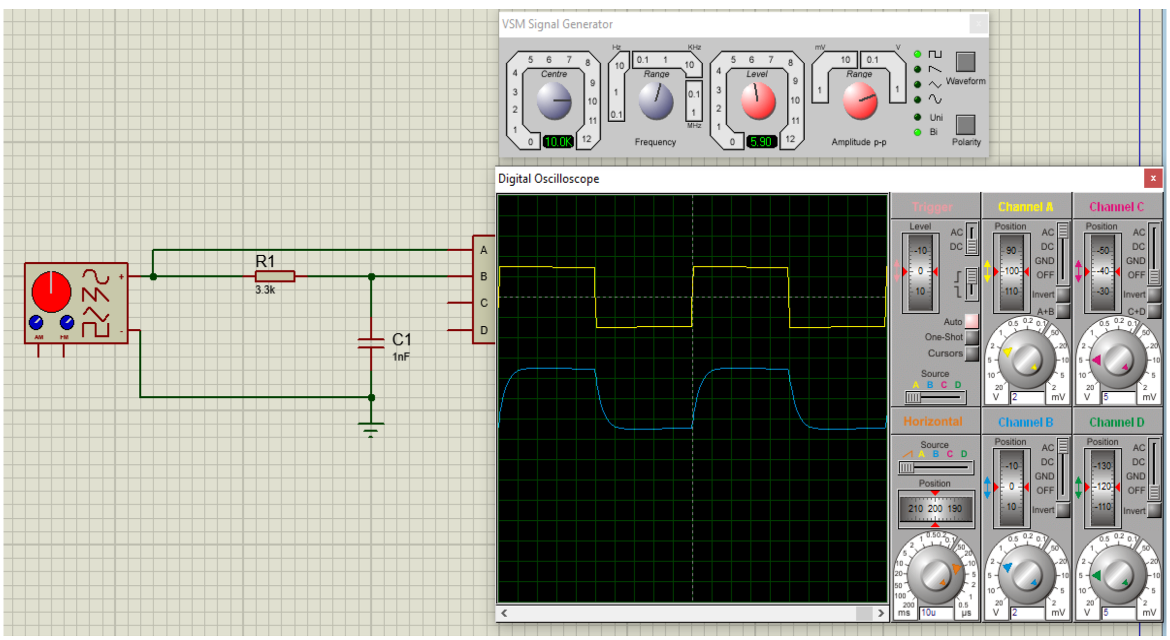


Figure 1.23: RC circuit using signal generator

Transient Response of an RL Circuit

Objective:

The objective of this experiment is to study the transient response of inductor circuits using a series RL configuration and understand the time constant concept.

Background:

This exercise is similar to the “Transient Response of an RC Circuit” exercise, except that the capacitor is replaced by an inductor. In this experiment, you will apply a square waveform to the *RL* circuit to analyze the transient response of the circuit. The pulse width relative to the circuit's time constant determines how it is affected by the *RL* circuit. Time Constant (τ): It is a measure of time required for certain changes in voltages and currents in *RC* and *RL* circuits. Generally, when the elapsed time exceeds five time constants (5τ) after switching has occurred, the currents and voltages have reached their final value, which is also called steady-state response. The time constant of an *RL* circuit is the equivalent inductance divided by the Thévenin resistance as viewed from the terminals of the equivalent inductor.

$$\tau = \frac{L}{R} \qquad \dots\dots\dots\dots\dots\dots(5)$$

A pulse is a voltage or current that changes from one level to another and back again. If a wave form's high time equals its low time, it is called a square wave. The length of each cycle of a pulse train is its period (T). The pulse width t_p of an ideal square wave is equal to half the time period.

The relation between pulse width and frequency for the square wave is given by:

$$f = \frac{1}{2f_p} \quad \dots\dots\dots\dots(6)$$

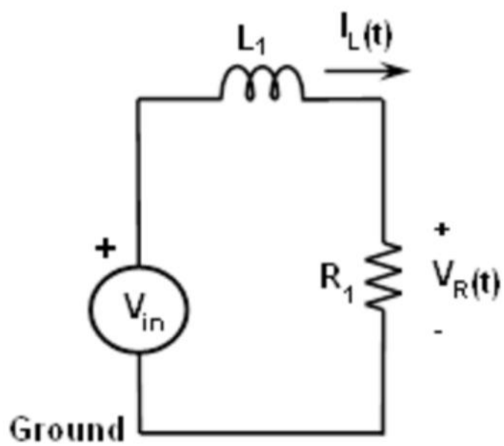


Figure 1.24: Series *RL* Circuit.

In an *RL* circuit, voltage across the inductor decreases with time while in the *RC* circuit the voltage across the capacitor increased with time. Thus, current in an *RL* circuit has the same form as voltage in an *RC* circuit: they both rise to their final value exponentially. The expression for the current in the inductor is given by:

$$i_L(t) = \frac{V}{R}\left(1 - e^{-t\frac{R}{L}}\right) for\ t \geq 0 \qquad \dots\dots\dots\dots\dots\dots(7)$$

Where *V* is the applied source voltage to the circuit for *t* = 0. The response curve is increasing and is shown in Figure 1.23.

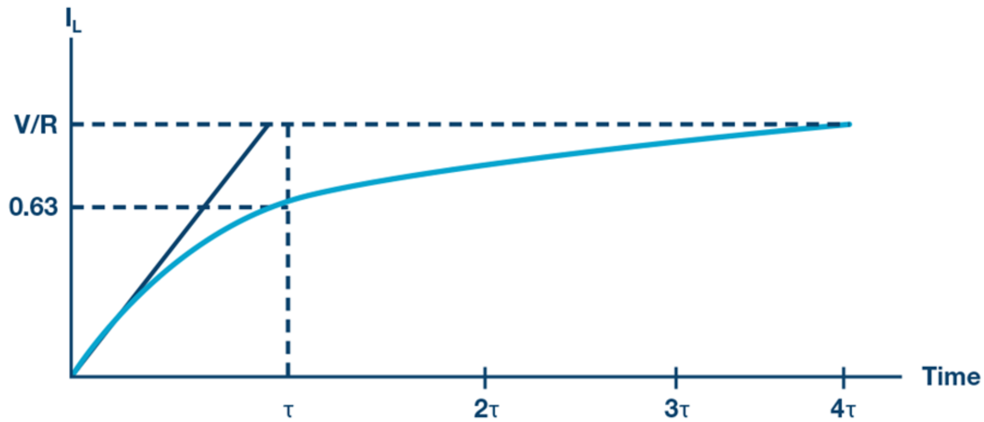


Figure 1.25: Current Increasing in an Inductor in a Series *RL* Circuit.
(Time Axis Normalized by , τ)

The expression for the current decay in the inductor is given by:

$$i_L(t) = I_0 e^{-t\frac{R}{L}} \; for \; t \geq 0 \qquad \ldots\ldots\ldots\ldots\ldots..(8)$$

Where, I_0 is the initial current stored in the inductor at $t = 0$, and , τ = (L/R) is the time constant. The response curve is a decaying exponential and is shown in Figure 3.

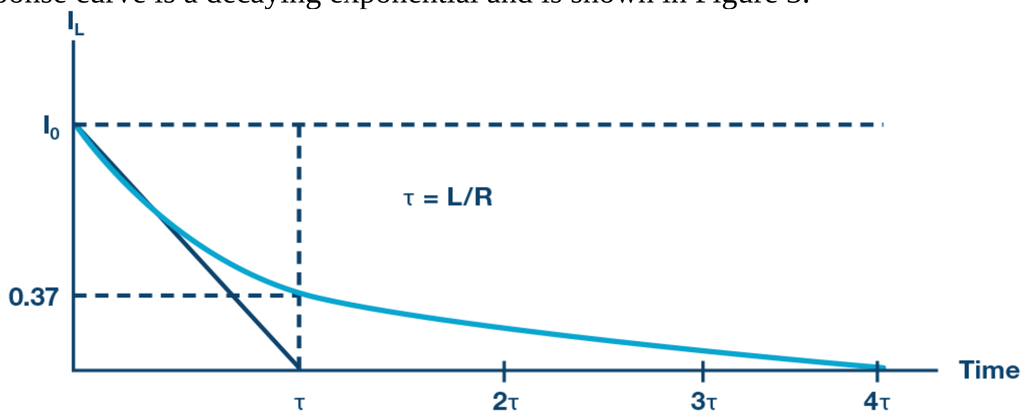


Figure 1.26: Current Decay Through the Inductor for a Series *RL* Circuit.

Since it is possible to directly measure the current through the inductor (current supplied by driving source) with the ADALM2000, we will measure and compare both the current and the output voltage across the resistor. The resistor waveform should be similar to the inductor current as, $V_R = R \, . I_L$. From the waveforms on the scope, we should be able to measure the time constant | which should be equal to, $\tau = \frac{L}{R}$.
Here, R_{TOTAL} is the total resistance and can be calculated from $R_{TOTAL} = R_L + R$.

Simulation in Proteus:

1. First construct a circuit in Proteus given in figure 1.28. Then add all the necessary connections.
2. The circuit Diagram will look like this.

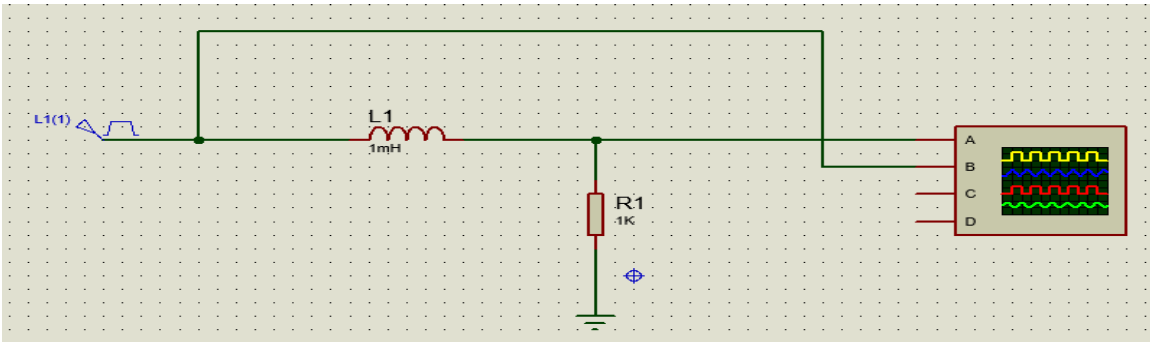


Figure 1.27: Circuit Diagram

3. Then you have to simulate the circuit pressing the run button. The simulation result will come out to be like this.

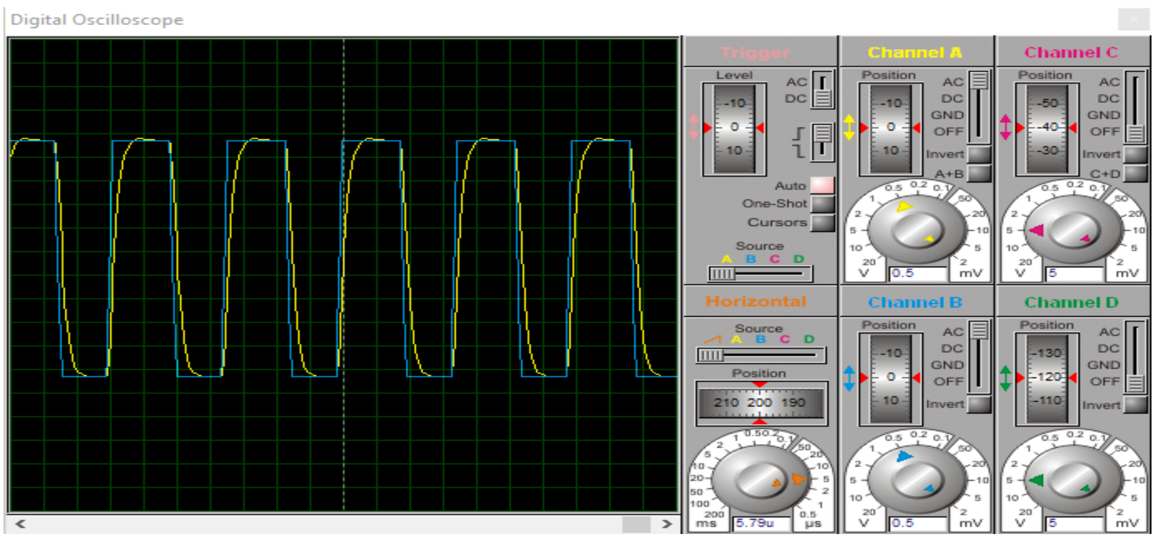


Figure 1.28: Simulation Result of RL circuit

4. Now you can change the values of L and R and see the changes in the waveforms and find out the reason behind it.

Knowledge test questions:

Pre-Lab Question

- Why do we need simulation in electrical engineering?
- List some circuit simulation software.

Post-Lab Question

- What is the resolution of Proteus Oscilloscope?
- How do you create sub-circuit?
- How many ways can you give a square pulse input in Proteus?

Experiment No.: 02

Name of the Experiment: Study of I-V Characteristics of the general-purpose diode and Zener diode.

Objective:

The objective of this experiment is

1. To plot I-V Characteristics of the Silicon P-N Junction Diode and Zener Diode
2. To find cut-in voltage for the Silicon P-N Junction diode
3. To find static and dynamic resistances in both forward and reverse-biased conditions
4. To plot I-V Characteristics of the Zener P-N Junction Diode

Course outcomes (COs), Program Outcomes (POs), and Assessment:

Exp. No.	**CO Statement**	**Corresponding PO**	**Domain / level of learning taxonomy**	**Delivery methods and activities**	**Assessment tools**
02	**CO1: I**nvestigate the performance of different digital devices, its characteristics and graphical analysis in electronic circuits**.**	**PO4–** Investigation	Affective domain/ analyzing level	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open-ended lab ☐ Project show & project presentation

For general-Purpose diode

Theory:

A p-n junction diode is a two-terminal device that acts as a one-way conductor. When a diode is forward biased the current flow from anode to cathode.

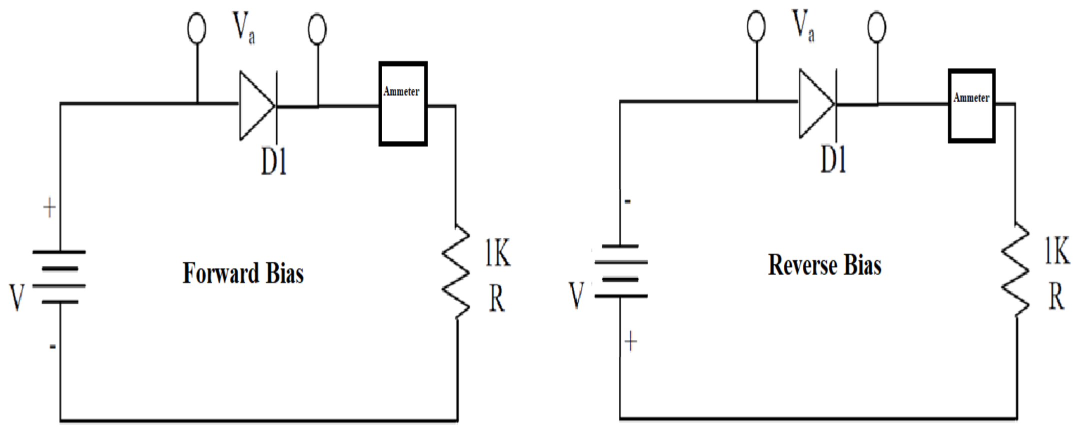


Figure 2.1: Typical Diode Circuit

When it is reverse biased as shown in figure, it is generally in pA (pico-amp) range, in many applications this current is neglected, and diode is considered open. The material for the p-n junction diode is silicon semiconductor. Semiconductors are a group of materials having electrical. Conductivity intermediate between metals and insulators.
Metals: Al (aluminium), Cu(copper), Au(gold).
Insulators: Ceramic, Wood, rubber.
Semiconductor: Si (silicon), Ge (germanium), GaAs (gallium-arsenide).
In semiconductor both holes and electrons contribute to current.

Current Voltage (I – V) Characteristics:

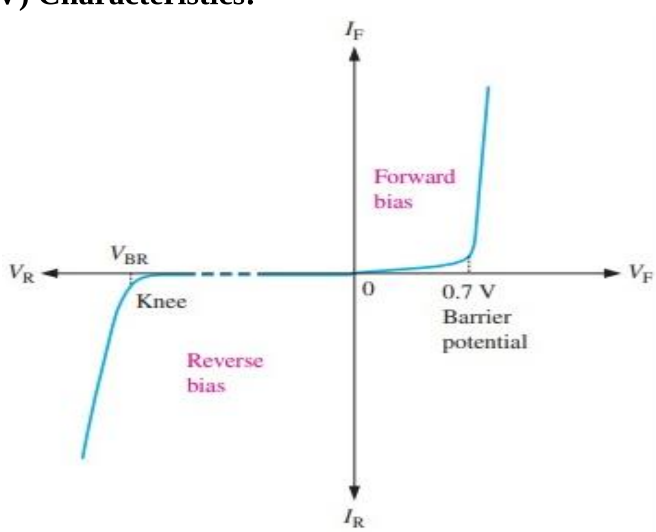


Figure 2.2: I -V Characteristics

- Forward voltage is measured across the diode and Forward Current is a measure of current through the diode.
- When the forward voltage across the diode equals 0 V, forward current I_F equals 0 mA.

- When the value starts from the starting point (0) of the graph, if V_F is progressively increased in 0.1-V steps, I_F begins to rise.
- When the value of V_F is large enough to overcome the barrier potential of the P-N junction, a considerable increase in I_F occurs. The point at which this occurs is often called the knee voltage $\mathbf{V_K}$. For germanium diodes, $\mathbf{V_K}$ is approximately 0.3 V, and 0.7 V for silicon.
- If the value of I_F increases much beyond $\mathbf{V_K}$, the forward current becomes quite large.

Equipment:

- Trainer Board/ Bread Board
- Variable DC power supply
- Oscilloscope
- Function generator
- Multimeter
- Resistors(1K)
- Connection Wires/Jumper Wires
- Probes(Function Connector, Oscilloscope)
- Silicon Diode (IN4007)

Procedure:

Forward Bias Condition:

- Connect the components as shown in the Figure 2.1.
- Vary the supply voltage such that the voltage across the Silicon diode varies from 0 to 0.6 V in steps of 0.1 V and in steps of 0.02 V from 0.6 to 0.76 V. In each step record the current flowing through the diode as I.

Reverse Bias Condition:

- Connect the diode in the reverse bias as shown in Figure 2.1.
- Vary the supply voltage such that the voltage across the diode varies from 0 to 10V in steps of 1 V. Record the current flowing through the diode in each step.
- Now plot a graph between the voltage across the diode and the current flowing through the diode in forward and reverse bias. This graph is called the I-V characteristics of the diodes.
- Calculate the static and dynamic resistance of each diode in forward and reverse bias using the following formulae.
- Static resistance, R = $\frac{V}{I}$
- Dynamic resistance, r = $\frac{\Delta V}{\Delta I}$

Table 2.1:

S.N	Forward Voltage Across the diode V_d (V)	Forward Current Through the diode I_d (mA)	Reverse Voltage Across the diode V_d (V)	Reverse Current Through the diode I_d (μA)

Graph:

- Take a graph sheet and divide it into 4 equal parts. Mark origin at the center of the graph sheet.
- Now mark +ve X-axis as V_F, -ve X-axis as V_R, +ve Y-axis as I_F and –ve Y-axis as I_R.
- Mark the readings tabulated for Si forward biased condition in first Quadrant and Si reverse biased condition in third Quadrant.

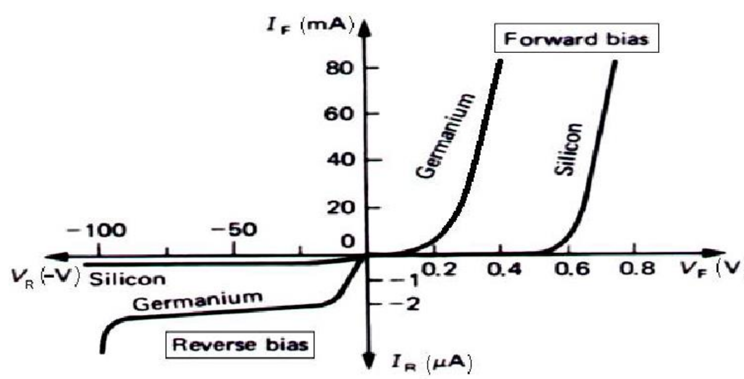


Figure: 2.3 (I-V Characteristics Curve)

Calculation from Graph:

Static forward Resistance, $(R_{dc}) = \frac{V_f}{I_f}, \Omega$

Dynamic Forward Resistance , $(r_{ac}) = \frac{\Delta V_f}{\Delta I_f}, \Omega$

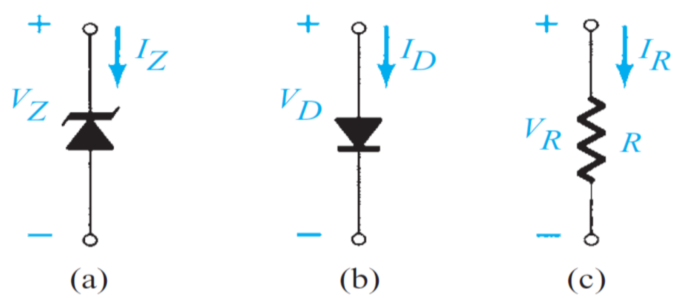
Static Reverse Resistance, $(R_{dc}) \quad = \frac{V_r}{I_r}, \Omega$

Dynamic Reverse Resistance, $\quad (r_{ac}) = \frac{\Delta V_r}{\Delta I_r}, \Omega$

For Zener Diode

<u>Theory:</u>

The very steep i–v curve that the diode exhibits in the breakdown region (as we have seen in the previous experiment) and the almost-constant voltage drop that this indicates suggest that diodes operating in the breakdown region can be used in the design of voltage regulators. From the previous experiment, we can say that ***voltage regulators are circuits that provide a constant dc output voltage in the face of changes in their load current and in the system power-supply voltage***. This in fact turns out to be an important application of diodes operating in the reverse breakdown region, and special diodes are manufactured to operate specifically in the breakdown region. Such diodes are called breakdown diodes or, more commonly, as noted earlier, Zener diodes. Figure 2.4 shows the circuit symbol of the Zener diode. In normal applications of Zener diodes, current flows into the cathode, and the cathode is positive with respect to the anode. Thus, I_Z and V_Z in Fig have positive values.



Conduction direction: (a) Zener diode;
(b) semiconductor diode;
(c) resistive element.

Figure 2.4: Symbol of Diode Types

Figure 2.5 shows details of the diode i–v characteristic in the breakdown region. We observe that for currents greater than the knee current I_{ZK} (specified on the data sheet of the Zener diode), the i–v characteristic is almost a straight line. The manufacturer usually specifies the voltage across the Zener diode V_Z at a specified test current, I_{ZT}. We can see these parameters in Figure as the coordinates of the point labeled Q. Thus a 6.8-V Zener diode will exhibit a 6.8-V drop at a specified test current of, say, 10 mA. As the current through the Zener deviates from I_{ZT}, the voltage across it will change, though only slightly. Figure 2.5 shows that corresponding to current change ΔI the zener voltage changes by ΔV, which is related to ΔI by,

$$\Delta V = r_z \Delta I \quad \text{………………(2.1)}$$

Where, r_z is the inverse of the slope of the almost-linear i–v curve at point Q. Resistance r_z is the incremental resistance of the Zener diode at operating point Q. It is also known as the dynamic resistance of the Zener, and its value is specified on the device data sheet. Typically, r_z is in the range of a few ohms to a few tens of ohms. Obviously, the lower the value of r_z is, the more constant the Zener voltage remains as its current varies, and thus the more ideal its

Performance becomes in the design of voltage regulators. In this regard, we observe from Figure 2.5 that while r_z remains low and almost constant over a wide range of current, its value increases considerably in the vicinity of the knee. Therefore, as a general design guideline, one should avoid operating the zener in this low-current region.

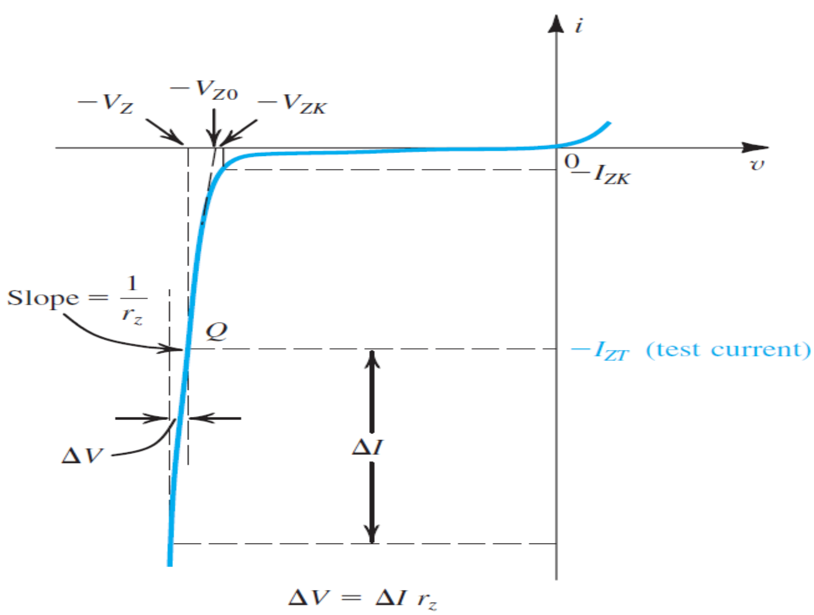


Figure 2.5: The diode i–v characteristic with the breakdown region shown in some detail

Equipment:

- Trainer Board/ Bread Board
- Variable DC power supply
- Oscilloscope
- Function generator

- Multimeter
- Resistors(1K)
- Connection Wires/Jumper Wires
- Probes (Function Connector, Oscilloscope)
- Silicon Diode (IN4007)
- Zener Diode (1N4735A/ FZ 5.1)

Circuit Diagram:

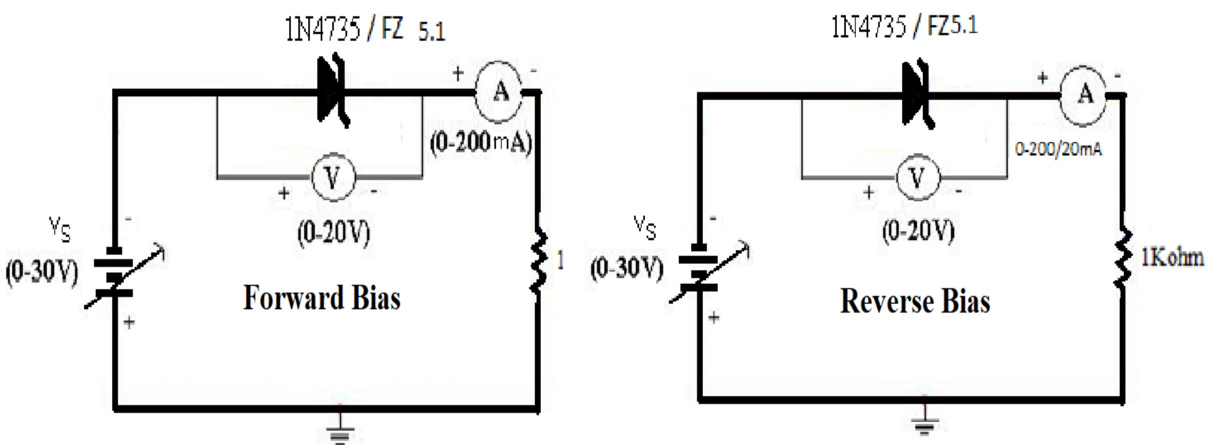


Figure 2.6: Circuit Diagram

Procedure:

Forward Bias Condition

- Connect the circuit as shown in Figure 2.*6*.
- Vary V_F gradually from 0 to 0.6 V in steps of 0.1 V and in steps of 0.02 V from 0.6 to 0.76 V. In each step record the current flowing through the diode as I_F.
- Tabulate different forward currents obtained for different forward voltages.

Reverse Bias Condition

- Connect the Zener diode in reverse bias as shown in the fig.2.6. Vary the voltage across the diode in steps of 1V from 0 V to 6 V and in steps 0.1 V till its breakdown voltage is reached. In each step note the current flowing through the diode
- Plot a graph between V and I. This graph will be called the I-V characteristics of the Zener diode. From the graph find out the breakdown voltage for the diode.

Observation:

S.N	Forward Voltage Across the diode V_F (V)	Forward Current Through the diode I_F (mA)	Reverse Voltage Across the diode V_R (V)	Reverse Current Through the diode I_R (μA)

Graph:

- Take a graph sheet and divide it into 4 equal parts. Mark origin at the center of the graph sheet.
- Now mark +ve X-axis as V_F, -ve X-axis as V_R, +ve Y-axis as I_F and –ve Y-axis as I_R.
- Mark the readings tabulated for forward biased condition in first Quadrant and reverse biased condition in third Quadrant.

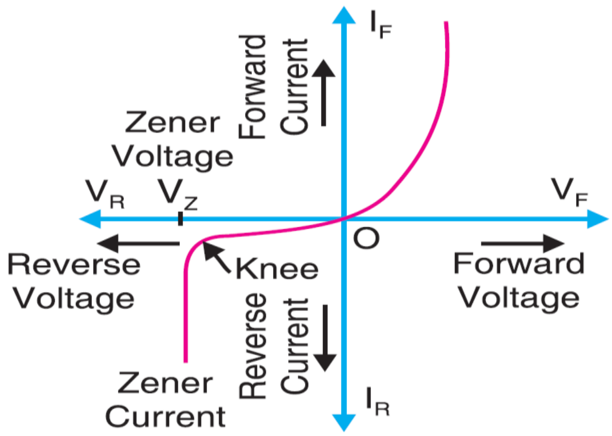


Figure 2.7: I-V Characteristics of Zener Diode

Knowledge Test Questions:

Pre Lab

1. What is slope of a graph?

2. Describe simple basic of a regulator?

3. What is the major difference between Electrical and Electronic circuits?

4. What is the significant of P-type and n-type?

5. Identify linear and nonlinear equations?

Post Lab

6.What is the difference between p-n Junction diode and Zener diode?

7. What is break down voltage?

8. What did you understand with I-V characteristics?

9. How can you define Q Point?

Experiment No: 03
Experiment Name: Design and analysis of half wave rectifier and full wave bridge rectifier using diode.

Objectives:

The objective of this experiment is:

- To simulate Diode Half Wave Rectifier
- To simulate Diode Full Wave Rectifier

Course outcomes (COs), Program Outcomes (POs), and Assessment:

Exp. No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
03	**CO1: I**nvestigate the performance of different digital devices, its characteristics and graphical analysis in electronic circuits**.**	**PO4–** Investigation	Affective domain/ analyzing level	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open-ended lab ☐ Project show & project presentation

Required equipment:

- Trainer Board/ Bread Board
- Variable DC power supply
- Oscilloscope
- Function generator
- Multimeter
- Resistors(1K)
- Connection Wires/Jumper Wires
- Probes (Function Connector, Oscilloscope)
- Silicon Diode (IN4007)
- Capacitor (10uF)

Design and analysis of half wave rectifier

Theory:

Diodes can be used to Rectify output from ac supply to produce a dc supply.

1. On + ve half cycle of input wave
   - Diode is forward biased.
   - Diode conducts.
2. On -ve half cycle of input wave
   - Diode is reverse biased.
   - Diode doesn't conduct.

Half Wave Rectifier

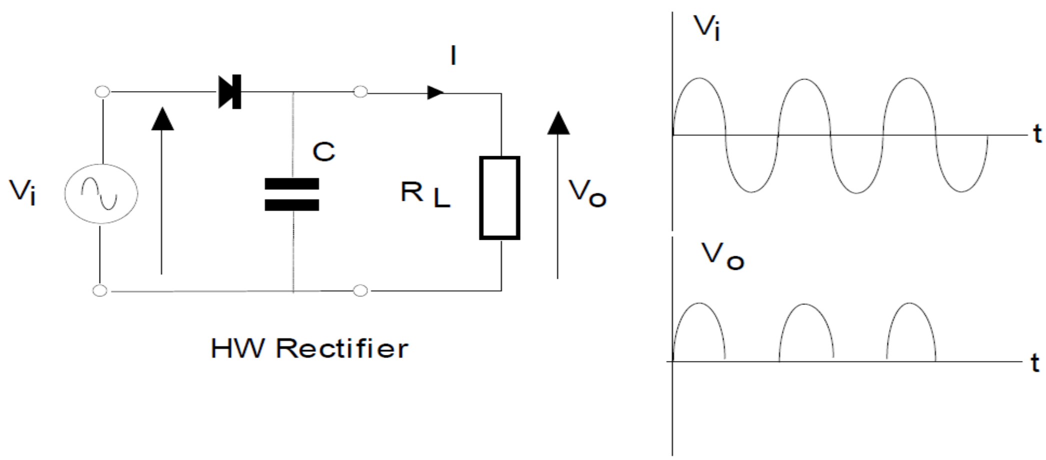


Figure 3.1: Half Wave Rectifier

Here,

- Average Voltage (as seen on dc voltmeter) $V_{avg} = \frac{V_p}{\pi}$
- RMS voltage (as seen on ac voltmeter) $V_{rms} = \frac{V_p}{\sqrt{2}}$

Smoothing:

If a capacitor is placed across output

- Capacitor charges on rising edge of + ve half-cycle.
- Discharges on falling edge.
- Output is smoothed.

Actual peak output will be reduced from peak input by value of forward bias, V_p (out) = V_p (in) – 0.7 V

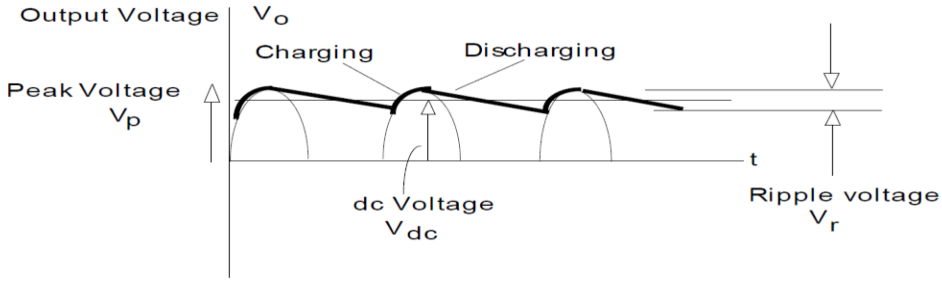


Figure 3.2: Smoothed Half Wave Rectifier Output

Ripple Voltage:

A finite load current I causes capacitor voltage to drop by V_r during ac cycle. Ripple in output is approx. sawtooth in shape (neglect charging time). Assuming discharge takes one complete period (T). Charge flowing from capacitor in time T, Q = IT. Fall in capacitor voltage = pk-pk ripple. Now, $V_r = \frac{Q}{C} = \frac{IT}{C}$. But $T = \frac{1}{f}$. So,

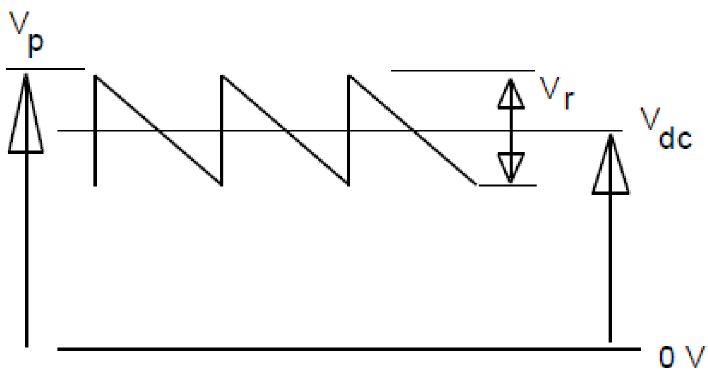


Figure: 3.3 with ripple voltage

V_r (HW) = $\frac{I}{CF}$, Half Wave Ripple Voltage

As the ripple voltage increases the average (dc) output voltage decreases.

$V_{dc} = V_p - \frac{1}{2\,V_r}$

$V_{dc} = V_p - \frac{1}{2}\,cf$, Half Wave DC Voltage

Ripple factor (r) defines magnitude of smoothing effect.

$r = \frac{V_r}{V_{dc}} \times 100\%$, Ripple Factor

Precautions:

- While doing the experiment do not exceed the readings of the diode. This may lead to damaging of the diode.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Circuit Diagram:

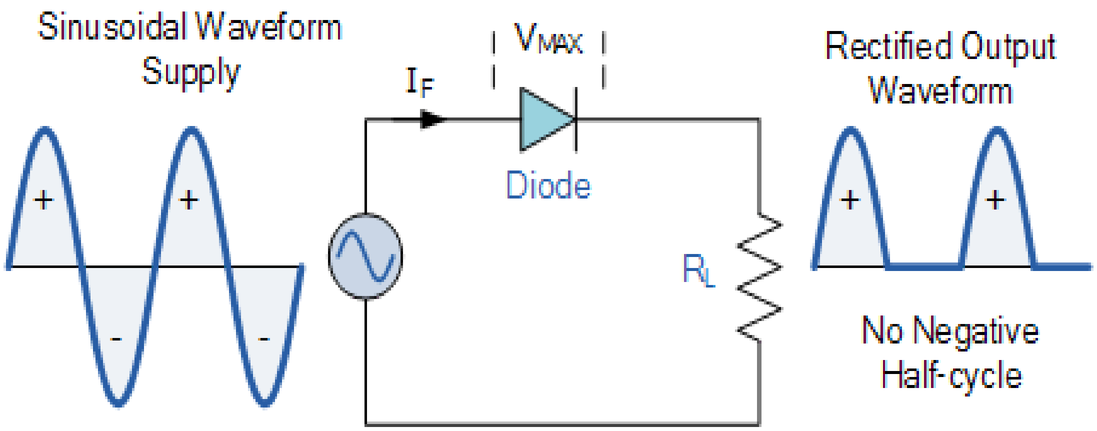


Figure 3.4: Circuit diagram of Half-wave rectifier

Procedure:

- Connect the circuit as shown in the figure 3.4
- Set the input voltage as 10 V_{p-p} and 1kz frequency.
- Connect the multimeter across the 1kΩ load.
- Measure the AC and DC voltages by setting multimeter to ac and dc mode respectively.
- Now calculate the ripple factor using the following formula.

Ripple Factor, r = $\frac{V_r}{V_{dc}} \times 100\%$

- Connect the oscilloscope channel-1 across input and channel-2 across output i.e load and
- Observe the input and output Waveforms.
- Now calculate the peak voltage of input and output waveforms and also the frequency.

Data Table:

Load Resistance (R_L)	V_{AC}(V)	V_{DC}(V)	Ripple Factor, r	Input Signal		Output Signal	
				V_m p-p(v)	Frequency (Hz)	V_m p-p(v)	Frequency (Hz)

Calculation:

- Ripple Factor, r = $\frac{V_r}{V_{dc}} \times 100\%$

Graph:

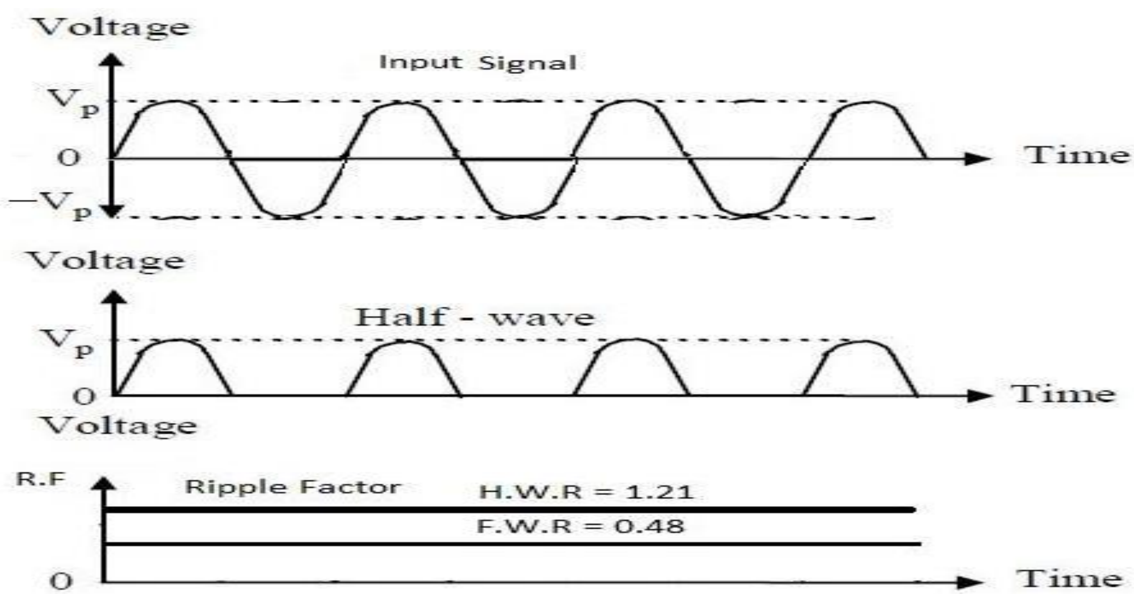


Figure 3.5: Input and Output Graph of HW Rectifier Circuit

Full Wave Rectifier

Theory:

Diodes can be used to Rectify output from ac supply to produce a dc supply.

- On + ve half cycle of input wave
  - Diodes D_1 and D_3 are forward biased.
  - Diodes D_1 and D_3 conducts.
- On -ve half cycle of input wave
  - Diodes D_2 and D_4 are in forward biased.
  - Diodes D_2 and D_4 aren conducted.

Full Wave Rectifier

Better rectification is obtained if circuit conducts on both input half-cycles.

On the first half cycle,

- When terminal A is + ve, D_1 conducts to make top end of load + ve.
- At same time terminal B is - ve, D_3 conducts to lower end of load.

On the next half cycle,

Terminal A is - ve and B is + ve, D_2 conducts to top end of load, D_4 conducts to lower end.

So, the average voltage, $V_{avg} = \frac{2\,V_p}{\pi}$ (Without Smoothing Capacitor)

For full wave rectification, ripple frequency is twice ac input frequency. So,

$V_r(FW) = \frac{1}{2}\,Cf$, Full Wave Ripple Voltage

$V_{dc} = \frac{1}{4}\,Cf$, Full Wave DC Voltage

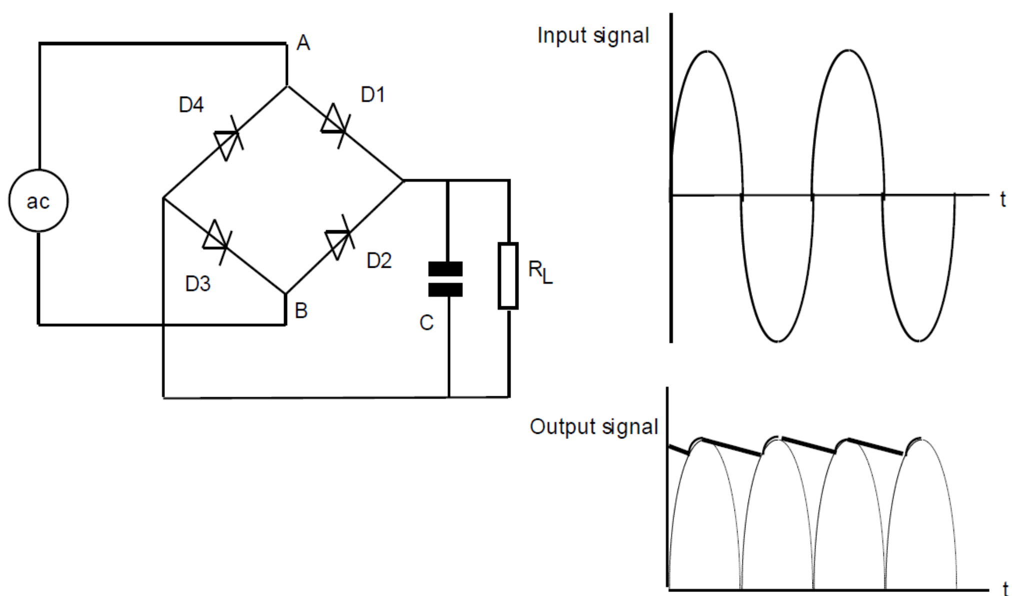


Figure 3.6: Full Wave Bridge Rectifier

Circuit Diagram:

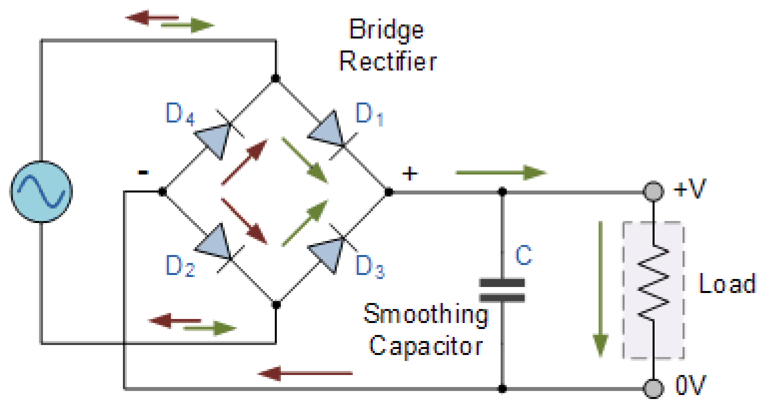


Figure 3.7: Full Wave Bridge Rectifier

Procedure:

- Connect the circuit as shown in the figure 3.7.
- Set the input voltage as 10 $V_{p\text{-}p}$ and 1kz frequency.
- Connect the multimeter across the 1kΩ load.
- Measure the AC and DC voltages by setting multimeter to ac and dc mode respectively.
- Now calculate the ripple factor using the following formula.

Ripple Factor, r = $\frac{V_r}{V_{dc}} \times 100\%$

- Connect the oscilloscope channel-1 across input and channel-2 across output i.e load and
- Observe the input and output Waveforms.
- Now calculate the peak voltage of input and output waveforms and also the frequency.

Data Table:

Load Resistance (R_L)	V_{AC}(V)	V_{DC}(V)	Ripple Factor, r	Input Signal		Output Signal	
				V_m p-p(v)	Frequency (Hz)	V_m p-p(v)	Frequency (Hz)

Calculation:

- Ripple Factor, r = $\frac{V_r}{V_{dc}} \times 100\%$

Graph:

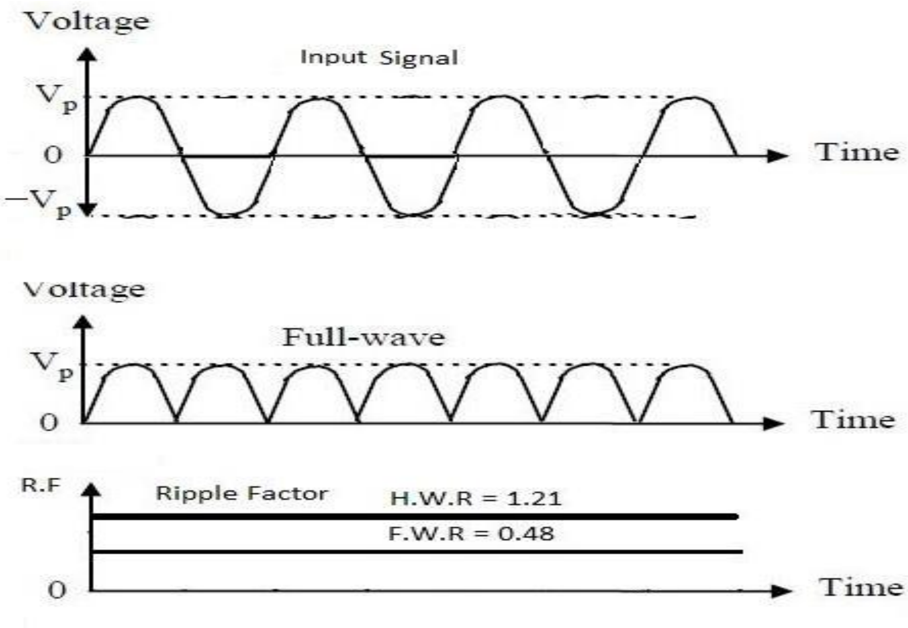


Figure 3.8: Input and Output Graph of FW Rectifier Circuit

Result and discussion:

We have studied.

- Full Wave characteristics are studied.
- Ripple factor of Full wave rectifier
- Regulation of Full wave rectifier

Knowledge Test Questions:

Pre-Lab Questions

1. What is a Wave?

2. Give some rectifications technologies?

Post Lab Questions

3. What is the difference between HW and FW?
4. What is the meaning of ripple?
5. Describe the impact of rectification technologies?

Experiment no: 04
Experiment name: Investigation and analysis of clipper and clamper circuits using a diode.

Objective:

- To observe the output of clipper and clamper circuits using diode

Course outcomes (COs), Program Outcomes (POs), and Assessment:

Exp. No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
04	**CO1: I**nvestigate the performance of different digital devices, its characteristics and graphical analysis in electronic circuits**.**	**PO4–** Investigation	Affective domain/ analyzing level	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open-ended lab ☐ Project show & project presentation

<u>Clipper Circuit</u>

Theory:

Clipper circuits have the ability to "clip" off a portion of the input signal without distorting the remaining part of the alternating waveform. The half wave rectifier of the next experiment will be an example of the simplest form of diode clipper. Depending on the orientation of thy diode, the positive or negative region of the input signal is "clipped" off. There are two general categories of clippers: series and parallel. The series clipper configuration is defined as one where the diode is in series with the load, while the parallel variety has the diode in branch parallel to the load. Clippers are useful for protecting circuits from exceeding various voltages (either positive or negative).

Clipper Circuits are three types

I. Non-Biased Clipper
   - Positive Clipper
   - Negative Clipper

II. Biased Clipper
- Biased Positive Clipper
- Biased Negative Clipper

III. Biased Combination Clipper

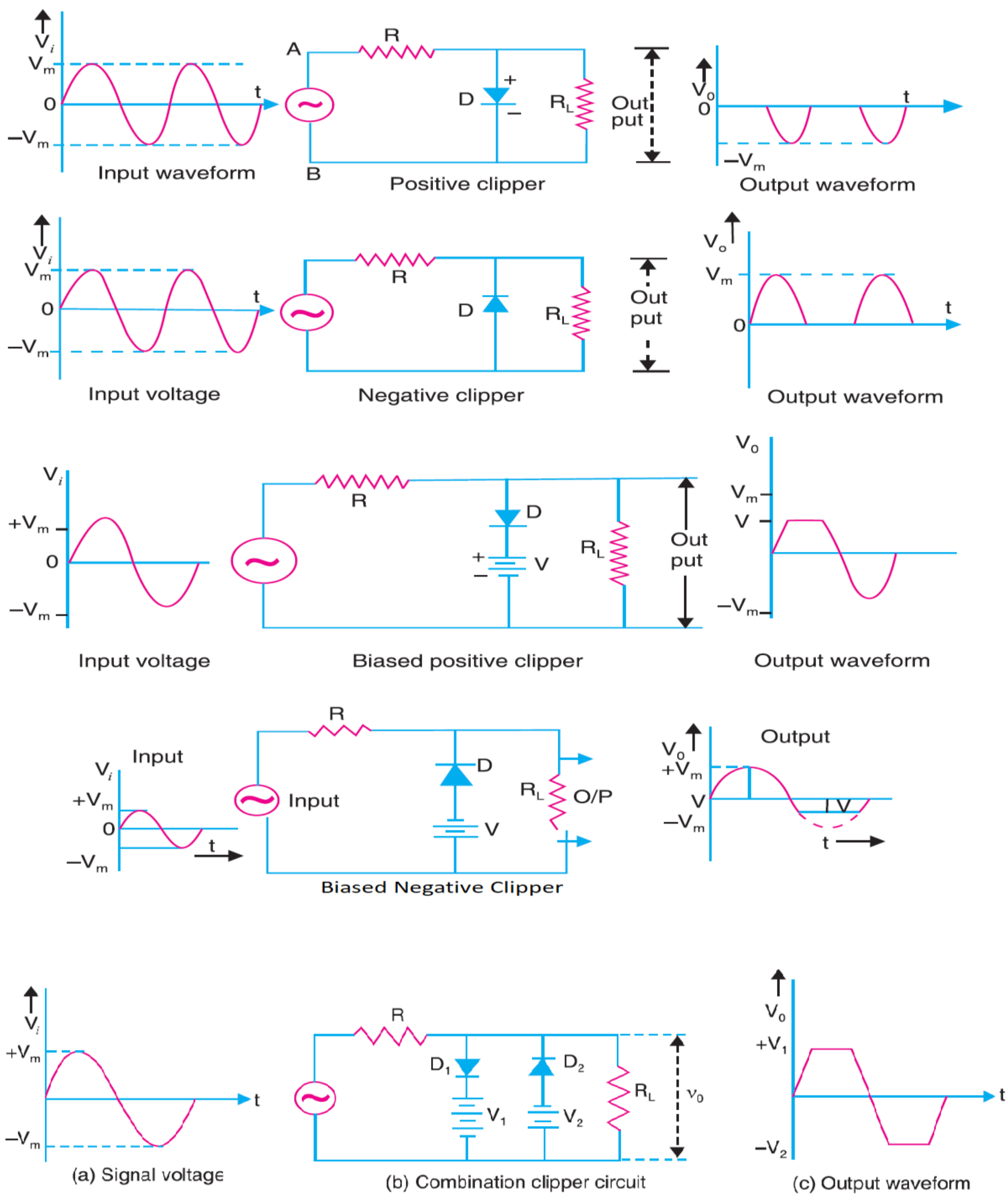


Figure 4.1: Clipper Circuit Types

Required Equipment:

- Trainer Board/ Bread Board
- Variable DC power supply
- Oscilloscope
- Function generator
- Multimeter
- Resistors(1K)
- Capacitor 0.1uF and 33uF
- Connection Wires/Jumper Wires
- Probes (Function Connector, Oscilloscope)
- Silicon Diode (IN4007)

Precautions:

- While doing the experiment do not exceed the readings of the diode. This may lead to damaging of the diode.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Circuit Diagram:

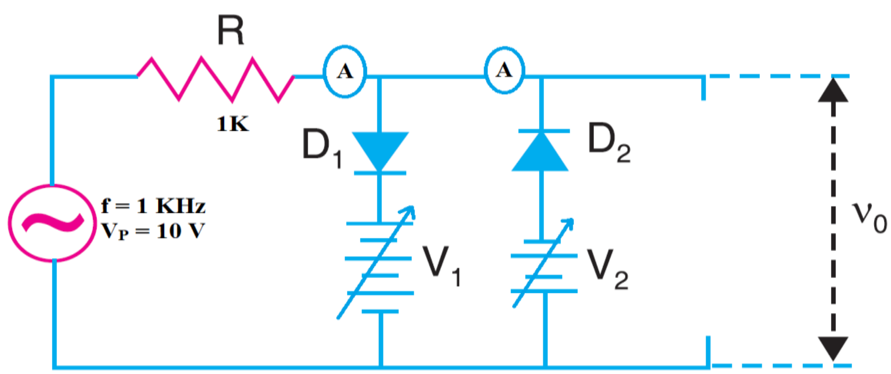


Figure 4.2: Combined Clipper Circuit

Procedure:

- Generate AC voltage in frequency generator as given in Figure 4.2.
- Make V_2 fixed to 2 voltage and take V_1 from 1 to 5 volt in 1 volt step.
- Take the reading of output voltage and current of loop 1 and loop 2.
- Make V_1 fixed to 5 voltage and take V_2 from 1 to 5 volt in 1 volt step.
- Take the reading of output voltage and current of loop 1 and loop 2.

Data Table 4.1:

S/N	V_o (When V_1 is fixed) (V)	V_o (When V_2 is fixed) (V)	I_{L1} (When V_1 is fixed) (mA)	I_{L2} (When V_2 is fixed) (mA)

Graph:

- Take a graph sheet and divide it into 2 equal parts.
- Now See the Graph of Input Voltage from the frequency generator in the oscilloscope.
- Also see the final graph when V_1 is 5 V and V_2 is 2 V on the other screen of oscilloscope.
- Compare the Graph

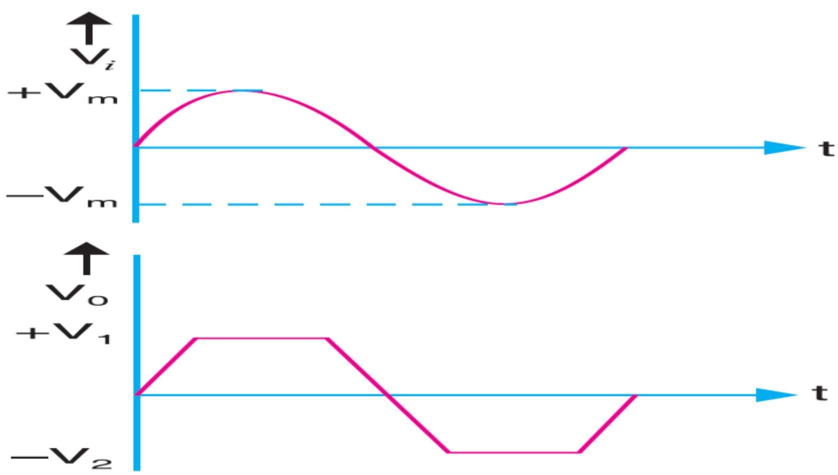


Figure 4.3: Input and Output Graph of Clipper Circuit

<u>Calculation from Graph:</u>

Find the Peak voltage of output voltage.

Clamper Circuit using Diode

Theory:

A clamper is a network constructed of a diode, a resistor, and a capacitor that shifts a waveform to a different dc level without changing the appearance of the applied signal. Additional shifts can also be obtained by introducing a dc supply to the basic structure. The chosen resistor and capacitor of the network must be chosen such that the time constant determined by τ = RC is sufficiently large to ensure that the voltage across the capacitor does not discharge significantly during the interval the diode is non-conducting. Throughout the analysis we assume that for all practical purposes the capacitor fully charges or discharges in five-time constants.
Clamping networks have a capacitor connected directly from input to output with a resistive element in parallel with the output signal. The diode is also in parallel with the output signal but may or may not have a series dc supply as an added element.

Clamper circuit is two types. One is **Positive Clamper** another is **Negative Clamper**.

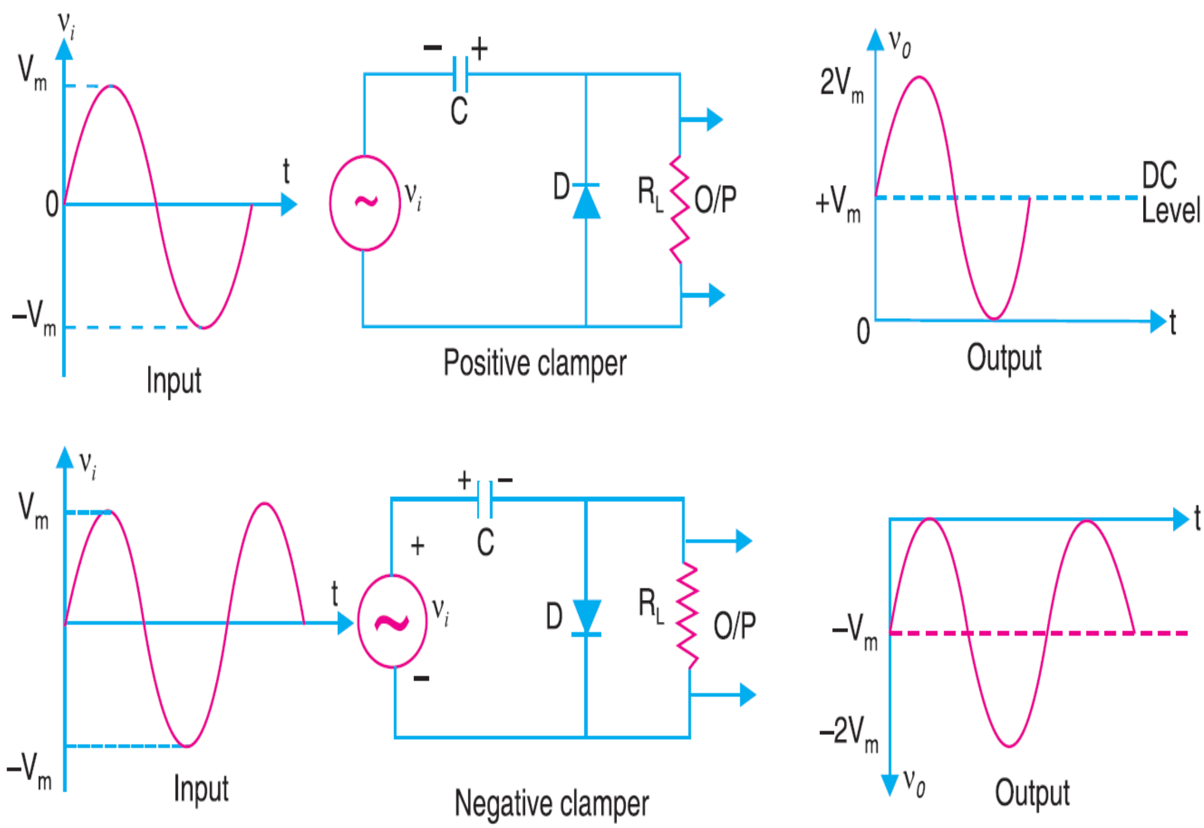


Figure 4.5: Clamper Circuit

Circuit Diagram:

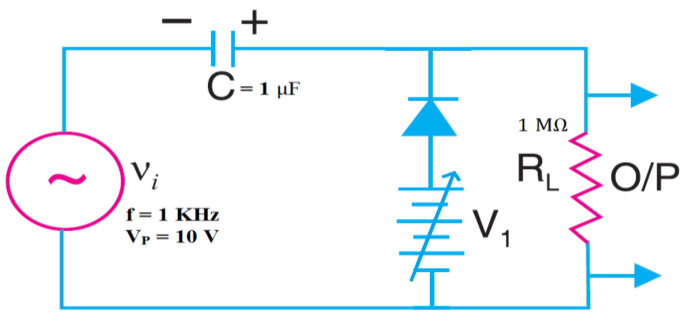


Figure 4.6: Clamper Circuit

Procedure:

- Generate AC voltage in frequency generator as given in figure 4.5
- Take V_1 from 1 to 5 volts in 1volt step.
- Take the reading of output voltage.
- Take the reading of capacitor voltage.
- Set R_L = 250 Ω, C = 1 μF and V_1 = 0V and observe the output.

Data Table 4.2:

S/N	V_o,(V)	Vc,(V)

Graph:

- Take a graph sheet and divide it into 2 equal parts.
- Now See the Graph of Input Voltage from the frequency generator in the oscilloscope.
- Also see the final graph when V_1 is 5 V on the other screen of oscilloscope.
- Compare the Graph.

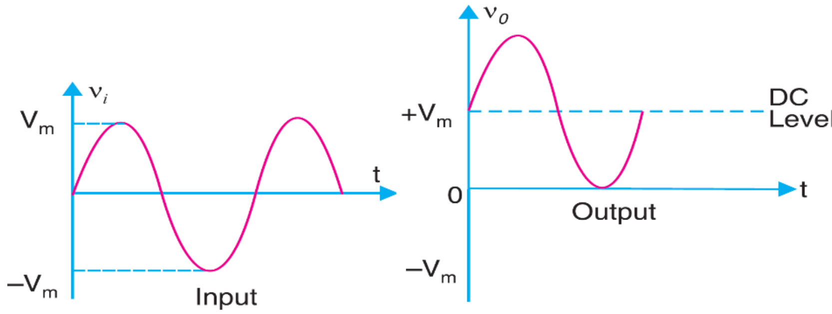


Figure 4.7: Input and Output Graph of Clamper Circuit

Calculation from Graph:

Find the Peak voltage of output voltage.

Result and Discussion:

Working principle of clipper and clamper circuit is learned from this experiment and also we observed that, during variable voltage how the output voltage is changed.

Knowledge Test Questions:

Pre-Lab Questions

1. Describe the difference between sinusoidal, pulse and sawtooth wave?
2. What is the significant of time dependent variable?
3. How to make regulator through potentiometer?
4. What is DC value of an AC signal?
5. What is the significant of multimeter?
6. What is frequency and angular frequency?

Post Lab Questions

7. Why we need clipper circuit?
8. What is the meaning of negative clipper?
9. What have you learned from this experiment?
10. Why we need clamper circuit?
11. What is the meaning of negative clamper?
12. sketch all the waveforms observed on the oscilloscope.

Experiment No: 05
Experiment Name: Study of Input-Output characteristics of CE/CB configuration BJT and different BJT biasing.

Objectives:

- The objective of this experiment is to simulate "DC characteristics of BJT in common emitter (CE) and common base (CB) configuration".
- To see the working process of the BJT biasing.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
05	CO2: Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.	**Investigation (PO4)** **Modern Tool Usage (PO5)**	**Affective domain/ analyzing level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Equipment list:

1. Trainer Board
2. Variable DC Power Supply
3. Oscilloscope
4. Function Generator
5. Multimeter
6. Resistor 250 KΩ,500 KΩ, 1 MΩ
7. Connecting Wires/ Jumper Wires (Male-Male, Male-Female and Female-Female)
8. Probes (Function Generator)

Precaution:

- While doing the experiment do not exceed the readings of the diode. This may lead to damaging of the diode.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Theory:

Input-Output characteristics of CE/CB configuration BJT

Transistor has two p-n junctions (see 5.1). One junction is called emitter junction and other is called collector function. When transistor is used as an amplifier, it is operated in active mode. In active mode, emitter junction is forward biased and collector junction is reverse biased. Emitter current is given by,

$$I_E = I_{nE} + I_{pE}$$

We can also write,

$$I_E = I_c + I_B = \left(\frac{\beta + 1}{\beta}\right) I_c$$

Where common emitter current gain, $\beta = \frac{I_c}{I_B}$

In good transistor $I_c \gg I_B$ i.e. $\beta \gg I_c$ can also be expressed by $I_c = \alpha I_E$

Where, $\alpha = \frac{\beta}{\beta+1}$ is called common base current gain. α is closed to unity.

Proper dc biasing of a transistor is a prerequisite for proper operation as an amplifier. The purpose of the biasing is to fix the I_c (dc) and V_{CE} (dc). But I_c is a function of temperature, V_{BE} and β. It is always desirable to design a biasing circuit where I_c is insensitive to change in β.

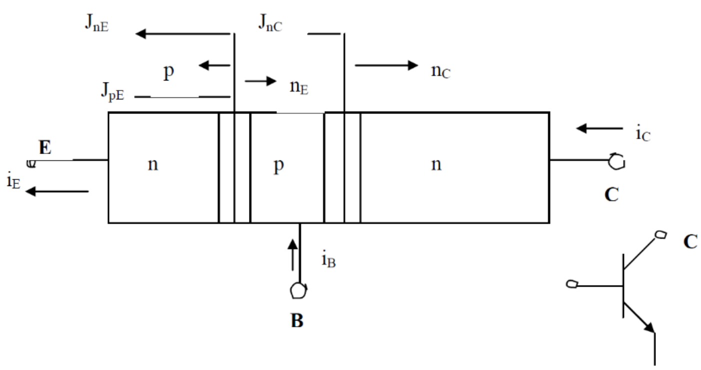


Figure 5.1 - p - n Junction Transistor

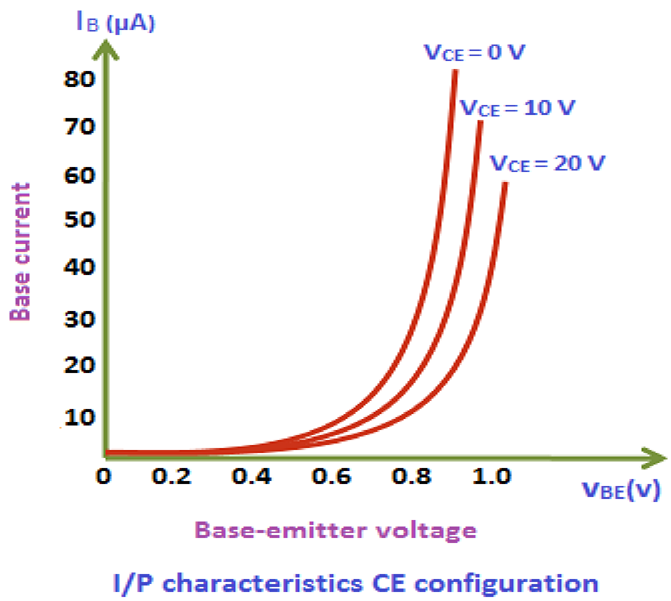


Figure 5.2: Input characteristics curve

Output Characteristics and Load Line

Static output characteristics define steady state conditions family of curves of I_c vs V_{CE} for increasing I_B. Graphical design procedure gives performance under non – linear conditions and start and finish points in analytical procedure. For $V_i = 0$, $I_B = 0$, no current flows through R_L thus $I_c = 0$. So, $V_S = V_{CE}$ (max). For $V_i >> V_{BE}$ (say $V_i = V_S$), I_B is high and I_C is at maximum and nearly all volts dropped across R_L. So, $V_{CE} = 0$ and $I_c\ (max) = \frac{V_S}{R_L}$. For that transistor is said to be saturated. A load line drawn from I_c (max) to V_{CE} in Figure and slope of the load line is $-\frac{1}{R_L}$.

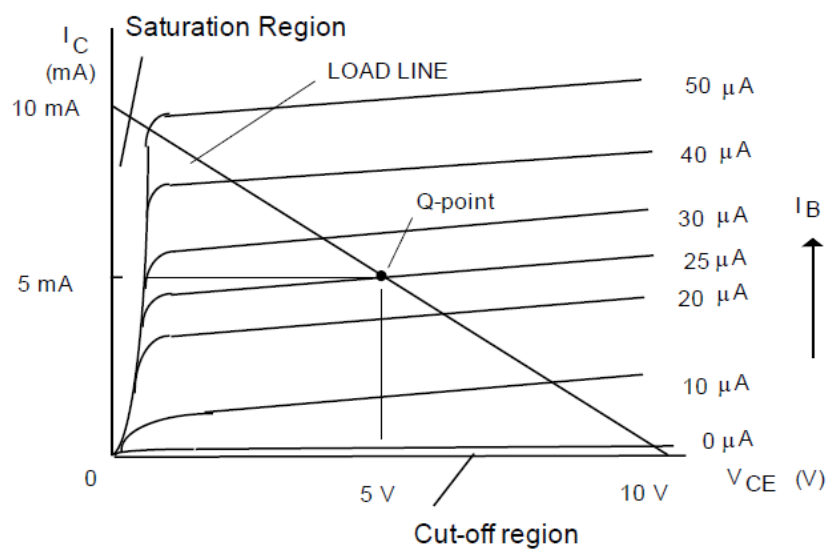


Figure5.3 - I - V Graph of BJT

Output Characteristics

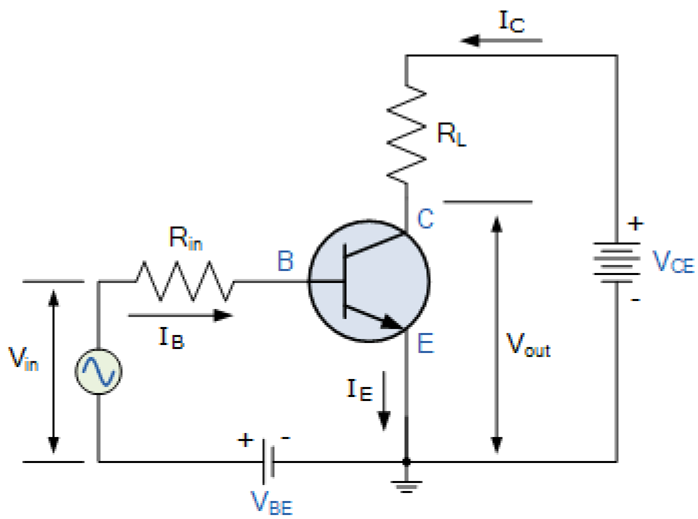


Figure 5.4 - Circuit for DC Analysis of BJT in CE Configuration

Procedure:

- Connect the circuit.
- Then set a fixed input voltage so that I_B is constant.
- Vary the V_{CE} voltage and take down notes for different V_{out} and I_C.
- Set the value of R_{in}= 500 KΩ when V_{in}=5V and vary V_{CE}.
- Again for the same circuit set the value of R_{in}= 250 KΩ when V_{in}=5V and vary V_{CE}.
- Plot the graph for I_C Vs. V_{CE} graph sheet.

Observation:

S.N	R_{in}= 500 KΩ		R_{in}= 250 KΩ	
	V_{CE} (Volts)	I_C (mA)	V_{CE} (Volts)	I_C (mA)

Study of Biasing BJT

Theory:

In order to characterize the operation of a particular transistor, a complete set of characteristic equations is needed. Typically, these curves look like those in Figure 5.5. These curves show that in the active region of operation, the collector current is constant and depends on the base current. These curves can be used to calculate the large signal current gain, β_{DC} (or h_{FE}) and the small signal current gain, β_{AC} (or h_{fe}). These values are in general calculated for a given bias point I_{CQ}, V_{CEQ} using the following equations:

$$\beta_{DC} = \frac{I_{CQ}}{I_{BQ}}$$

$$\beta_{AC} = \left| \frac{I_{CQ} - I_{CQ'}}{I_{BQ} - I_{BQ'}} \right|$$

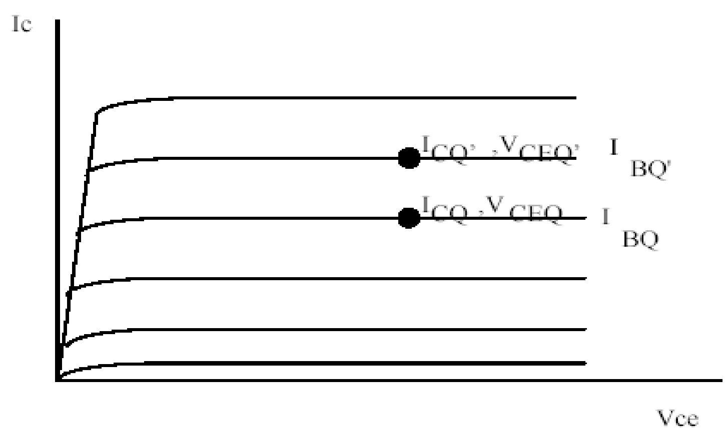


Figure 5.5: Output Characteristics Curve for the BJT

From this, one can see that the large signal gain depends only on the Q point and the small signal gain depends only on small deviations around the Q point.

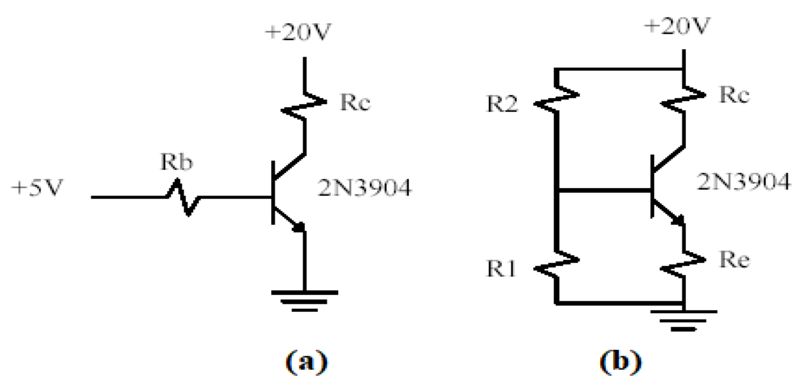


Figure 5.6: (a) Fixed Biased and (b) Voltage Divider Biased Circuits

In order to use a transistor in an amplifying circuit it has to be biased. In other words, a Q point has to be set in order to place the device in the active region of operation. There are several methods which can be used bias a transistor. Figure 5.6 demonstrate two possibilities. The first scheme (Figure 5.6(a)) is called a fixed bias scheme. In a fixed biasing the base current is set through a base resistor and the emitter of the transistor is grounded. This scheme is not used in practice since the Q point depends very strongly on β. A second possibility, which is commonly used, is the self-biasing scheme. Here the base voltage is set through a voltage divider and the emitter is tied to ground through a resistor. If designed correctly, this scheme is relatively independent of β.

Circuit Diagram:

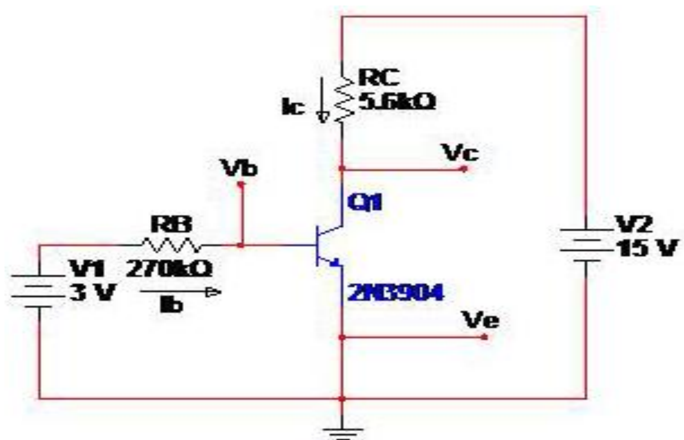


Figure 5.7: Biasing of BJT

Procedure:
Saturation Mode

- Connect the base collector in forward bias and the base emitter in forward bias to prove the relation of the current through the collector and emitter. The relation that has proven is:

$$I_C = I_E$$

- Connect 4.4V from breadboard to resistor (270k) to the bias of transistor.
- Connect the red wire of the function that carry 15V to the resistor (5.6k) and then connect to the collector.
- Connect the black wire from the function generator(ground) and wire from the ground of the breadboard to the emitter.
- Use Multimeter to measure the currents and voltages.

Observations:

V_{CC}	V_B	V_E	V_C	V_{CE}	I_B	I_E	I_C	β

Graph:

Using x axis as a V_{CC} and y axis as a I_C and determine the graph.

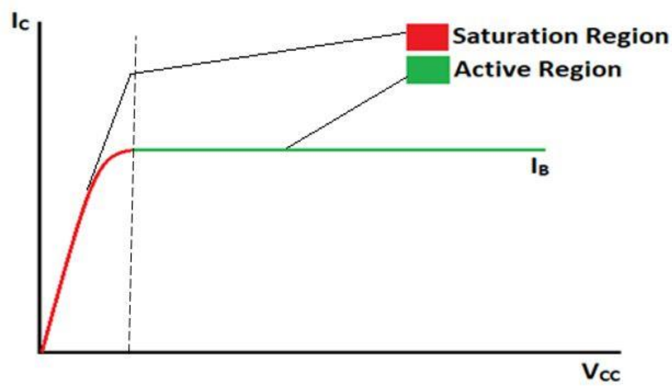


Figure 5.8: Transistor Graph from Data

Knowledge Test Questions

Pre-Lab Questions

1. What is the equation for voltage gain?
2. What is cut off frequency?
3. What is active region?

Post Lab Questions

1. What are the applications of CE amplifier?
2. What is Bandwidth?
3. Describe the impact of amplifier?

Experiment No: 06
Experiment Name: Design of a CE BJT amplifier and analysis of frequency response and h-parameters of the same amplifier

Objective:

The objective of this experiment is

- To Measure the voltage gain of a CE amplifier
- To draw the frequency response curve of the CE amplifier

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
06	CO2: Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.	**Investigation (PO4)** **Modern Tool Usage (PO5)**	**Affective domain/ analyzing level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Required Equipment:

1. Trainer Board
2. Oscilloscope
3. Function Generator
4. Multimeter
5. Resistance 33 KΩ, 3.3 KΩ, 330 KΩ , 1.5 KΩ, 1 KΩ , 2.2 KΩ, 4.7 KΩ
6. Capacitors 10µF, 100 µF, 1K pF
7. Transistor: BC 107
8. Connecting Wires/ Jumper Wires (Male-Male, Male-Female and Female-Female)
9. Probes (Function Generator)

Precautions:

- While doing the experiment do not exceed the readings of the diode. This may lead to damaging of the diode.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Theory:

The CE amplifier provides high gain & wide frequency response. The emitter lead is common to both input & output circuits and is grounded. The emitter-base circuit is forward biased. The collector current is controlled by the base current rather than emitter current. When a transistor is biased in active region it acts like an amplifier. The input signal is applied to base terminal of the transistor and amplifier output is taken across collector terminal. A very small change in base current produces a much larger change in collector current. When positive half-cycle is fed to the input circuit, it opposes the forward bias of the circuit which causes the collector current to decrease; it decreases the voltage more negative. Thus, when input cycle varies through a negative half-cycle, increases the forward bias of the circuit, which causes the collector current to increases thus the output signal is common emitter amplifier is in out of phase with the input signal. An amplified output signal is obtained when this fluctuating collector current flows through a collector resistor Rc. The capacitor across the collector resistor Rc will act as a bypass capacitor. This will improve high frequency response of amplifier.

Circuit Diagram:

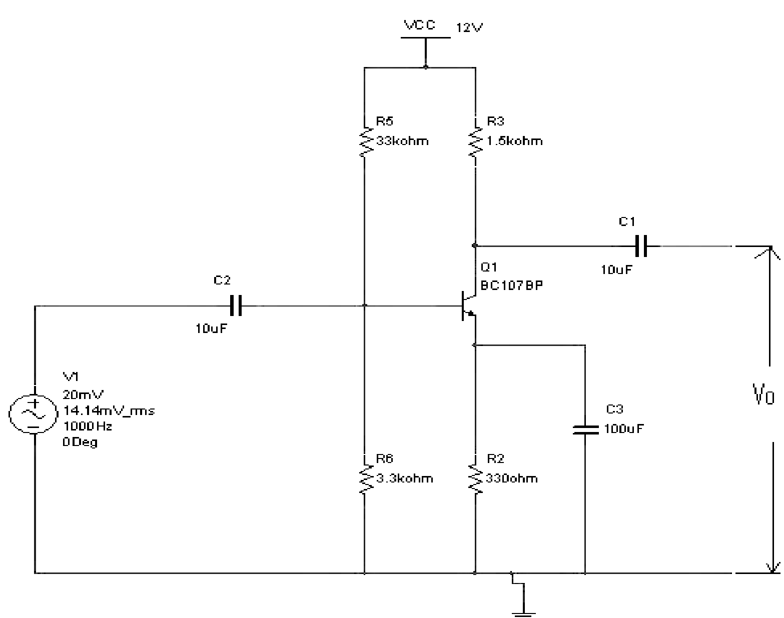


Figure: 6.1

Procedure:

- Connect the circuit as shown in circuit diagram.
- Apply the input of 20mV peak-to-peak and 1 KHz frequency using Function Generator
- The voltage gain can be calculated by using the expression, $Av = (V_0/V_i)$
- For plotting the frequency response, the input voltage is kept Constant at 20mV peak-to-peak, and the frequency is varied from 100Hz to 1MHz Using function generator
- Note down the value of output voltage for each frequency.
- All the readings are tabulated and voltage gain in dB is calculated by Using The expression Av = 20 log10 (V0/Vi)
- A graph is drawn by taking frequency on x-axis and gain in dB on y-axis On Semi log graph.
- The band width of the amplifier is calculated from the graph using the expression, Bandwidth, BW = f_2-f_1
- Where f_1 lower cut-off frequency of CE amplifier, and where f_2 upper cut-off frequency of CE amplifier
- The bandwidth product of the amplifier is calculated using the Expression
  Gain Bandwidth product = 3 dB midband gain X Bandwidth

Observation:

Make a table to determine output voltage, V_o and gain,G like below -

S/N	V_i (V)	Frequency, f (KHz)	Output Voltage, Vo (V)	Gain, G (dB)=20log10(V_o/V_i)
1				
2				
3				
4				
5				

Graph:

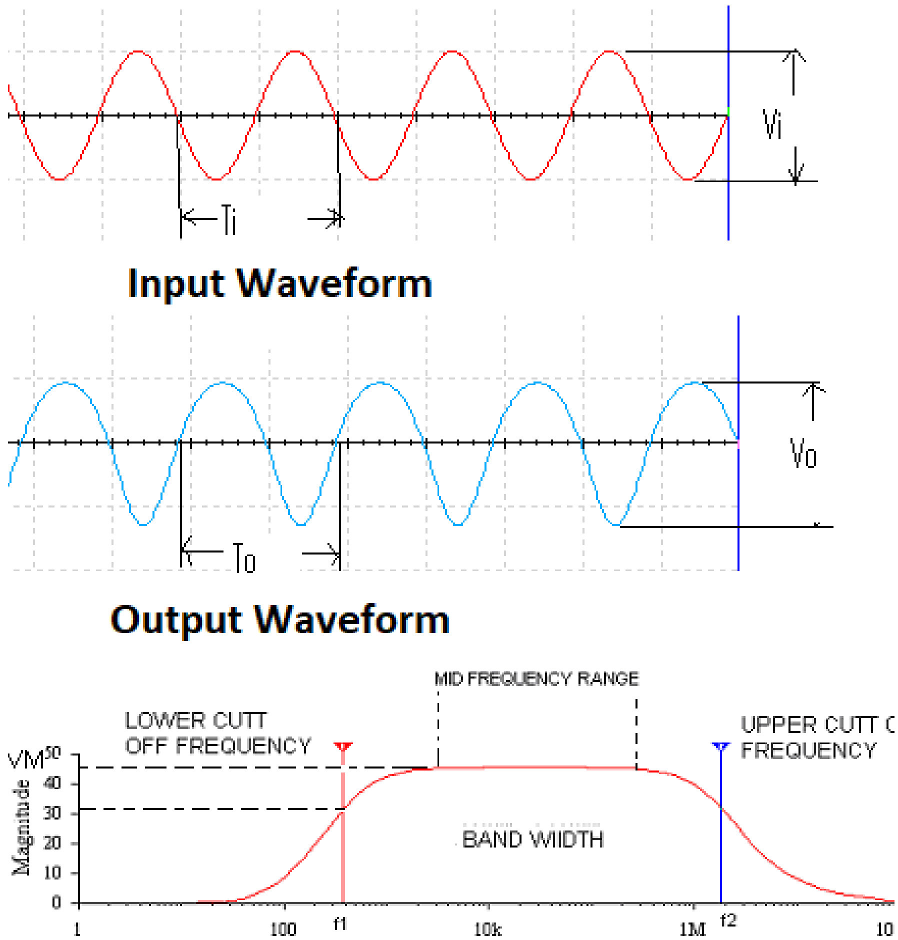


Figure: 6.2

Discussion:

We have to study.

- Gain
- Frequency Response

Knowledge Test Questions

Pre-Lab Questions

1. What is the operating point?
2. What type of amplifier is used in BJT??
3. What is active region?

Post Lab Questions

1. What is the base current amplification factor β ?
2. What is Bandwidth?
3. Describe the impact of amplifier?

Experiment No: 07
Experiment Name: Investigation of drain and transfer characteristics of MOSFET.

Objective:
The objective of this experiment is to simulate "**Study on DC characteristics of Enhancement-type Metal-Oxide-Semiconductor Field-Effect Transistor (MOSFET)**."

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
07	CO2: Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering, and oscillation.	**Investigation (PO4)** **Modern Tool Usage (PO5)**	**Affective domain/ analyzing level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Equipment List:

1. Trainer Board
2. Variable DC Power Supply
3. Oscilloscope
4. Function Generator
5. Multimeter
6. Resistor 74 KΩ, 15 KΩ, 4.7 KΩ , 1 KΩ, 2.2 KΩ , 8.2 KΩ
7. MOSFET
8. Connecting Wires/ Jumper Wires (Male-Male, Male-Female and Female-Female)
9. Probes (Function Generator)

Precaution:

- While doing the experiment do not exceed the readings of the MOSFET. This may lead to damaging of the device.
- Find all the terminal of the MOSFET before making the circuit. Wrong Selection of terminal will damage the MOSFET, or desire output will not be obtained.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Theory:

It is a three-terminal device.

- Source: source of charge carriers (current)
- Drain: sink of charge carriers (current)
- Gate: potential (voltage) on gate controls current flow

The basic component in a CMOS circuit is the MOSFET transistor. This transistor consists of a drain contact and a source contact with a channel in between which is doped such that it will not conduct when a potential is applied between the drain and source. It cannot conduct because it has no charge carriers of the type produced by the source electrode. If a suitable potential (voltage) is applied between the gate electrode and the source then charge is induced in the channel by the electric field set up across the gate oxide layer. This charge consists of the same type of carriers as in the source and hence the device can conduct.

By using n-type impurities in the source (electron donors), we need to induce a negative charge in the channel (initially p type = material). This gives an n-channel device. On the other hand, we can dope the source with p type impurities and produce holes which act as positive charge carriers. We thus need a positive charge induced in the channel (originally n-type material) to make it conduct. This is a p channel device.

As the device is normally off and a voltage is required on the gate to make it conduct this type of MOSFET is known as an enhancement mode device. Depletion mode devices exist (normally conduct until turned off by a voltage on the gate) and these can be useful in logic circuits but not as switching elements. Here we see the basic construction of a MOSFET.

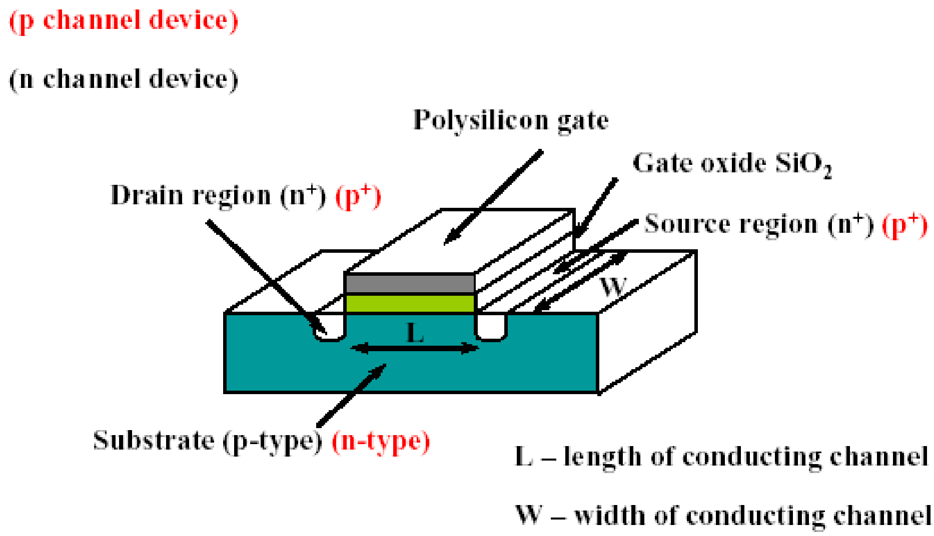


Figure 7.1 - Basic Structure of MOSFET

Note the gate electrode sits above the channel and is separated from it by a thin layer of oxide. When a positive voltage is applied to the gate a negative charge is induced in the channel. If the source and drain are n-type material then they need a negative charge in the channel to make the device conduct. This is an n channel device and the channel will initially be p-type and hence have no negative charge carriers to allow it to conduct.

The opposite is true for p-channel devices as indicated in the diagram. The major geometrical parameters (dimensions) of importance are the length of the channel (L), the width of the channel (W) and the thickness of the gate oxide t_{ox}. If we vary the voltage between the drain and source electrode whilst keeping the gate electrode at some fixed voltage, we get the current flow shown in the graphs like Figure . Note that the n channel device needs a positive voltage on its gate relative to the source to make it conduct. On the other hand, the p-channel device needs a negative voltage on its gate to make it conduct. Its charge carriers are positive and hence the drain needs to be negative relative to the source in order to attract the carriers and produce a current flow. The complementary nature of these characteristics is what we exploit in a CMOS gate (Complementary Metal Oxide Silicon).

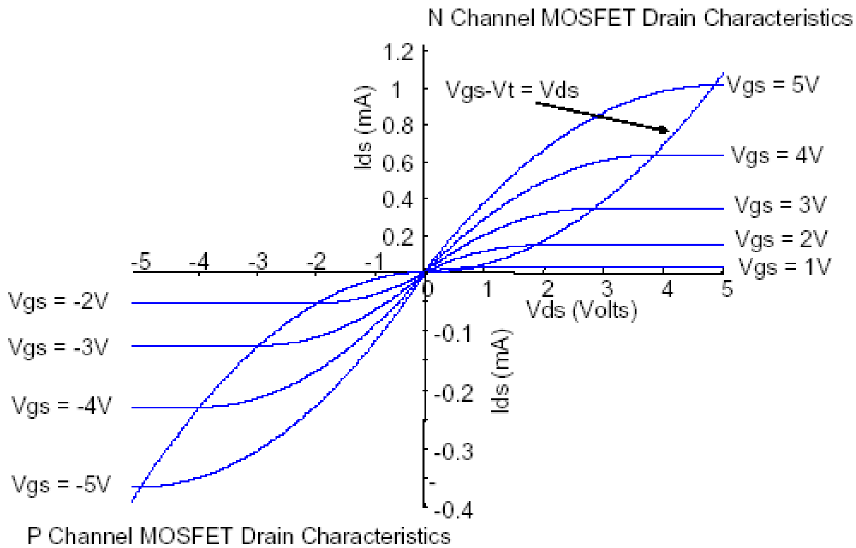


Figure 7.2 - MOSFET Characteristics

When $V_{GS} > V_T$, the transistor is ON. Now dependent on the value of drain voltage V_{DS}, MOSFET either operates in triode or saturation region. Before saturation, $(V_{GS} - V_{Ds}) > V_T$ and

$$I_D \approx \mu_n C_{ox} \left(\frac{W}{L}\right) \left[(V_{GS} - V_T)\, V_{DS} - \frac{1}{2} V_{DS}^2\right]$$

In saturation region $(V_{GS} - V_{DS}) < V_T$ and

$$I_D = \mu_n C_{ox} \left(\frac{W}{L}\right) (V_{GS} - V_T)^2 (1 + \lambda V_{DS})$$

Where, $\lambda = \frac{1}{V_A}$. V_A is the early voltage. μ_n is the electron mobility $(\frac{cm^2}{V.s})$, C_{ox} is the oxide capacitance per unit, W and L are width and length of the gate respectively.

Circuit Diagram:

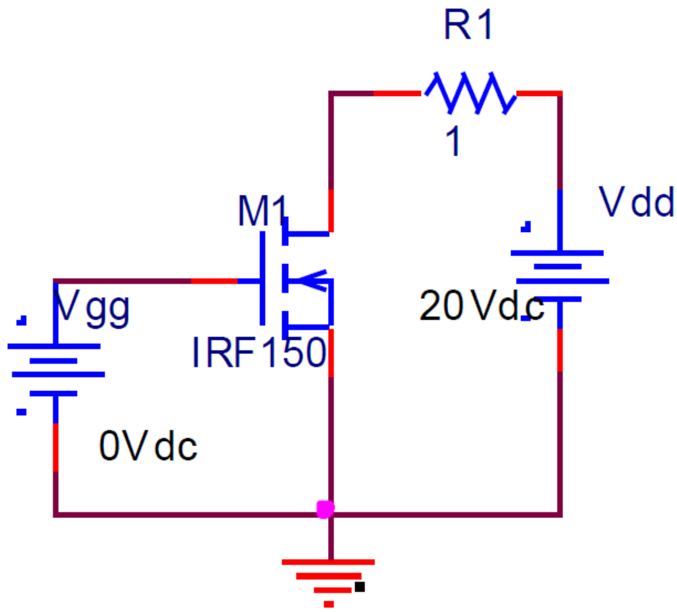


Figure 7.3 - Circuit for DC analysis of Enhancement type (n) MOSFET

Procedure:

1. Build the circuit on the breadboard shown in the figure.
2. Use the variable DC voltage source to supply the voltage for the biasing of MOSFET.
3. Calibrate the Oscilloscope before use it.
4. Measure the input and output voltage.
5. Find out the entire terminal voltage using multimeter.
6. Turn off the power to all the equipment after taking data.

Observation:

V_{GS} = 1.1 V		V_{GS} = 1.2 V		V_{GS} = 1.3 V		V_{GS} = 1.4 V		V_{GS} = 1.5 V	
V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)	V_{DS} (V)	I_D (mA)

CALCULATION:

1. Threshold voltage V_T: Gate to source voltage at which, drain current starts flowing.

2. Transconductance g_m: Ratio of small change in drain current (ΔI_D) to the corresponding Change in gate to source voltage (ΔV_{GS}) for a constant V_{DS}. $G_m = \Delta I_D / \Delta V_{GS}$, at constant V_{DS}.

3. Output drain resistance: It is given by the relation of small change in drain to source voltage (ΔV_{DS}) to the corresponding change in Drain Current (ΔI_D) for a constant V_{GS}.
r_d or $r_o = \Delta V_{DS} / \Delta I_D$ at a constant V_{GS}.

Graph: Plot I_D vs V_{DS} for different values of V_{GS}.

Result and Discussion:

V_T =
G_m =
R_o=

Conclusion:

Knowledge Test Questions

Pre-Lab Questions

1. What is threshold voltage?
2. What is transconductance?
3. What is active region?

Post Lab Questions

1. What are the types of MOSFET?
2. Write down the advantages of MOSFET over BJT?
3. Is the MOSFET a voltage controlled or current controlled device?

Experiment No: 08
Experiment Name: Study of a 741 Operational Amplifier as Inverting amplifier, non-inverting amplifier, differentiator, integrator circuit and passive filter.

Objective:

- To study op-amp as an amplifier in both inverting and non-inverting configurations.
- To study op-amp as an amplifier in both differentiator, and integrator configurations.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
08	CO2: Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.	**Investigation (PO4)** **Modern Tool Usage (PO5)**	**Affective domain/ analyzing level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Equipment List:

1. Trainer Board
2. Oscilloscope
3. Function Generator
4. Multimeter
5. Resistance 100 KΩ
6. IC 741
7. Connecting Wires/ Jumper Wires (Male-Male, Male-Female and Female-Female)
8. Probes (Function Generator)

Precaution:

- While doing the experiment do not exceed the readings of the OP AMP. This may lead to damaging of the device.
- Find all the pins of the OP AMP before making the circuit. Wrong Selection of pin will damage the OP AMP, or desire output will not be obtained.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Theory:

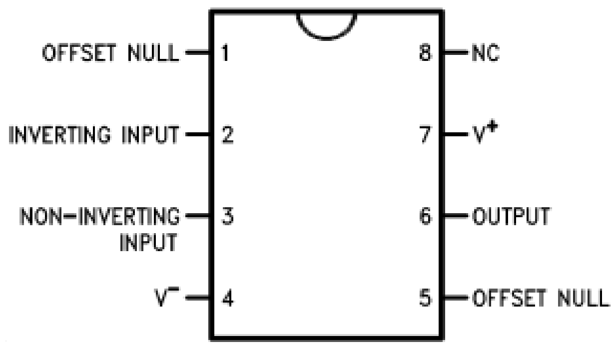


Figure 8.1: Op-amp IC. (LM 741)

An amplifier is a circuit that receives a signal of some amplitude at its input and delivers an undistorted larger amplitude of the signal at its output. The operational amplifier package is a high gain amplifier to which feedback is added to control its overall response characteristic.

An Operational Amplifier, or op-amp for short, is fundamentally a voltage amplifying device designed to be used with external feedback components such as resistors and capacitors between its output and input terminals. These feedback components determine the resulting function or "operation" of the amplifier and by virtue of the different feedback configurations whether resistive, capacitive or both, the amplifier can perform a variety of different operations, giving rise to its name of "Operational Amplifier".

Equivalent Circuit of an Ideal Operational Amplifier where:

- $\mathbf{Z_{in}}$ **=** infinite
- $\mathbf{Z_{out}}$ **=** Zero ohms
- **Open loop Gain =** 10^4 to 10^6

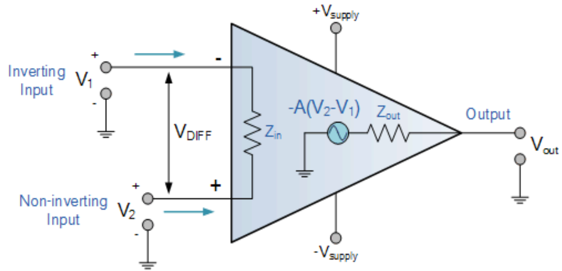


Figure 8.2: Open Loop Gain

The **Voltage Gain** (A_v) of the operational amplifier can be found using the following formula:

$$\text{Voltage Gain, (A)} = \frac{V_{out}}{V_{in}}$$

and in **Decibels** or (dB) is given as:

$$20\log(A) \text{ or } 20\log\frac{V_{out}}{V_{in}} \text{ in dB}$$

In this experiment, we will be studying the closed-loop configuration of a typical op-amp. In a closed-loop configuration, an external resistance is connected between the output and the inverting input terminal as negative feedback. Here, the gain can be controlled by changing the value of the resistor. Two widely used amplifier configurations using op-amp are given below.

- Inverting Amplifier
- Non-inverting Amplifier

In both cases, the closed-loop gain of the amplifier is determined by the input resistance R_i and the feedback resistance R_f.

- **Inverting Amplifier:** In an inverting amplifier, the input is applied to the inverting input terminal. The output obtained here is inverted with respect to the input. The closed-loop gain for this type of amplifier is given below.

$$\text{Closed Loop Gain, } A_{cl} = \frac{Vo}{E1} = -\frac{Rf}{Ri}$$

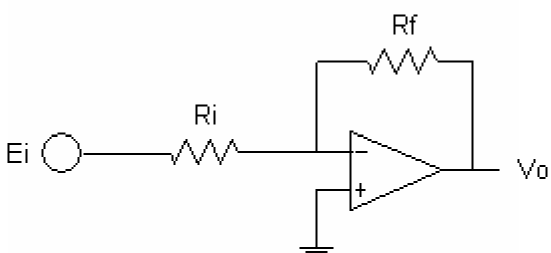


Figure 8.3: Closed Loop Gain for inverting Op-Amp

The minus sign indicates that the polarity of the output is inverted with respect to the input. That is why this circuit is called an inverting amplifier.

▪ **Non-inverting Amplifier:** In this type of amplifier the input is applied to the non-inverting input terminal and the output is not inverted with respect to the input.

$$Gain = \frac{Vo}{Vi} = 1 + \frac{Rf}{Ri}$$

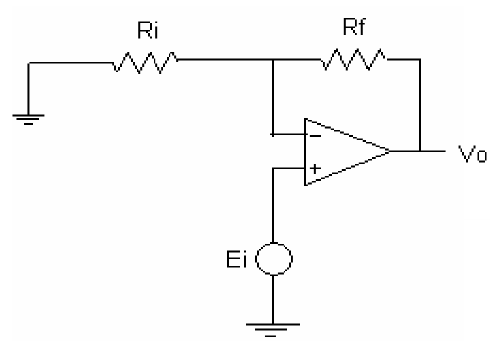


Figure 8.4: Non-inverting Op-Amp

Since the gain is positive, this amplifier is called a non-inverting amplifier. The above equation shows that the voltage gain of the amplifier is always greater than 1.

Inverting Amplifier:

1. Accurately measure the resistance of the resistors by multimeter and record.
2. Construct the following circuit.
3. Set biasing voltages to +15V – Pin 7 and –15V – Pin 4.
4. Set the AC voltage source (Function Generator) to sinusoidal wave mode. Set the sinewave to 1.0 V amplitude and 500Hz frequency.
5. Observe the waveform in the oscilloscope. Take readings for different frequencies e.g., 50Hz, 500Hz, 1KHz, 5KHz, 10KHz, 50KHz and 100KHz. For the 2 channels of the oscilloscope use one to monitor the input waveform from the source and the other the output (Pin 6 of the Op-amp) waveform.
6. Apply input signal of 1V amplitude to Pin 2. Apply the input probe to Source voltage terminal.
7. Fill in the table.
8. Draw the input and output waveforms.
9. Plot the Gain in dB vs the frequency in a semilog graph paper with Y axis Linear and X Axis Log

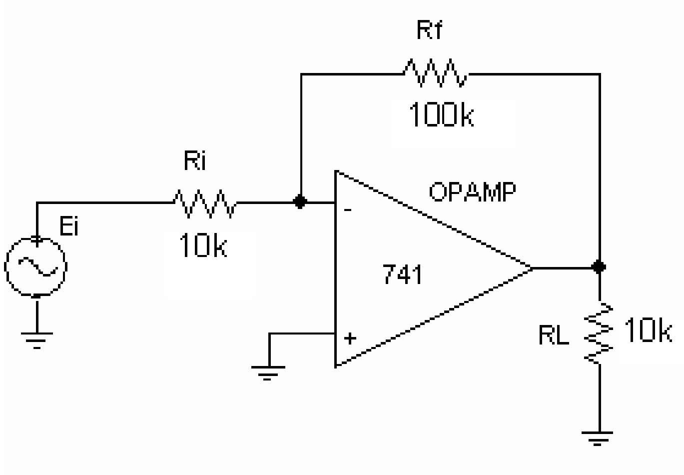


Figure 8.5:

Observation:

f (Hz)	R_i (Ω)	R_f (Ω)	Actual Gain $A_{CL} = - R_f / R_i$	E_i (Amplitude)	V_o (p-p)	Observed Gain $A_o = V_o / E_i$	Gain (dB)
50 Hz				1V			
500 Hz				1V			
1 KHz				1V			
5 KHz				1V			
10 KHz				1V			
50 KHz				1V			
500 KHz				1V			

Graph:

- Draw in a semi log paper (x –axis) the frequency response curve (f vs Gain in dB) and mark the – 3dB point.
- Now change the input resistance R_i to 1K and R_f to 10K and observe the waveforms with a 1 KHz waveform. The Gain is still 10 but can you see some distortions in the waveforms
- Also try to monitor the signal input at Pin 2 of the Op Amp. Check the result? Calculate the gain-bandwidth product GBW.

Non-inverting Amplifier:

1. Accurately measure the resistance of the resistors by multimeter and record.

2. Construct the following circuit.
3. Set biasing voltages to +15V – Pin 7 and –15V – Pin 4.
4. Set the AC voltage source (Function Generator) to sinusoidal wave mode. Set the sinewave to **1.0 V amplitude and 500Hz frequency.**
5. Observe the waveform in the oscilloscope. Take readings for different frequencies e.g. 50Hz, 500Hz, 1KHz, 5KHz, 10KHz, 50KHz and 500KHz. For the 2 channels of the oscilloscope use one to monitor the input waveform (Pin 2) from the source and the other the output (Pin 6) waveform.
6. Apply input signal of 1V amplitude to Pin 3 of the op amp. If you apply the input probe to the probe to the op-amp input terminal (pin 2)
7. Fill in the table.
8. Draw the input and output waveforms.
9. Plot the Gain in dB vs frequency in a semilog graph paper with Y axis Linear and X Axis Log

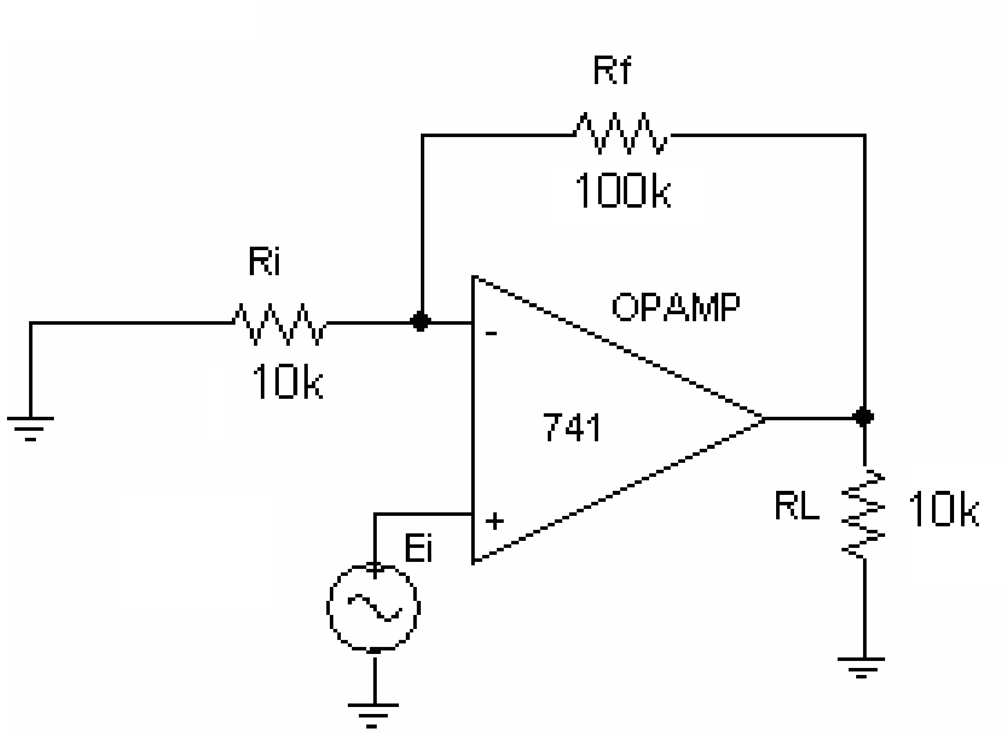


Figure:8.6

Observation:

f (Hz)	R_i	R_f	$A_{CL} = 1 + R_f / R_i$	E_i (Amplitude)	V_o (p-p)	Gain $A_o = V_o / E_i$	Gain (dB)
50 Hz				1V			
500 Hz				1V			
1 KHz				1V			
5 KHz				1V			
10 KHz				1V			
50 KHz				1V			
500 KHz				1V			

Integrator (Using Op-amp): (Low Pass Active Filter)

Operational amplifiers can also be used to form or change the circuit frequency response by making the filter's output bandwidth narrower or even wider by generating a more selective output reaction.

A first-order active low pass filter is a simplistic filter that is composed of only one reactive component Capacitor accompanying with an active component Op-Amp. A resistor is utilized with the capacitor or inductor to form RC or RL low pass filter respectively. In a passive circuit, the output signal amplitude is smaller than the input signal amplitude.

To surmount this problem, active circuit designs were introduced. When a passive low pass filter is connected to an Op-Amp either in inverting or non-inverting condition, it gives an active low pass filter design.

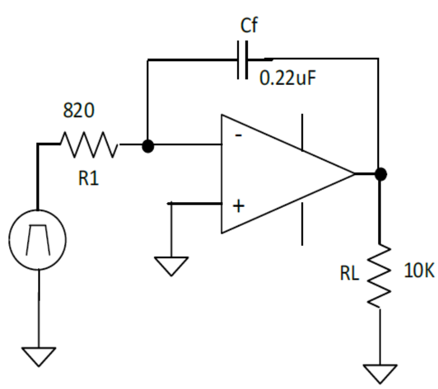


Figure:8.7

$$T_i > R_1 C_f$$

$$V_0(t) = -\frac{1}{R_1 C_f}\int V_i(t)\,dt$$

Procedure for Integrator circuit:

- Construct the above circuits.
- Set biasing voltages to +15V and -15V.
- Set the function generator to square wave mode for the integrator. Set the sine wave to 5V amplitude (10 Volts peak to peak) and 500Hz frequency.
- Draw the input and output waveforms.

Op-Amp as a Differentiator;

$$R_f C_i \geq \frac{10}{\pi f_h}$$

$$V_o(t) = -R_f C_i \, {}^{dv}/_{dt}$$

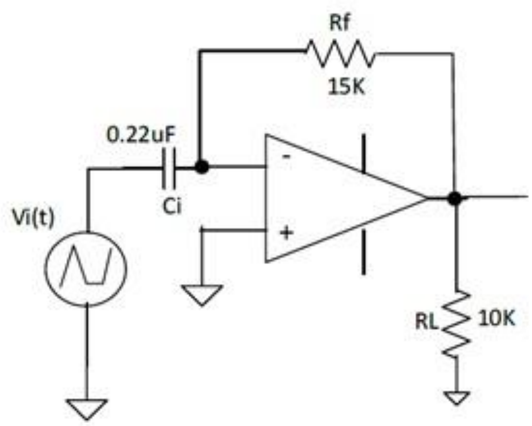


Figure:8.8

Procedure for Differentiator Circuit

- Construct the above circuits.
- Set biasing voltages to +15V and -15V.
- Set the function generator to triangular wave mode for the differentiator. Set the wave to 5V amplitude and 500Hz frequency.
- Observe output and draw the input and output waveforms.

Table 01: Active Lowpass filter

Frequency	$V_{i(P-P)}$	$V_{O(P-P)}$	Phase (°)	A_V = 20log v_o/v_i

Table 2: Active High pass filter

Frequency	$Vi_{(P-P)}$	$Vo_{(P-P)}$	Phase (°)	A_V = 20log log vo/vi

Result and Discussion:

Conclusion:

Knowledge Test Questions

Pre-Lab Questions

1. What are the characteristics of an Ideal Op Amp?
2. Write down the noninverting closed-loop op-amp circuit gain factor?
3. Write down the gain equation for inverting and non-inverting amplifier?

Post Lab Questions

1. Which pins are used for biasing?

2. In the experiments why did we only use the capacitors not inductors?
3. Describe the impact of amplifier?

Experiment No: 09
Experiment Name: Design and analysis of -40 dB/decade active high pass and low pass Butterworth filter using 741 Op-amp.

Objective:
- To study op-amp as active High-pass Active filter and plot the frequency response curve.
- To study op-amp as active Low-pass Butterworth filter.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
01	CO1: Study of microwave components including cable connectors, waveguides and passive devices and their applications.	**Engineering Knowledge (PO1)** **Individual and Team Work(PO9)**	**Cognitive domain/ Understanding level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Theory:

High Pass Filter:
As the name implies, a high pass filter is a filter that passes the higher frequencies and rejects those at lower frequencies. This can be used in many instances, for example when needing to reject low frequency noise, hum, etc. from signals. This may be useful in some audio applications to remove low frequency hum, or within RF to remove low frequency signals that are not required.
Op amp high pass filters are very easy to design and have straightforward equations for the Butterworth response. This first-order high pass filter consists simply of a passive filter followed by a non-inverting amplifier. The frequency response of the circuit is the same as that of the passive filter, except that the amplitude of the signal is increased by the gain of the amplifier
Although any form of filter response can be chosen, the Butterworth response simplifies the equations and it is possible to calculate the values required in seconds.
Despite the ease of design, op amp high pass filters are able to provide a high level of performance from just a relatively small number of components. Op amp high pass filters are easy to implement using a few components and they are required in a variety of different circuits to eliminate hum and other noise.

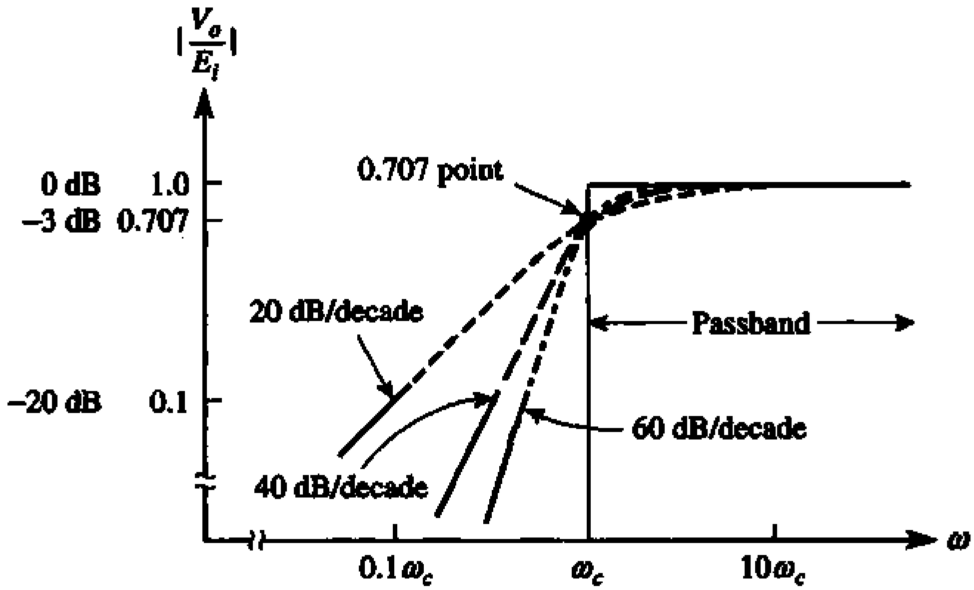


Figure 9.1 (High Pass Filter Curve)

The shape of the curve is of importance. One of the most important features is the cut-off frequency. This is normally taken as the point where the response has fallen by 3dB. Another important feature is the final slope of the roll-off. This is generally governed by the number of 'poles' in the filter. Normally there is one pole for each capacitor or inductor in a filter.
When plotted on a logarithmic scale the ultimate roll-off becomes a straight line, with the response falling at the ultimate roll-off rate. This is 6dB/octave or 20dB/decade per pole within the filter.

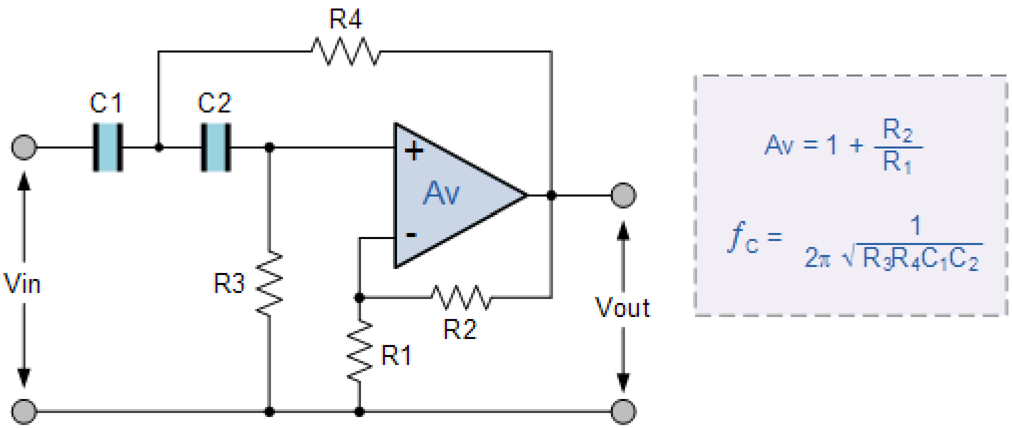


Figure 9.2: Second order high pass filter

Circuit Diagram of high pass filter (in PSpice):

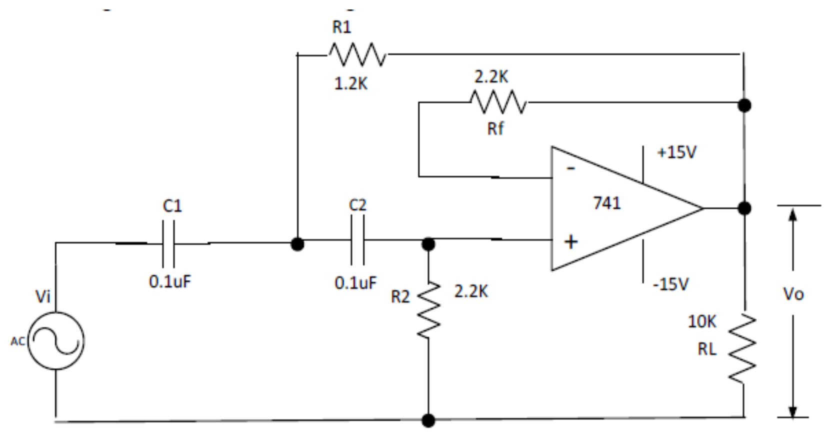


Figure 9.5: Second order high pass filter (PSpice)

1. f_c = 1KHz
2. C = 0.1 uF
3. $R_2 = \frac{1.1414}{2\pi f_c C}$
4. $R_1 = \frac{R_2}{2} = 1.13\ K = 1.2\ K$
5. $R_f = R_2 = 2.2K$
6. $A_{v,dB} = 20 \log_{10} \frac{V_o}{V_i}$

$$V_i = Sinusoidal\ Waveform = 1V_{p-p}$$

Draw the Frequency Response Plot for below data (Av – f) curve in a semi log graph paper with x axis log scale.

Low pass filter:

As the name implies, a low pass filter is a filter that passes the lower frequencies and rejects those at higher frequencies. Operational amplifiers or op-amps provide a very effective means of creating low pass filters without the need for inductors. By incorporating the filter elements into the feedback loop of an op amp, high performance low pass filters are easily created with a minimum number of components and without the need for inductors. Op amp low pass filters can be used in many areas power supplies to the outputs of digital to analogue converters to remove alias signals and many more applications.

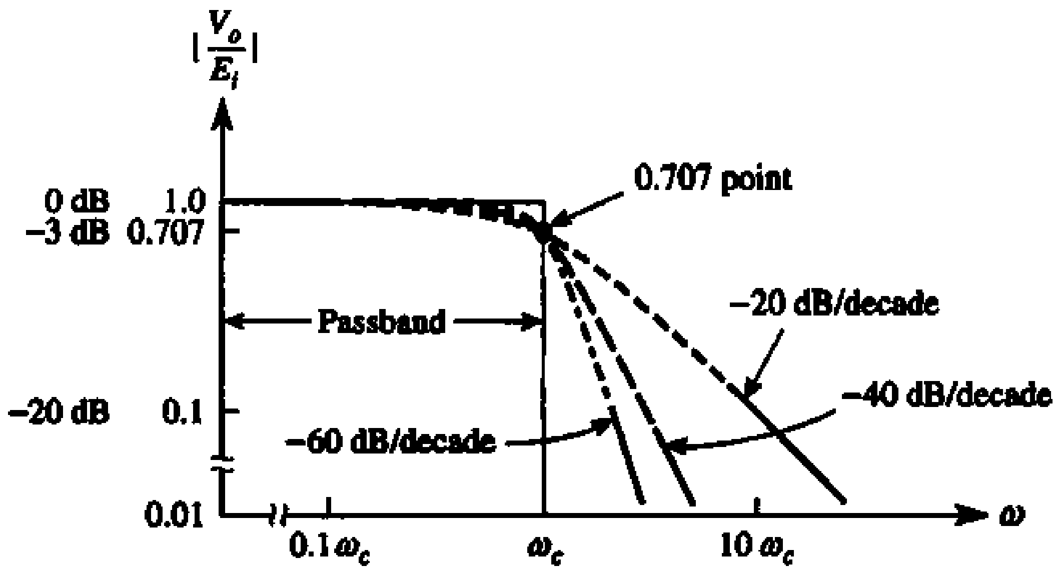


Figure 9.3: Low pass filter response curve

As with the passive filter, a first-order low-pass active filter can be converted into a second-order low pass filter simply by using an additional RC network in the input path. The frequency response of the second-order low pass filter is identical to that of the first-order type except that the stop band roll off will be twice the first-order filters at 40dB/decade (12dB/octave). Therefore, the design steps required of the second-order active low pass filter are the same.

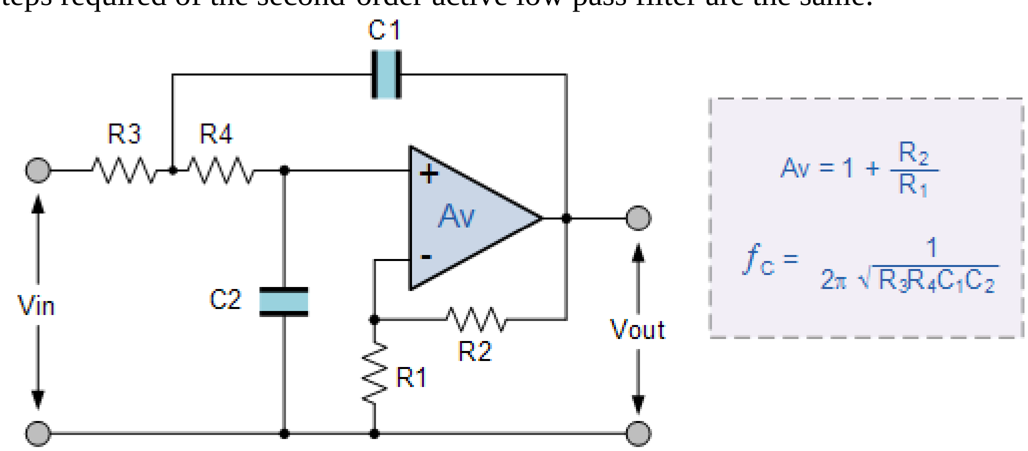


Figure 9.4: (Second order active low pass filter with amplification)

When cascading together filter circuits to form higher-order filters, the overall gain of the filter is equal to the product of each stage. For example, the gain of one stage may be 10 and the gain of the second stage may be 32 and the gain of a third stage may be 100. Then the overall gain will be 32,000, (10 x 32 x 100) as shown below. Second-order (two-pole) active filters are important because higher-order filters can be designed using them. By cascading together first and second-order filters, filters with an order value, either odd or even up to any value can be constructed. In the next tutorial about *filters*, we will see that Active High Pass Filters, can be constructed by reversing the positions of the resistor and capacitor in the circuit.

Circuit Diagram for Active Low Pass Filter (PSpice):

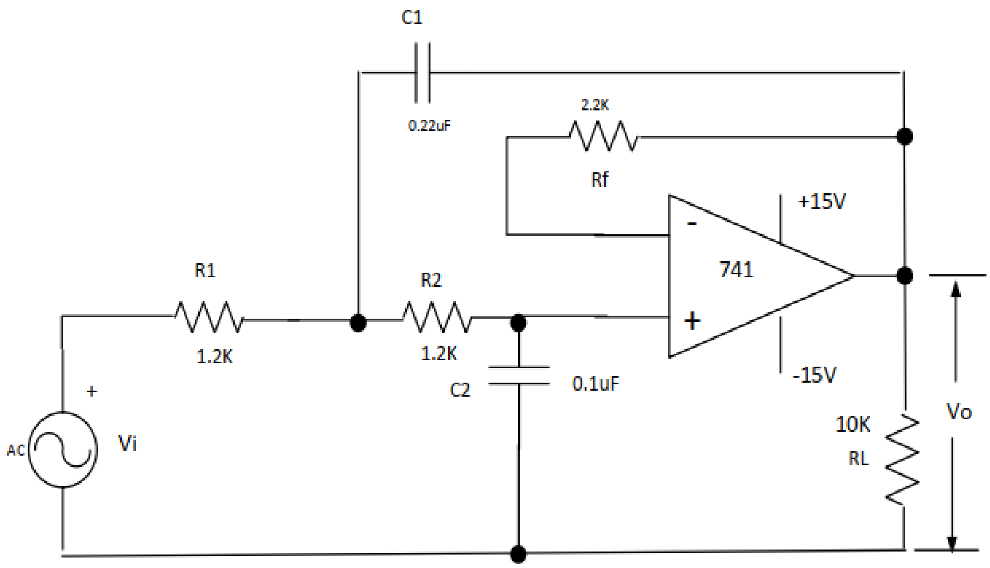


Figure 9.5: Second Order Active Low Pass Filter

1. F_c = 1 KHz
2. C_2 = 0.1 uF
3. C_1 = 0.2 uF
4. $R = \frac{0.707}{2\pi FcC2}$ = 1.13 K = 1.2K (R1 = R2 = R)
5. R_f = 2.4 K = 2.2 K (R_f = 2R)
6. A_v, dB = $20 \log \frac{Vo}{Vi} = 20 \log \frac{Vo,a}{Vi,a}$

Input Voltage Vi = Sine wave Amplitude = 1 Volt (p-p)

Equipment List:

- Op-Amp IC: 741 (1 Piece)
- Resistor: 10k (1 Piece), 1.2K (1 Piece), 2.2K 2 (Pieces)
- Capacitor: 0.1uF 2 Pieces, 0.01uF (2 Pieces), 0.22 uF (2 pieces)
- AT-7000 Trainer Board (1 Unit)
- DC Power Supply (1 Unit)
- Oscilloscope (1 Unit)
- Multimeter (1 Unit)
- Wires
- RLC meter (1 Unit)

Precautions:

- While doing the experiment do not exceed the readings of the Op-Amp. This may lead to damaging of the IC.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Working Procedure:

For high pass filter:

1. Measure the resistance of the resistors with a multimeter and record exact values.
2. Measure the capacitance value with a RLC meter
3. Adjust function generator frequencies as accurately as possible. Set Function generator to Sine wave with 1Vpk-pk amplitude
4. Construct the above circuit. Set biasing voltages to +15V and –15V. (Caution: Do not exceed ± 15 volts DC
5. Draw the Gain vs Frequency curve of the high pass filter. Use semi log graph paper with X axis denoting Frequency (log scale) and determine the lower cutoff or corner frequency, what is the value of the output voltage at the -3dB point.
6. Draw the Gain Vs Frequency in the same graph paper with the component parameter as written on the components or as per the color codes.
7. What is the lower cutoff frequency with the 0.1uF and 0.01 uF capacitor in the graph. Drawboth frequency response curves in one graph paper to compare the change in frequency response.

For low pass filter:

1. Measure the resistor values with a multimeter and record exact values.
2. Adjust function generator frequencies as accurately as possible. Set Function generator to Sinewave with 1Vpk-pk.
3. Construct the above circuit. Set biasing voltages to +15V and –15V.
4. Draw the Gain vs Frequency curve of the low pass filter. Use semi log graph paper with x–Axis denoting Frequency (log scale) and determine the upper cutoff frequency, what is the value of the output voltage at the -3dB point

Observations:

For high pass filter:

S/N	Frequency	C	R	Vo(Volt)	Gain, $A_v = \frac{Vo}{Vi}$	$Av, dB = 20Log(\frac{Vo}{Vi})$
1	100					
2	500					
3	1k					
4	5k					
5	10k					
6	50k					
7	100k					

For low pass filter:

S/N	Frequency	C	R	Vo(Volt)	Gain, $A_v = \frac{Vo}{Vi}$	$Av, dB = 20Log(\frac{Vo}{Vi})$
1	100					
2	500					
3	1k					
4	5k					
5	10k					
6	50k					
7	100k					

Results and Discussion:

1. Plot the Voltage gain vs frequency in semi log graph paper for both High pass and Low pass filter.
2. Calculate the cut-off frequency in both instances and justify with theoretical value.

Knowledge Test Questions:

Pre-Lab Questions

- What is Filter in electronics?
- How to plot semi log graph of two variables?
- How many different filters are there?
- What is Frequency Response Curve?

Post Lab Questions

- Why is second order filter preferable than first order filter?
- What do you understand by 30 dB/decade change in Frequency response curve?

Experiment No: 10
Experiment Name: Investigation of adder circuit and Wien Bridge oscillator using 741 Op-amp.

Objective:

- To learn and simulate adder circuit.
- To learn about Wien Bridge oscillator using 741 Op-amp

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
12	**CO2:** Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.	**PO4–** Investigation	**Affective domain/analyzing level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Theory:

Adder circuit/Summer amplifier:

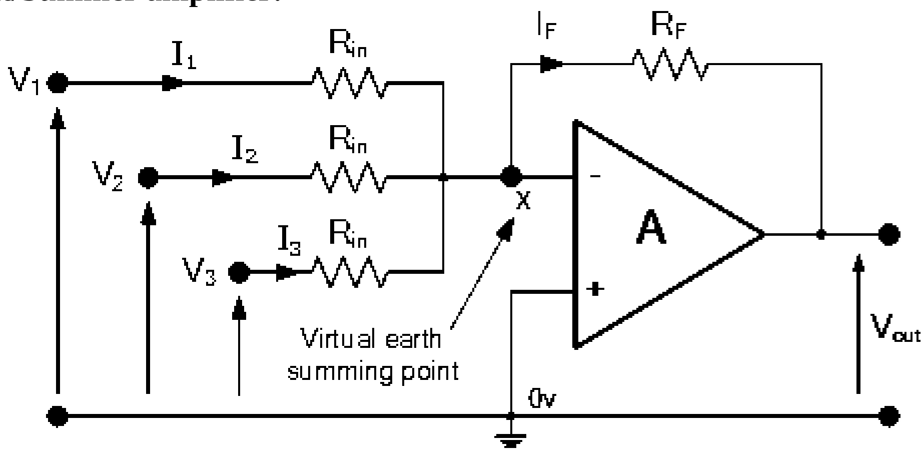


Figure 10.1: Adder circuit

In this simple summing amplifier circuit, the output voltage, (V_{out}) becomes proportional to the sum of the input voltages, V_1, V_2, V_3, etc. Then we can modify the original equation for the inverting amplifier to take account of these new inputs thus:

$$I_F = I_1 + I_2 + I_3 = -R_F(\frac{V_1}{R_{in}} + \frac{V_2}{R_{in}} + \frac{V_3}{R_{in}})$$

This is the equation of Summing amplifier or Op-amp adder circuit. The Amplification factor can change based upon the input resistance with each voltage source.

Wein bridge oscillator:

Any circuit that generates an alternating voltage is called an oscillator. The Wien bridge oscillator is used to generate voltage with frequency in the range of 10Hz to about 1MHz typically. It is used in all commercial audio generators. If $R_2 > 2R_4$, then the output will be a square wave. Its frequency is found out by the following formula.

$$f = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$$

If, $R_1=R_2=R$ and $C_1=C_2=C$ then we can write,

$$f = \frac{1}{2\pi RC}$$

Circuit Diagram to be realized:

- Adder circuit:

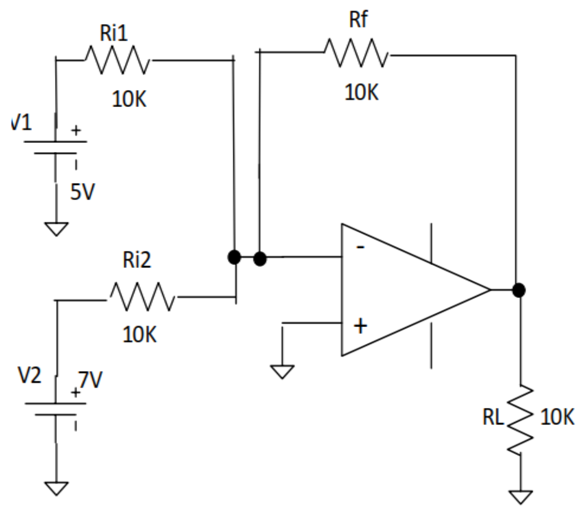


Figure 10.2: Adder Circuit

- Wein bridge oscillator:

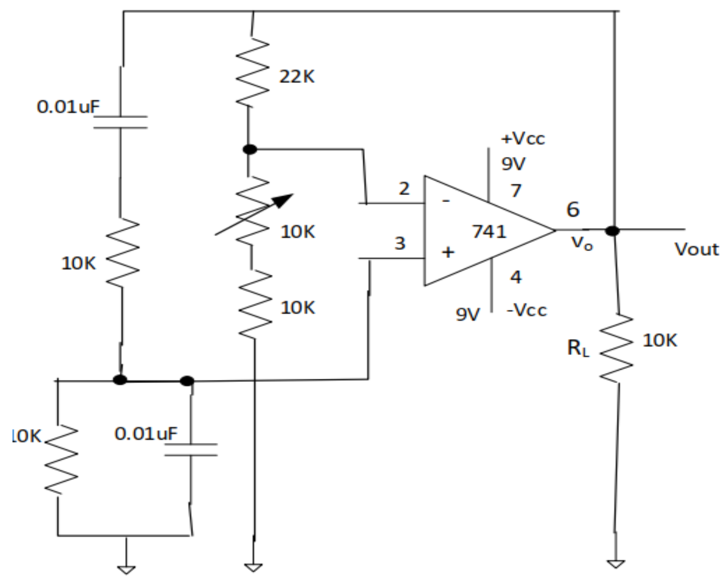


Figure 10.3: Schematic of Wine bridge oscillator

Required Equipment of inverting adder circuit:

- IC: 741 1 Piece
- Resistor: 820 Ohms 1 Piece

- Resistor: 10K 4 Pieces
- Resistor: 15K 1 Piece
- Capacitor: 0.22uF 2 Pieces
- AT-7000 Trainer Board 1 Unit
- DC Power Supply 1 Unit +15 volts, -15 Volts
- Oscilloscope 1 Unit
- Multimeter 1 Unit
- RLC meter
- Wires

Precaution:

- While doing the experiment do not exceed the readings of the Op-Amp. This may lead to damaging of the IC.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Procedure:

For constructing the adder circuit:

1. Construct the above circuits.
2. Set biasing voltages to +15V and -15V.
3. Set input DC voltages as accurately as possible to 5 volts and 7 volts DC.
4. Calculate the output voltage of the inverting adder using a multimeter.
5. Vary the 7 Volt supply through 3 Volts - 9 volts in increments of 2V and observe output with multimeter.

For constructing Wein bridge circuit:

1. Measure accurately the value of the resistors with a multimeter and record.
2. Construct the above circuit. Set biasing voltages to +9V and –9V. Do not exceed ±9 Volts.

3. Keep the potentiometer midway. Once you have got a waveform from the output on the oscilloscope, slowly rotate the shaft of the potentiometer to get a perfect sine wave. Calculate the frequency of the sine wave and calculate the peak to peak voltage t that frequency.
4. Compare the calculated value of the frequency and that from the reading of the oscilloscope.
5. Discuss how would you make a continuously variable frequency generator? Show with a diagram.

Result and Discussion:

1. Change the input voltage and resistance and determine the output voltage. Justify your observation theoretically.
2. Determine the frequency of the sin-wave from wein bridge oscillator.

Knowledge Test Questions:

Pre-Lab Questions

- Why does an Op-amp have infinite input voltage?
- What is common ground in Op-amp?
- What is inverting and non-inverting operation of Op-amp?

Post-Lab Questions

- Why does a Wein bridge oscillator produce sin wave?
- When can a Wein bridge oscillator produce square wave?
- What is Wein bridge oscillator used for?

Experiment No: 11

Experiment Name: Study of 555 Timer IC Internal Circuit and Design an Astable/Free running Multivibrator, Monostable Multivibrator and Bistable Multivibrator Circuit Using 741 Op-amp.

Objective:

To design and study the following circuits using IC 555:

- An astable multivibrator
- A monostable multivibrator
- A bistable multivibrator

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

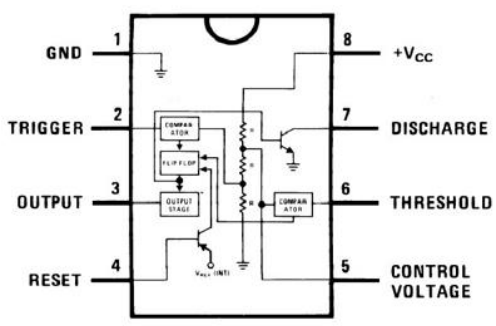
Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
11	**CO2:** Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.	**PO4–** Investigation	**Affective domain/analyzing level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Theory:

Multivibrators: Individual Sequential Logic circuits can be used to build more complex circuits such as Counters, Shift Registers, Latches or Memories etc., but for these types of circuits to operate in a "Sequential" way, they require the addition of a clock pulse or timing signal to cause them to change their state. Clock pulses are generally square shaped waves that are produced by a single pulse generator circuit such as a Multivibrator which oscillates between a "HIGH" and a "LOW" state and generally has an even 50% duty cycle, that is it has a 50% "ON" time and a 50% "OFF" time. Sequential logic circuits that use the clock signal for synchronization may also change their state on either the rising or falling edge, or both of the actual clock signal. There are basically three types of pulse generation circuits depending on the number of stable states,

- Astable - has NO stable states but switches continuously between two states. This action produces a train of square wave pulses at a fixed frequency.
- Monostable - has only ONE stable state and if triggered externally, it returns to its first stable state.
- Bistable - has TWO stable states that produce a single pulse either positive or negative in value.

IC 555 TIMER: The 555 timer IC was first introduced around 1971 by the Signetics Corporation as the SE555/NE555 and was called "The IC Time Machine" and was also the very first and only commercial timer IC available. It provided circuit designers with a relatively cheap, stable, and user-friendly integrated circuit for timer and multivibrator applications. These ICs come in two packages, either the round metal-can called the 'T' package or the more familiar 8-pin DIP 'V' package as shown in figure below. The IC comprises of 23 transistors, 2 diodes and 16 resistors with built-in compensation for component tolerance and temperature drift.



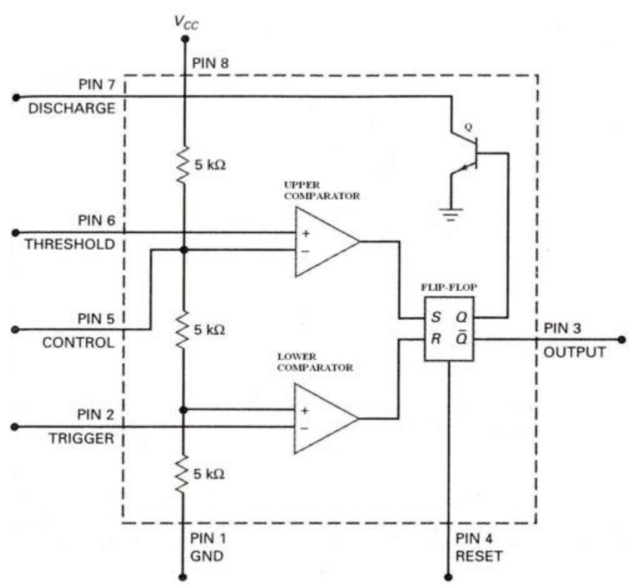


Figure 11.1: 555 and its construction.

The pin connection are as follows:

1. Ground.
2. Trigger input.
3. Output.
4. Reset input.
5. Control voltage.
6. Threshold input.
7. Discharge.
8. $+V_{CC}$. +5 to +15 volts in normal use.

Pin1: Ground. All voltages are measured with respect to this terminal.

Pin2: Trigger. The output of the timer depends on the amplitude of the external trigger pulse applied to this pin. When a negative going pulse of amplitude greater than 1/3 V_{CC} is applied to this pin, the output of the timer high. The output remains high as long as the trigger terminal is held at a low voltage.

Pin3: Output. The output of the timer is measured here with respect to ground. There are two ways by which a load can be connected to the output terminal: either between pin 3 and ground or between pin3 and supply voltage $+V_{CC}$. When the output is low the load current flows through the load connected between pin3 and $+V_{CC}$ into the output terminal and is called sink current. The current through the grounded load is zero when the output is low. For this reason the load connected between pin 3 and $+V_{CC}$ is called the normally on load (we will use this for our circuit) and that connected between pin 3 and ground is called normally off-load. On the other hand, when the output is high the current through the load connected between pin 3 and $+V_{CC}$ is zero. The output terminal supplies current to the normally off load. This current is called source current. The maximum value of sink or source current is 200mA.

Pin4: Reset. The 555 timer can be reset (disabled) by applying a negative pulse to this pin. When the reset function is not in use, the reset terminal should be connected to $+V_{CC}$ to avoid any possibility of false triggering.

Pin5: Control Voltage. An external voltage applied to this terminal changes the threshold as well as trigger voltage. Thus by imposing a voltage on this pin or by connecting a pot between this pin and ground, the pulse width of the output waveform can be varied. When not used, the control pin should be bypassed to ground with a 0.01μF Capacitor to prevent any noise problems.

Pin6: Threshold. When the voltage at this pin is greater than or equal to the threshold voltage 2/3 V_{CC}, the output of the timer is low.

Pin7: Discharge. This pin is connected internally to the collector of transistor Q. When the output is high Q is OFF and acts as an open circuit to external capacitor C connected across it. On the other hand, when the output is low, Q is saturated and acts as a short circuit, shorting out the external capacitor C to ground.

Pin8: $+V_{CC}$. The supply voltage of +5V to + 18V is applied to this pin with respect to ground.

Operation: The functional block diagram shows that the device consists of two comparators, three resistors and a flip-flop. A comparator is an OPAMP that compares an input voltage and indicates whether an input is higher or lower than a reference voltage by swinging into saturation in both the direction. The operation of the 555 timer revolves around the three resistors that form a voltage divider across the power supply to develop the reference voltage, and the two comparators connected to this voltage divider. The IC is quiescent so long as the trigger input (pin 2) remains at $+V_{CC}$ and the threshold input (pin 6) is at ground. Assume the reset input (pin 4) is also at $+V_{CC}$ and therefore inactive, and that the control voltage input (pin 5) is unconnected.

The three resistors in the voltage divider all have the same value (5K in the bipolar version of this IC and hence the name 555), so the trigger and threshold comparator reference voltages are 1/3 and 2/3 of the supply voltage, respectively. The control voltage input at pin 5 can directly affect this relationship, although most of the time this pin is unused. The internal flip-flop changes state when the trigger input at pin 2 is pulled down below $+V_{CC}/3$. When this occurs, the output (pin 3) changes state to $+V_{CC}$ and the discharge transistor (pin 7) is turned off. The trigger input can now return to $+V_{CC}$; it will not affect the state of the IC.
However, if the threshold input (pin 6) is now raised above $+(2/3)V_{CC}$, the output will return to ground and the discharge transistor will be turned on again. When the threshold input returns to ground, the IC will remain in this state, which was the original state when we started this analysis. The easiest way to allow the threshold voltage (pin 6) to gradually rise to $+(2/3)V_{CC}$ is to connect it externally to a capacitor being allowed to charge through a resistor. In this way we can adjust the R and C values for almost any time interval we might want.

IC 555 Timer as Multivibrator: The 555 can operate in either mono/bi-stable or astable mode, depending on the connections to and the arrangement of the external components. Thus, it can either produce a single pulse when triggered, or it can produce a continuous pulse train as long as it remains powered.

Astbale Multivibrator:

These circuits are not stable in any state and switch outputs after predetermined time periods. The result of this is that the output is a continuous square/rectangular wave with the properties depending on values of external resistors and capacitors. Thus, while designing these circuits following parameters need to be determined:

1. Frequency (or the time period) of the wave.
2. The duty cycle of the wave.

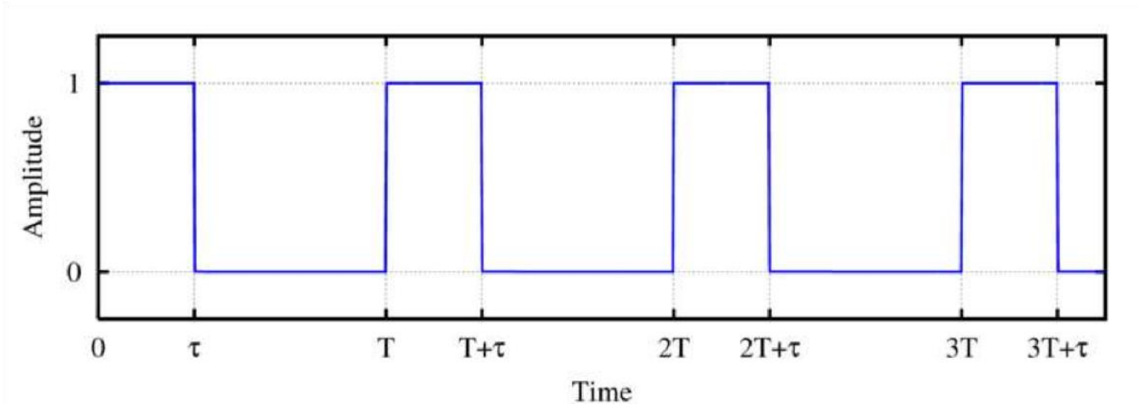


Figure 11.2: A rectangular waveform

Referring to the above figure of a rectangular waveform, the time period of the pulse is defined as T and duration of the pulse (ON time) is τ. Duty cycle can be defined as the On time/Period that is, τ/T in the above figure. Obviously, a duty cycle of 50% will yield a square wave.
The key external component of the astable timer is the capacitor. An astable multivibrator can be designed as shown in the circuit diagram (with typical component values) using IC 555, for a duty cycle of more than 50%. The corresponding voltage across the capacitor and voltage at output is also shown. The astable function is achieved by charging/discharging a capacitor through resistors connected, respectively, either to V_{CC} or GND. Switching between the charging and discharging modes is handled by resistor divider R1-R3, two Comparators, and an RS Flip-Flop in IC 555. The upper or lower comparator simply generates a positive pulse if V_C goes above 2/3 V_{CC} or below 1/3 V_{CC}. And these positive pulses either SET or RESET the Q output.

The time for charging C from 1/3 to 2/3 Vcc, i.e, ON Time = 0.693 (R_A + R_B). C The time for discharging C from 2/3 to 1/3 Vcc, i.e. OFF Time = 0.693 R_B. C To get the total oscillation period, just add the two:

$$T_{osc} = 0.693\cdot(R_A+R_B)\cdot C + 0.693\cdot(R_B)\cdot C = 0.693 \cdot (R_A + 2\cdot R_B) \cdot C$$

Thus,

$$f_{osc} = 1/ T_{osc} = 1.44/(R_A + 2\cdot R_B).C$$

$$\text{Duty cycle} = R_A+R_B/ R_A + 2\cdot R_B$$

Circuit diagram:

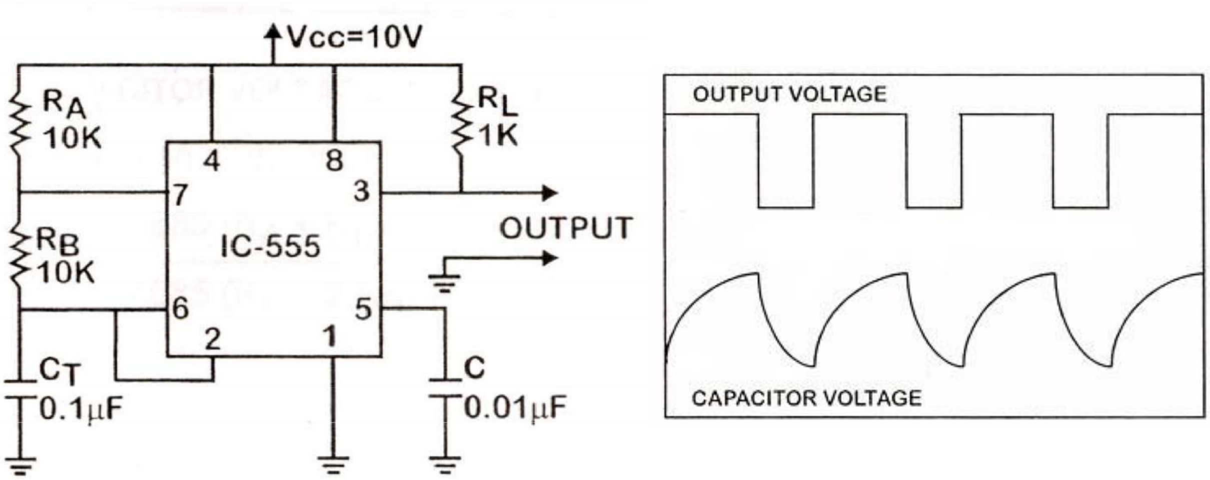


Figure 11.3: Circuit Diagram of astable multivibrator with duty cycle less than 50%

Astable multivibrator with duty cycle less than 50%:

Generally astable mode of IC 555 is used to obtain the duty cycle between 50 to 100%. But for a duty cycle less than 50%, the circuit can be modified as per the circuit diagram. Here a diode D1 is connected between the discharge and threshold terminals (as also across R_B). Thus the capacitor now charges only through R_A (since R_B is shorted by diode conduction during charging) and discharges through R_B. Another optional diode D2 is also connected in series with R_B in reverse direction for better shorting of R_B. Therefore, the frequency of oscillation and duty cycle are

$$f_{osc} = 1/\ T_{osc} = 1.44/(R_A + R_B).C$$

$$\text{Duty Cycle} = R_A/\ (R_A + R_B)$$

Monostable multivibrator:

Monostable multivibrator often called a one shot multivibrator is a pulse generating circuit in which the duration of this pulse is determined by the RC network connected externally to the 555 timer. In a stable or standby state, the output of the circuit is approximately zero or a logic-low level. When external trigger is applied (See circuit diagram) output is forced to go high (□ V_{CC}). The time for which output remains high is determined by the external RC network connected to the timer. At the end of the timing interval, the output automatically reverts back to its logic-low stable state. The output stays low until trigger pulse is again applied. Then the cycle repeats. The monostable circuit has only one stable state (output low) hence the name monostable.

Initially when the circuit is in the stable state i.e, when the output is low, transistor Q in IC 555 is ON and the capacitor C is shorted out to ground. Upon the application of a negative trigger pulse to pin 2, transistor Q is turned OFF, which releases the short circuit across the external capacitor C and drives the output high. The capacitor C now starts charging up towards V_{CC} through R.

When the voltage across the capacitor equals 2/3 V_{CC}, the upper comparator's (see schematics of IC 555) output switches from low to high, which in turn drives the output to its low state via the output of the flip-flop. At the same time the output of the flip-flop turns transistor Q ON and hence the capacitor C rapidly discharges through the transistor. The output of the monostable remains low until a trigger pulse is again applied. Then the cycle repeats. The pulse width of the trigger input must be smaller than the expected pulse width of the output waveform. Also the trigger pulse must be a negative going input signal with amplitude larger than 1/3 V_{CC} (Why?). The pulse width can be calculated as (How?): T= 1.1 R.C.

Once triggered, the circuit's output will remain in the high state until the set time, T, elapses. The output will not change its state even if an input trigger is applied again during this time interval. The circuit can be reset during the timing cycle by applying negative pulse to the reset terminal. The output will remain in the low state until a trigger is again applied. The circuit is designed as shown in the circuit diagram, the left part of which shows how to generate negative a trigger pulse from a square wave signal.

Circuit Diagram:

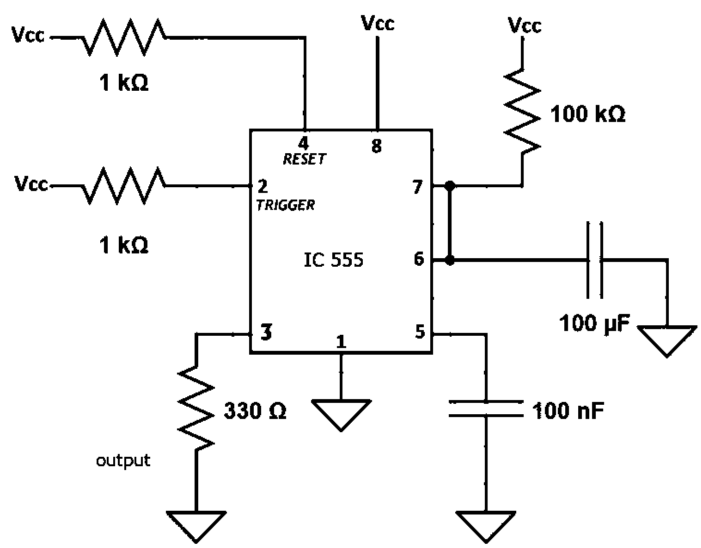


Figure 11.4: Circuit Diagram of monostable multivibrator.

Bistable Multivibrator:

In these circuits, the output is stable in both the states. The states are switched using an external trigger but unlike the monostable multivibrator, it does not return back to its original state. Another trigger is needed for this to happen. This operation is similar to a flip-flop. There are no RC timing network and hence no design parameters. The following circuit can be used to design a bistable multivibrator. The trigger and reset inputs (pins 2 and 4 respectively on a 555) are held high via pull-up resistors while the threshold input (pin 6) is simply grounded. Thus configured, pulling the trigger momentarily to ground acts as a 'set' and transitions the output pin (pin 3) to

Vcc (high state). Pulling the threshold input to supply acts as a 'reset' and transitions the output pin to ground (low state). No capacitors are required in a bistable configuration.

Circuit Diagram:

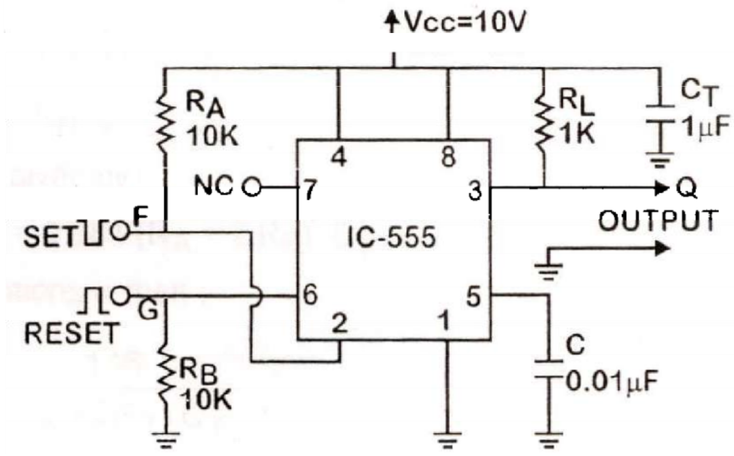


Figure 11.5: Circuit Diagram of Bistable Multivibrator

Equipment:

- IC 555 (1 No.)
- Resistors :1KΩ (2 pieces); 10KΩ (2 pieces) ; 2.7KΩ (1 pieces)
- Potentiometer: 10 KΩ, (1 piece)
- Capacitors:0.01 μF, 0.047 μF, 0.1 μF, 1 μF; 1 piece each)
- Diodes 1N 4148 (2 pieces)
- D.C. Power supply (10V)
- Function Generator
- Oscilloscope
- Connecting wires
- Breadboard

Precautions:

- While doing the experiment do not exceed the readings of the Op-Amp. This may lead to damaging of the IC.
- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.
- Check the DC voltage input for 555 Timer IC.

Procedure:

- Astable Multivibrator: (For duty cycle more than 50%)

1. Configure the circuit as per the circuit diagram.
2. Use $R_A = R_B = 10\ k\Omega$, $R_L = 1\ k\Omega$ and $C_T = 0.1\ \mu F$, $C = 0.01\ \mu F$. Using the power supply set $V_{CC} = 10$ V.
3. Compute the expected values of f_{osc} and duty cycle (%).
4. Connect the output terminal (pin 3) to channel 1 of the oscilloscope. Also feed the voltage across capacitor to channel 2.
5. Power on your circuit and observe and save the output. Determine the values of f_{osc} and duty cycle (%) from your observations and compare with the theoretical values.
6. When you are done, turn off the power to your experimental circuit.

- Monostable Multivibrator

1. Configure the circuit as per the circuit diagram. set $V_{CC} = 10$ V
2. Connect the output terminal (pin 3) to the oscilloscope.
3. Power on your circuit and trigger the circuit using momentarily connecting jumper wire from 2 to ground and observe the output.
4. Observe time for which output is high, compare with RC value. While output is high, try triggering repeatedly and observe the output.

- Bistable Multivibrator

1. Configure the circuit as per the circuit diagram.
2. Use $R_A = R_B = 10\ k\Omega$, $R_L = 1\ k\Omega$ and $C_1 = 1\ \mu F$, $C_2 = 0.01\ \mu F$. Using the power supply set $V_{CC} = 10$ V.
3. Connect the output terminal (pin 3) to the oscilloscope in DC COUPLING mode.
4. Power on your circuit.
5. Connect the point F to ground momentarily. This will set the output Q in the oscilloscope to 1 or HIGH level. This state will be permanently stable state and the operation is called "SET".
6. Now connect the point G to V_{CC} momentarily. This will set the output Q in the oscilloscope to 0 or LOW level. This is called "RESET" operation.
7. When you are done, turn off the power to your experimental circuit

Observations:

I. Astable Multivibrator:

$R_A = R_B =$ __________ $k\Omega$, $C_T =$ _________ μF

Output waveform and capacitor voltage as observed in oscilloscope: (use data here)

Parameters	Calculated value	Observed value	Error
fosc			
Duty cycle (%)			

II. Monostable Multivibrator:

R_A = _________ kΩ, C_T = _________ μF

Parameter	Calculated value (ms)	Observed value (ms)	Error
Output high duration			

III. Bistable Multivibrator:

Point	Connected to	Output
F	Ground	
G	VCC	

Knowledge Test Questions:

Pre-lab Questions

- What is a multivibrator circuit?
- Write the truth table of SR flipflop.
- What is a comparator circuit?

Post-Lab Question

- How can you vary duty cycle from 0% to 100% with an astable multivibrator?
- What is the function of discharge pin in 555 Timer?
- What are applications of Monostable Multivibrator?

Experiment No: 12

Experiment Name: Design and investigation of Class A and B power amplifiers.

Objective:

- To learn how to design and analyze the Class A and Class B Power Amplifier using TIP31 BJT power transistor.

Course Outcomes (COs), Program Outcomes (POs) and Assessment:

Expt No.	CO Statement	Corresponding PO	Domain / level of learning taxonomy	Delivery methods and activities	Assessment tools
12	**CO2:** Design and develop of different electronic devices using Diode, BJT, MOSFET and op-amp for the procedure of amplification, rectification, mathematical operation, filtering and oscillation.	**PO4–** Investigation	**Affective domain/analyzing level**	☐ Simulation ☑ Experiment ☑ Practice lab ☑ Group discussion ☑ Tutorial	☑ Lab tests ☑ Lab reports ☑ Final lab test ☐ Open ended lab ☐ Project show & project presentation

Theory:

The power amplifiers are said to be Class A amplifiers if the Q point and the input signal are selected such that the output signal is obtained for a full input signal cycle. For all values of the input signal, the transistor remains in the active region and never enters into the cut-off or saturation region. When an A.C signal is applied, the collector voltage varies sinusoidally hence the collector current also varies sinusoidally. The collector current flows for 3600 (full cycle) of the input signal, i.e. the angle of the collector current flow is 3600.

Class A Power amplifier:

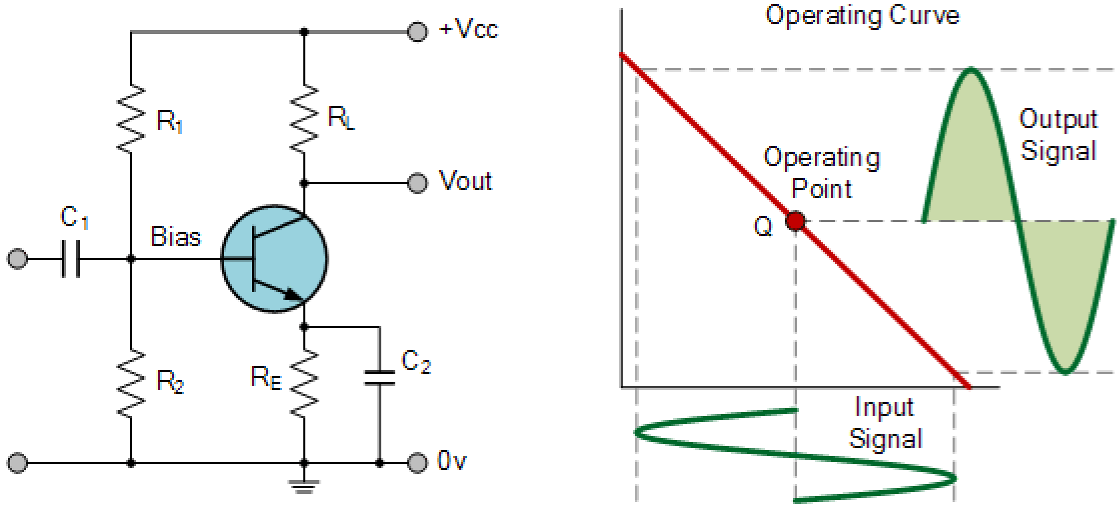


Figure 12.1: Class A power Amplifier Using a npn Transistor

To achieve high linearity and gain, the output stage of a class A amplifier is biased "ON" (conducting) all the time. Then for an amplifier to be classified as "Class A" the zero signal idle current in the output stage must be equal to or greater than the maximum load current (usually a loudspeaker) required to produce the largest output signal. As a class 'A' amplifier operates in the linear portion of its characteristic curves, the single output device conducts through a full 360 degrees of the output waveform. Then the class 'A' amplifier is equivalent to a current source. Since a class 'A' amplifier operates in the linear region, the transistors base (or gate) DC biasing voltage should by chosen properly to ensure correct operation and low distortion. However, as the output device is "ON" at all times, it is constantly carrying current, which represents a continuous loss of power in the amplifier.

Due to this continuous loss of power class 'A' amplifiers create tremendous amounts of heat adding to their very low efficiency at around 30%, making them impractical for high-power amplifications. Also due to the high idling current of the amplifier, the power supply must be sized accordingly and be well filtered to avoid any amplifier hum and noise. Therefore, due to the low efficiency and overheating problems of Class A amplifiers, more efficient amplifier classes have been developed.

Circuit diagram of Class A power amplifier to be realized:

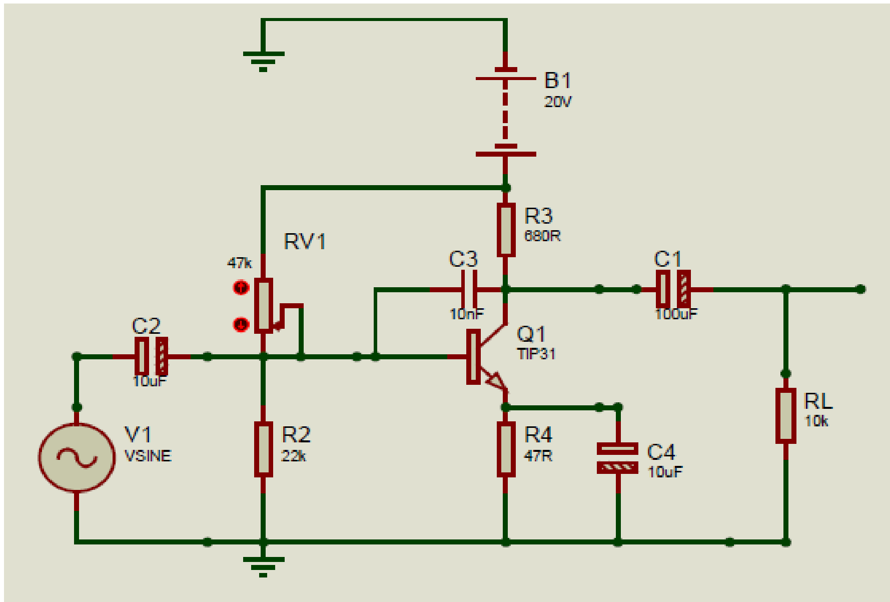


Figure 12.2: Circuit Diagram for Class 'A' amplifier circuit with Data

Class 'B' Power Amplifier:

When an amplifier is biased at cut-off such that it operates in a linear region for 180 degrees of the input cycle and it is in cut-off for 180 degrees it is a class 'B' amplifier. To avoid this we use two transistors and it turns on only when input is applied this essentially has no bias and the transistor conducts current for one-half of the signal cycle. To obtain o/p for the full cycle of the signal, it is necessary to use two transistors and have each conduct on opposite half-cycles, the combined operation provides a full cycle of the output signal. Since one part of the circuit pushes the signal high during one-half cycle and the other part pulls the signal low during the other half cycle, the circuit is referred to as a push-pull amplifier circuit.

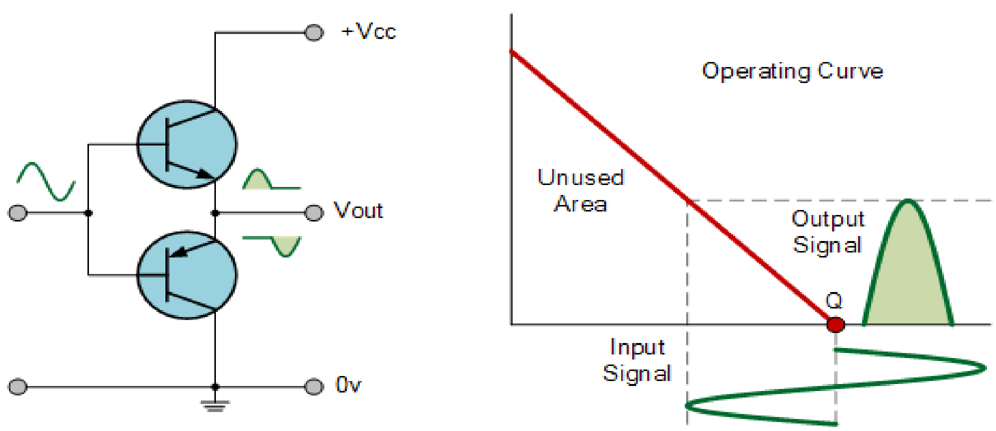


Figure 12.3: Operating Curve

Circuit Diagram of Class 'B' amplifier to be realized:

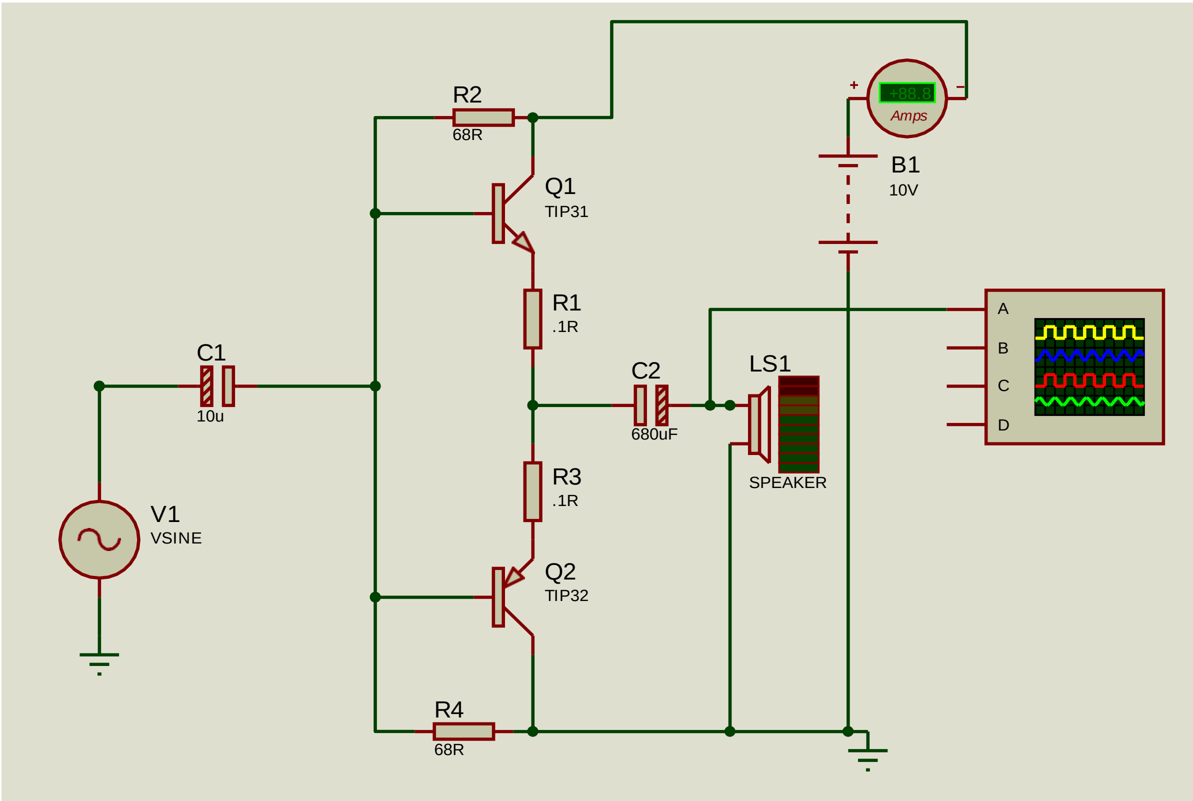


Figure 12.4: Circuit Diagram of Class 'B' Amplifier

Equipment:

- Regulated Power Supply (RPS); Dual / (0-30)V; 1 piece;
- Function Generator, Analog/(0-3)MHz; 1 piece;
- CRO, Analog/(0-30)MHz; 1 piece;
- Connecting Wires, Single Strand; As required;
- Transistor; TIP 131 & TIP 132; 1 piece;
- Resistors; 0.1 Ohms 2W(2 piece), 820 Ohms(1 piece), 330K(1 piece), 68K(1 piece);
- Capacitor; 10uF(1 piece), 680uF(1 piece), 150pF(1 piece);
- Diode; IN4148; 3 pieces;
- Potentiometer; 10K; 1 piece;

Precaution:

- While doing the experiment do not exceed the readings of the BJT IC. This may lead to damaging it.

- Connect voltmeter and ammeter in correct polarities as shown in the circuit diagram.
- Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

Procedure:

1. Connect the circuit as per the circuit diagram.
2. Set V1 using the signal generator to **50 mVp-p**.
3. Set RV1 to middle position (initially), adjust it to get the maximum voltage output with no distortion
3. Keeping the **input voltage constant**, vary the frequency from 50 Hz to 500 KHz in regular steps and note down the corresponding output voltage.
4. Plot the gain (gain in dB) vs Frequency curve in a semi-log graph paper to find the bandwidth. Mark the -3dB points and the lower and upper cutoff frequencies.

Observation:

For Class ‘A’ amplifier:

Frequency	Input voltage	Output voltage	Gain in dB = 20 $\log_{10}$(Vout/Vin)
50	50mV		
100	50mV		
200	50mV		
500	50mV		
1k	50mV		
10k	50mV		

For class ‘B’ amplifier:

Frequency	Input voltage	Output voltage	Gain in dB = 20 $\log_{10}$(Vout/Vin)
50	50mV		
100	50mV		
200	50mV		
500	50mV		
1k	50mV		

10k	50mV		

Calculation and Result:

Calculate the output power, DC input power and efficiency of both amplifiers.
For class 'A' amplifier,

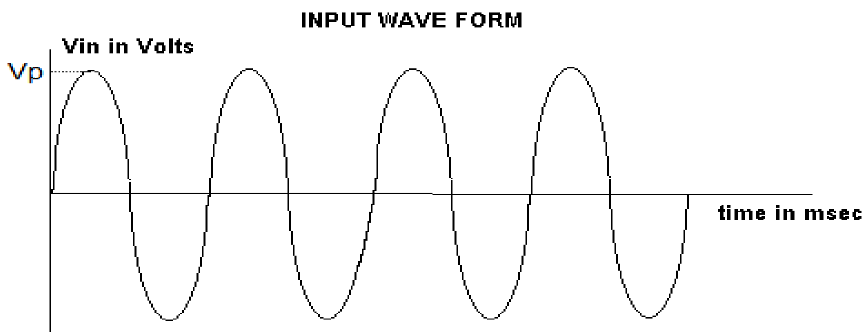
$$\eta = \frac{P_{OUT}}{P_{DC}} * 100\%$$

Do, determine P_{OUT} we can be measured using multimeter in the output side. For calculating P_{DC}, we can use the equation below,

$$P_{DC} = V_{supply} * I_C$$

Graph:

Output graph from class 'A' and 'B' amplifiers.



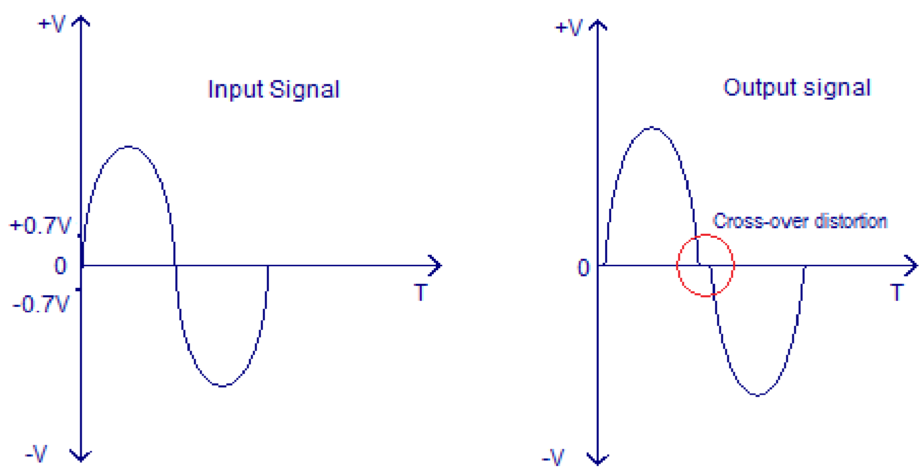


Figure12.5: Class 'A' and Class 'B' amplifier output graph.

Discussions: There are some advantages and disadvantages of using class 'B' amplifier. Those are, Advantages:

- High efficiency when compared to the Class A configurations.
- Push-pull mechanism avoids even harmonics.
- No DC components in the output (in ideal case).

Disadvantages:

- The major disadvantage is the cross-over distortion.
- Coupling transformers increases the cost and size.
- It is difficult to find ideal transformers.
- Transformer coupling causes hum in the output and also affects the low frequency response.
- Transformer coupling is not practical in case of huge loads.

Knowledge Test Questions:

Pre-Lab Questions	Post-Lab Questions
• What is an amplifier? • Why do we use amplifier? • Why do we have different classes of amplifier?	• What is the efficiency of Class 'A' and class 'B' amplifier? • How can amplifier efficiency be increased?